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Ministry of Health Malaysia

Malaysian DIETARY GUIDELINES FOR OLDER PERSONS

National Coordinating Committee on Food and Nutrition (NCCFN)

Ministry of Health Malaysia

2023

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Malaysian DIETARY GUIDELINES FOR OLDER PERSONS

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Message by Minister Ministry of Health Malaysia

The increasing trend of ageing population at the global level, including Malaysia is inevitable. The aged population has its own unique problems and will pose challenges and demands on the healthcare services. The prevalence of morbidity and mortality associated with various forms of malnutrition among this aged population is indeed on the rise. This could be attributed to the shifts in dietary patterns, physiological changes such as dental health status, changes in lifestyles, quality and quantities of food consumed and low socioeconomic status. Non-communicable diseases and malnutrition are important causes of mortality and morbidity among older persons.

The government has continously been planning and implementing various strategies to increase nutrition and health status among older persons. Thus, it is a shared responsibility between government, private sectors, non-governmental agencies and the community in ensuring the nutritional well-being for older persons. The Ministry of Health Malaysia is committed to improving the health of its population by providing appropriate guidance on nutrition and food preferences.

The establishment of the Malaysian Dietary Guidelines for Older Persons 2023 (MDGOP) is timely, with the objectives of supporting older persons and assisting them in achieving the Malaysian recommended nutrient intakes. This dietary guideline focuses on the disease prevention of older persons via the adoption of dietary habits and practices.

Therefore, it is anticipated that all the relevant ministries, agencies, academic institutions, hospitals, the healthcare industries, professional organisations, and the food industries will adopt and apply the MDGOP 2023 to meet the demands of the older persons looking for dietary guidance. The great efforts of the Technical Working Group on Nutritional Guidelines, under the auspices of the National Coordinating Committee on Food and Nutrition (NCCFN), in establishing this guideline for older persons in the country, is commendable.

YB Dr. Zaliha Mustafa
Minister
Ministry of Health Malaysia

Foreword by
Director-General of Health
Ministry of Health Malaysia

Malaysia would be an ageing nation by 2030, with approximately 15% of the population are above 60 years old. There is an increasing trend of individuals aged 60 and above with a declining percentage of young people aged 14 and below. This demographic shift underscores the changing of the age distribution in the country which is going faster than initially expected.

As people age, their intrinsic capacity such as their physical and mental capacities tends to decline while their health issues become more chronic and complex. They are prone to having decreased muscle mass, diminished organ function, and a greater susceptibility to illnesses, food and nutrition insecurity, as well as poor nutritional status. Older persons could also develop an empty nest syndrome that is portrayed by feelings of loss, sadness, anxiety, grief, irritability and fear.

Therefore, addressing these challenges requires a holistic approach, such as accessibility to a good health-care system, including updated health and nutrition information, social support, and community resources to help them navigate the challenges, ensure better quality of life for older persons, and at the same time they could contribute to society. Among older people, proper nutrition is a key strategy to prevent or delay onset of chronic disease and functional decline, and to support quality of life.

Thus, I believe that the publication of the Malaysian Dietary Guidelines for Older Persons (MDGOP) is crucial in our effort to adopting healthy eating habits. Health-care professionals and related stakeholders also require reliable nutrition education resources could obtain from this guideline.

It is my fervent hope that all the healthcare professionals need to ensure vital nutrition information is effectively communicated and not overlooked. I would extend my congratulations to the Technical Working Group on Nutritional Guidelines and sub-committee of the MDGOP for their endless efforts for the completion of this guideline.

Datuk Dr. Muhammad Radzi Bin Abu Hassan
Director-General of Health
Ministry of Health Malaysia

Preface by Deputy Director-General of Health (Public Health) Ministry of Health Malaysia

Good nutrition is a fundamental component of health and quality of life of older persons. Malnutrition is one of the underlying causes that may deteriorate health problems such as an increased risk of morbidity for chronic diseases, and low quality of life (QoL). The National and Health Morbidity Survey (NHMS) 2018 reported that the prevalence of malnutrition was 30.8%, meanwhile the prevalence of self-reported diabetes, hypertension, hypercholesterolemia and overweight/obesity among older persons, was 23.3%, 42.2%, 35.6%, 58.4%, respectively. In addition, 10.3% of them were experiencing food insecurity. These non-communicable diseases (NCDs) could diminish their quality of life, raise health-care costs, and increase pressures on family members who are responsible for their care. Thus, adhering to a healthy diet plays an essential role in preventing and controlling various nutrition-related diseases.

In this respect, the Malaysian Dietary Guidelines for Older Persons (MDGOP) has been established to provide appropriate dietary guidance, in ensuring a well-balanced diet for these populations. My sincere appreciation goes to the Technical Working Group on Nutritional Guidelines and the sub-committee of MDG for Older Persons for their tremendous efforts for the completion of this guideline. My heartfelt gratitude also goes out to all the individuals who have contributed in various ways, directly or indirectly towards the development of the Malaysian Dietary Guidelines for Older Persons 2023.

Datuk Dr. Norhayati Binti Rusli
Deputy Director-General of Health (Public Health)
Ministry of Health Malaysia

Preface by Chairman Malaysian Dietary Guidelines for Older Persons

The population of older persons in Malaysia has increased over the last decade in parallel with the increasing lifespan. Finding form the National Health Morbidity Survey (NHMS), 2018 have reported emerging dietary issues and diet related health problems among these older persons. Adequate food intake ensures the nutritional well-being of older persons as they are at risk of decreased food intake. Inadequate food intake may be due to physiological changes such as maldigestion, malabsorption, dry mouth, poor appetite, chronic diseases and illnesses. In giving dietary advice and recommendations to the older persons, the food-based dietary guideline is more applicable than a nutrient-based guideline in view of the fact that culturally sensitive and familiar foods can be incorporated into the recommendations for the older persons.

The Malaysian Dietary Guidelines for Older Persons 2023 first of its kind, takes into consideration the studies reporting the dietary intake, food consumption pattern and their nutritional as well as health related problems. The MDG 2023 comprise of 6 key messages with 33 key recommendations and 196 how to achieve prepared by the sub-committee of the MDG for Older Persons. This MDG will provide useful information to related stakeholders to guide and advise older persons in making healthier and wholesome food choices.

I would like to thank the writers, Focus Group Discussion members, the Editorial team, the Consensus Workshop participants, the TWG secretariat and all those who assisted in developing this valuable document for their hard work and dedication.

Emerita Prof. Dr. Norimah A Karim
Chairman
Malaysian Dietary Guidelines for Older Persons

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Individuals from the various Departments and Institutes, the Ministry of Health Malaysia. Academicians from local universities, Nutritionists, Dietitians, representatives from related professional bodies, representatives from the food manufacturing and trading industry, and consumer bodies are all acknowledged by the Technical Working Group on Nutritional Guidelines. Their invaluable contributions and dedication to complete this document are sincerely appreciated.

A word of gratitude is also conveyed to the:

- Director, Food Safety and Quality Division, MOH
- Directors, State Department of Health (all over Malaysia), MOH
- Director, Kuala Lumpur Hospital
- Director, Sultanah Aminah Johor Bharu Hospital
- Director, Tengku Ampuan Afzan Hospital, Kuantan
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- Dean, Faculty of Social Sciences and Humanities, Universiti Teknologi Malaysia
- Dean, Faculty of Applied Sciences, UCSI University
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for their generous support and cooperation.

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Coordination and Documentation

The documentation of the Malaysian Dietary Guidelines (MDG) for Older Persons was coordinated by the Technical Working Group (TWG) on Nutritional Guidelines, which is under the purview of the National Coordinating Committee on Food and Nutrition (NCCFN), Ministry of Health Malaysia. The Nutrition Division, Ministry of Health Malaysia served as the secretariat for the Malaysian Dietary Guidelines for Older Persons.

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Executive Summary

The Malaysian Dietary Guideline (MDG) for Older Persons (MDGOP) 2023 is the first MDG of its kind for the older persons population (60 years old and above) in Malaysia. As of the 2020 consensus, the older persons make up 10.3% of the total Malaysian population. Just as in the adult population, the older persons are facing double burden of malnutrition by the coexistence of undernutrition problems (underweight, muscle wasting, at risk of malnutrition, vitamin D deficiency) along with overnutrition problems (overweight, obesity, abdominal obesity) and diet related non communicable diseases (hypertension, diabetes mellitus and cardiovascular diseases). The recent NHMS (2018) which focused on the older persons population revealed that 5.2% are underweight, 37% overweight, 17.6% obese and 36.4% are centrally obese (abdominal obesity). With regards to dietary intake, only 10.8% and 10.9% of older persons met the recommended intake of fruits (2 servings per day) and vegetables (at least 3 servings per day) respectively. Traditional diets have been replaced by diets high in fats, salt and sugar. There is an increased sugar sweetened beverages intake and inadequate consumption of fruits and vegetables. Hence, it is very timely that the MDGOP 2023 is developed to provide specific recommendations and advice on healthy and wholesome diets and lifestyle.

The MDGOP 2023 is written by a group of experts from the academia, Ministry of Health Malaysia and related professional bodies who have extensive knowledge and experience of Nutrition and Health Sciences. Several drafts were reviewed, validated and approved by the TWG Nutritional Guidelines. The Key Messages, Key Recommendations and How To Achieve were then vetted through Focus Group Discussion (FGD) held on 23th November 2022, comprising of Public Health Personnel (end users) to determine the relevance and clarity of the statements. The final draft was presented in a Consensus Meeting held on 9th August 2023 involving various ministries, government agencies, academia, professional bodies, industries and Consumer Associations for revision and approval.

The MDGOP 2023 are intended for health professionals, policy makers, educators, food manufacturers and researchers. It applies to all healthy older persons as well as those with common health conditions such as malnutrition. They do not apply to older persons who needs special dietary advice for a medical condition.

The MDGOP 2023 features some of the following keypoints:

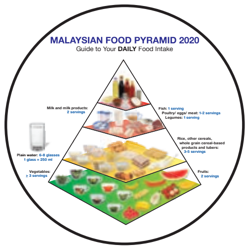
- The MDGOP 2023 uses the Recommended Nutrient Intakes for Malaysia (2017) that provide the nutrient intake requirement for older persons while combination of various food groups were used to formulate diets to meet the nutrient requirements.
- The MDGOP 2023 is based on foods that are available, accessible culturally appropriate and sustainable for the older persons population. These are used to develop recommended diet patterns that fulfil nutrient intake requirements and address emerging nutrition concerns and problems of the older persons.
- It is developed for older persons population aged 60 years old and above.
- The MDGOP 2023 uses the Malaysian Food Pyramid 2020 and Malaysian Healthy Plate to help with diet plan and guides to create healthy, balanced and wholesome meal.
- The MDGOP 2023 has 6 Key Messages and each key message is supported with recent scientific evidence and current nutritional and health status of older persons.
- The MDGOP 2023 provides recommendations that include many choices. The guidelines focuses on dietary patterns that promote health and wellbeing. It has 33 Key Recommendations and 196 ways on "how to achieve" to guide users to make informed choices towards healthier eating habits.

Key Messages of the Malaysian Dietary Guidelines for Older Persons 2023

Key Message 1

Key Message 1 :

Eat a variety of wholesome foods within your recommended servings



Key Message 2

Key Message 2 :

Maintain an optimal body weight and body composition for healthy ageing

Key Message 3

Key Message 3 :

Be physically active for optimal health

Key Message 4

Key Message 4 :

Drink plenty of water daily

Key Message 5

Key Message 5 :

Consume foods and beverages with low fat, salt and sugar

Key Message 6

Key Message 6 :

Practise food safety when purchasing, preparing, cooking and storing food

Key Message 1

Eat a variety of wholesome foods within your recommended servings

KM1

Eat a variety of wholesome foods within your recommended servings

Key Message 1

Eat a variety of wholesome foods within your recommended servings

Prof. Dr. Suzana Shahar, Prof. Dr. Sakinah Harith, Assoc. Prof. Dr. Siti Nur'Asyura Adznam, Assoc. Prof. Dr. Tanti Irawati Rosli, Dr. Megan Chong Hueh Zan, and Ms. Rohida Saleh Hudin

1.1 Terminology

Adequate diet
An adequate diet provides enough energy, nutrients and fibre to maintain an individual's health. A diet that is adequate for one individual may not be adequate for another.

Ageing
Ageing is mostly defined as an age-dependent or age-progressive decline in intrinsic physiological function, leading to an increase in age specific mortality rate (i.e., a decrease in survival rate) and a decrease in age-specific reproductive rate.

Balanced diet
A balanced diet is a diet that contains a combination of foods that provide a proper balance of nutrients. The body needs many types of foods in varying amounts to maintain health. The right balance of nutrients needed to maintain health can be achieved by eating proper balance of all healthy foods including fruit, vegetable, cereal, fish, meat, legume and milk.

Calcium rich food
The foods rich in calcium are dairy products (milk, cheese and yogurt) and non-dairy sources (seafood, leafy greens, legumes, dried fruits, tofu) and various foods that are fortified with calcium.

Community
The community in the context of this guidelines refers to the people living in the same place (neighbours) or within the vicinity of the housing area.

Food groups

A food group puts together foods of similar nutrient content and function. There are five food groups which are vegetables; fruits; rice, other cereals, whole grain cereal-based products and tubers; fish, poultry, eggs, meat, and legumes; milk and milk products. These food groups contain foods that are similar in calories, carbohydrate, protein and fat contents.

Health supplement

Health supplement refers to any form of product i.e. capsules, tablets, powder or liquids used to supplement a diet and to maintain, enhance and improve the health function of the human body. Health supplement may contain one or more ingredients derived from natural or synthetic sources including vitamins, minerals, enzymes, probiotics and other bioactive substances.

Healthy diet

A healthy diet is a diet which provides the proper combination of energy and nutrients. The four characteristics of a healthy diet are varied, adequate, balanced and moderate.

Main meal

The type of meal served or eaten includes prepared food at a certain time. In most modern cultures, three main meals are breakfast, lunch and dinner.

Malaysian Food Pyramid 2020

A food pyramid is a visual tool that is used as a guide to your DAILY food intake in achieving a healthy diet. It is developed to provide a guide for the types and amounts of food that can be eaten in combination to provide a balanced diet. A food pyramid consists of four levels that represent five food groups. The recommended number of servings per day for each food group is indicated next to it. From the bottom to the top of the food pyramid, the number of servings of each food group becomes smaller indicating that an individual should eat more of the foods at the base of the pyramid and less of the foods at the top of the pyramid.

Malaysian Healthy Plate

Malaysian Healthy Plate (MOH, 2016) is a visual guide to show the total food in each food group that needs to be consumed in a meal to achieve a healthy and balanced diet based on the principle of quarter, quarter, half. It is used to translate recommendations from the Malaysian Dietary Guidelines and Malaysian Food Pyramid to help Malaysian practise healthy eating habits by planning their daily meal.

Moderation

Moderation is a key to healthy diet. Moderation refers to eating the right amount of foods to maintain a healthy weight and to optimise the body's metabolic process.

Nourishing formula

Nourishing formula are pre-prepared formulas intended to supplement nutrients and fluid when an individual experiences low appetite and/ or poor oral intake. These are not meant to be used as replacement of main meals.

Nutrient-dense foods

Nutrient density is a measure of the amount of nutrients a food contains in comparison to the number of calories. A food is more nutrient dense when the level of nutrients is higher in relation to the number of calories the food contains. By eating these high nutrient density foods, a person will get more of the essential nutrients needed for excellent health, including vitamins, minerals, phytonutrients, essential fatty acids, fibres and lesser amount of calories.

Nutritious snacks

Nutritious snacks are lower in fat, sugar, sodium and calories and include wholegrain foods, fruits, vegetables, nuts and seeds, low-fat dairy products and high-fibre foods.

Processed foods

Edible parts of plants and animals after separation from nature or modified or preserved by minimal processes or modified with the addition of salt, sugar, oils or fats to preserve and enhance their sensory qualities. These include canned or bottled vegetables or legumes (pulses) preserved in brine; whole fruit preserved in syrup; tinned fish preserved in oil; some types of processed animal foods such as ham, nuggets, sausage, and smoked fish; most freshly baked breads; and simple cheeses to which salt is added (Monteiro et al., 2019).

Serving size

In the Malaysian Food Pyramid, serving size is the recommended amount of foods consumed daily in household measures used for foods and drinks, for example cup, plate, bowl, tablespoon, teaspoon and glass. However, serving size defined in the Malaysian Food Pyramid may not be equal to a serving size defined in a food label.

Ultra-Processed Foods (UPFs)
Ultra-processed foods are characterised by NOVA as industrial formulations generated through compounds extracted, derived or synthesized from food or food substrates. Ultra-processed foods also commonly contain artificial substances such as colours, sweeteners, flavours, preservatives, thickeners, emulsifiers and other additives used to promote aesthetics, enhance palatability and increase shelf life. Ready to eat foods and beverages, spreads, packaged snacks, pastries, cakes, instant noodles, pre-prepared ready to heat food products are some examples of ultra-processed foods high in sugar, salt, fat and artificial substances (Monteiro et al., 2019b; Lane et al, 2020).

Variety
Variety refers to eating many different types of foods each day and to ensure better selection of healthier foods. By selecting a variety of foods, the chances of consuming the multitude of nutrients the body needs are optimised.

Whole grains
Whole grains consist of the intact, ground, cracked or flaked caryopsis, whose principal anatomical components are the starchy endosperm, germ, and bran, which are present in the same relative proportions as they exist in the intact caryopsis (AACCI, 1999; Jones, 2010).

Wholesome foods
Foods that are clean, healthy and nutritious to be consumed.

1.2 Introduction

Adequate food intake ensures the nutritional well-being of older persons. The older persons are particularly vulnerable to decreased food intake due to physiological changes such as maldigestion, malabsorption, dry mouth (xerostomia), poor appetite, chronic diseases and illnesses. They are also at risk of psychological stress such as bereavement, social isolation and loneliness. In providing dietary advice to the elderly, a food-based dietary guideline is more relevant than a nutrient-based guideline because of the ability to incorporate culturally sensitive and familiar foods to the older persons. This first message of the food-based dietary guidelines consists of six key recommendations:

i. Choose your daily foods from each food group

ii. Eat at least three main meals every day, with additional 1 or 2 times nutritious snacks

iii. Eat more vegetables (including *ulam*), fruits and adequate amount of legumes

iv. Consume adequate protein, calcium, vitamin D and iron rich foods

v. Be involved in the purchasing and preparation of food and ensure food security

vi. Eat together with others

The best way to meet daily requirements is to eat a varied diet that combines cereals and grains, fruits and vegetables, fish, meat, poultry, legumes, dairy products and also to drink plenty of plain water. The guideline also incorporated recommendations which are not only pertaining to the types of foods but also the psychological aspects which have some influences on food consumption. In this key message, in addition to the use of the food pyramid to provide guidance on the concept of variety and serving sizes in general, the different food groups are also presented as a food plate in order to increase understanding towards portion sizes at each meal.

These recommendations have also been suggested by Dietary Guidelines from various countries such as Australia (Nutrition Australia Victorian Division, 2013), New Zealand (Ministry of Health, 2013) and Singapore (Lee, 2011).

1.3 Scientific Basis

This section presents the scientific basis for the six key recommendations in this key message. In addition, the scientific basis for the additional recommendation of this key message are also provided, namely that are related to healthy dentition and consumption of dietary supplement (including herbal medicine).

1.3.1 Choose your daily foods from each food group

The three essential macronutrients, namely protein, carbohydrate and fat provide energy (calories) to the body. The Recommended Nutrient Intakes for Malaysia (RNI), 2017 recommends that total carbohydrate should contribute around 50% to 65%, total fat 25% to 30%, and protein 10% to 20% of total calorie intake per day for adults (NCCFN, 2017).

The total recommended caloric intake for adult man is reduced from 2460 kcal/day at age 30 to 59 years old to 2030 kcal/day for individuals aged 60 years old and above. The corresponding figures for women are 2180 kcal/day and 1770 kcal/day, respectively (PAL 1.6) (NCCFN, 2017). The reduction is necessary because lean body mass, which is the metabolically active tissue, decreases with ageing. If energy intake is maintained at levels recommended for adults, weight gain will occur. As energy intake is reduced, it is essential for the elderly to consume nutrient-dense foods. Unrefined or minimally processed cereals and grains (whole grain or whole meal) are preferable as these complex carbohydrate food sources are higher in fibre and nutrients. Foods such as fish, meat and poultry as well as legumes, nuts, dairy products, fruits and vegetables provide nutrients which are vital for health, particularly in older persons. In aging, adequate protein intake of at least 1 to 1.5 g/ kg body weight/ day is in combination of resistance exercise is recommended for prevention and management of sarcopenia (a condition of low muscle mass together with low muscle function) (Bauer et al., 2019). A systematic review highlighted a few micronutrients commonly reported as inadequate in the diet of older adults (Borg et al., 2015). These included vitamin D, thiamin, riboflavin, calcium, magnesium and selenium. Rich food sources of these nutrients need to be emphasized in planning for meals for older persons. Specific dietary patterns such as Mediteranean Diet and MIND Diet (Mediterranean and DASH) have been recommended to lower risk of cardiovascular diseases and cognitive impairment (Kheirori & Alizadeh, 2021). However, there is a need to recommend food sources culturally acceptable and affordable to be consumed by Malaysian aging population.

1.3.2 Eat at least 3 main meals every day, with additional 1 or 2 times nutritious snacks

Appetite and food intake often decline with ageing. Older people tend to be consistently less hungry than younger people, eat smaller meals, have fewer snacks between meals and also eat more slowly. Between age 20 and 80 years, there is on average, a decrease in energy intake of approximately 30%. When this decline in energy intake is more than the decrease in energy use that is also normal with ageing, then there is loss of weight (Oydis et al., 2022).

Healthier choices of foods can delay or even reverse many of the problems associated with aging, helping older persons to continue living independently and enjoy a good quality of life. Therefore, the older persons have to eat at least three meals every day, including a variety of vegetables and fruits. Eating at least 3 main meals a day can provide sufficient energy required and also maintain the blood glucose level. Skipping a meal usually makes blood sugars drop causing dizziness and also individuals tend to overeat at the following meal. The older persons should be encouraged to eat small meals to avoid chest and breathing discomfort associated with eating large meals. Small and frequent meals should be encouraged for those having poor appetites (Malek Rivan et al., 2022).

Eating a snack between meals can make the older persons feel more energetic. Helping aging individuals make the right food choices at each eating occasion will help improve nutrient intakes and potentially lead to positive health outcomes. Healthy snacks include fruit, low-fat yogurt or a low-fat muffin (Jessica et al., 2019), biscuits, less-sugar and low-fat *kuih* and dessert soup.

1.3.3 Eat more vegetables (including *ulam*), fruits and adequate amount of legumes

Fruits and vegetables are generally low in energy. In contrast, they are great sources of vitamins and minerals including calcium, folate, iron, magnesium, potassium, vitamin A, vitamin C, vitamin E and vitamin K. These food groups are also high in fibre and other bioactive compounds such as phytonutrients.

High fibre intake has been shown to be associated with lower mortality from circulatory, digestive, and non-cardiovascular disease (CVD) and non-cancer inflammatory diseases (Youngyo & Youjin, 2016). The cardio-protective effects of dietary fibre consumption that is important in the older persons, is actually a result of lifetime fibre intake. A longitudinal cohort study demonstrated that subjects who had stiffer carotid arteries at age 36 years had a lower lifetime consumption of fruits, vegetables and whole grains during the course of young age (van de Laar et al., 2012).

Dietary fibre comprises non-digestible carbohydrates and lignin from plant sources. It has crucial role in health and disease outcome. High-fibre foods, such as fruits, vegetables and whole grains, have consistently been highly beneficial to health and effectively reduced the risk of disease (Corina-Bianca et al., 2022). The Institute of Medicine (2002) recommends 30 g/day of total fibre for men aged over 50 years old and 21 g/day for women over 50 years old. After considering the recommendations from the IOM, the Academy of Nutrition and Dietetics, and WHO, the Malaysian Technical Sub-Committee for Energy and Macronutrients recommended 20 g/day to 30 g/day of dietary fibre across all age groups in Malaysia (NCCFN, 2017). Up to 50% of Malaysian adults were estimated to be consuming less than 20 g/day of fibre (Ng et al., 2010). One local study pointed out that a majority of Malaysians were knowledgeable about the source and role of dietary fibre in human health, including its role as a laxative assisting in the reduction of body weight and cholesterol level. However, most of them did not know about the amount of recommended intake (Daud et al., 2018). Stephen et. al (2017) and colleagues reported a comprehensive review of dietary fibre intake in European countries, including data from nearly 140,000 individuals covering a broad age range from early childhood to older adulthood. Overall, dietary fibre intake for adults living in European countries was 18 g–24 g per day for men and 16 g–20 g per day for women. Comparison with data from the National Health and Nutrition Examination Survey (NHANES) showed that on average, dietary fibre intake in European countries was higher than in North America. Based on these data, it appears that within Europe and the US, dietary fibre intake is around one third below the recommended level (Stephen et al., 2017).

Fibre is also protective towards other diseases such as periodontitis or gum disease. Nielsen et al. (2016) found a relation between periodontal disease and dietary fibre intake among adults aged above 30 years old in the United States. Although an association between low fibre consumption and periodontal disease was observed, this relation was predominately a result of the low whole-grain consumption in the sample rather than from low fruit and vegetable consumption. These findings are important because periodontal disease, which is a chronic inflammatory disease, has been found to be associated with vascular diseases and hypertension. However, periodontitis increases with age. In Malaysia, the National Oral Health Survey of Adults Year 2010 showed that only 0.4% of individuals aged 65 years old and older have healthy gingivae and 9.4% with deep periodontal pockets when screened for periodontal disease (NOHSA, 2010). Further, 19.8 % and 6.9 % of individuals aged 75 years and above were found to have shallow pockets and deep pockets when screened for periodontal disease (NOHSA, 2010).

Phytonutrients is a broad name for a wide variety of compounds produced by plants. Phytonutrients comprise many different chemicals, including carotenoids, indoles, glucosinolates, organosulfur compounds, phytosterols, polyphenols, and saponins. Each phytonutrient comes from a variety of different plant sources and has different propose effects on, and benefits for the body. A balanced eating pattern that includes different forms and colours of fruits and vegetables, provide all beneficial compounds including phytonutrients. Clinical data has supported the epidemiological findings which link associated consumption of lycopenes and bone healths among post-menopausal women. These data, along with in vitro work showing that lycopenes attenuates the production of osteoclast cells, suggest that lycopenes protects bone mass during aging by attenuating bone resorption (Monjotin et al., 2022). A more recent Malaysian study indicated high consumption of traditional vegetables or ulam of at at least one serving daily is desirable as it seems to be associated with a better nutritional status, mood state and cognitive status (You et al., 2020).

1.3.4 Consume adequate protein, calcium, vitamin D and iron rich foods

Musculoskeletal health is one of the key for healthy ageing, thus, it is essentials to ensure optimal intake of nutrients to prevent undesirable outcomes such as sarcopenia, frailty and osteoporosis. Among the dietary recommendations for prevention of sarcopenia is consumption of adequate food, with special emphasis of protein of ≥1.0 g/kg BW for healthy older adult and ≥1.2 g/kg BW for those with sarcopenia and/or frailty. This target protein intake should be achieved primarily by diet, and where that is not possible, then protein supplementation can be considered (Chen et al., 2022). A meta analysis of four studies involving older adults with mean age 73 years old indicated that older adults with sarcopenia consumed significantly less protein than their peers with no sarcopenia. Thus, an inadequate protein intake might be associated with sarcopenia in older adults (Coelho-Junior et al., 2022). In addition to protein, vitamin D status as assessed using serum 25-OH vitamin D levels can be considered in older adults at risk of malnutrition or sarcopenia. Oral vitamin D supplementation (800–1000 IU/day) may be beneficial for older persons with vitamin D insufficiency. Higher doses may be required for those who are deficient in 25-OH vitamin D (Chen et al., 2022). Generally, in Malaysia, infants, obese patients and older persons are at risk of vitamin D deficiency, however, there is no clear guidelines in treating vitamin D deficiency in Malaysia. Neverheless, this condition can be prevented with taking adequate vitamin D in food resources, sun exposure, or supplementation (Md Isa et al., 2022; Lee et al., 2023).

Calcium is an essential nutrient that is necessary for many functions in human health. Research has shown that adequate calcium intake can reduce the risk of fractures, osteoporosis, and diabetes in some populations (Beto, 2015). Osteoporosis is the most common skeletal disorder worldwide affecting the older persons, primarily postmenopausal women. Adequate calcium intake is essential for achieving peak bone mass and maintaining bone health. A study carried out by Sakinah et al. (2013) among Kelantanese women showed that the mean calcium intake was 492.9 ± 316.51mg/day, with 86.3% of the pre- and 91.9% of the postmenopausal subjects having calcium intakes lower than the recommended daily intake for Malaysian women (800mg/day and 1000 mg/day respectively). The intake of calcium was affected by calorie intake and education level. Thus, a diet plan must be made carefully to achieve the calcium requirements through diet. Dietary supplements could be considered when necessary with advice and monitoring by the medical and health professionals. However, it should be noted that minerals in supplements are not that easily absorbed in the body.

On the other hand, minerals consumed from diet are more easily and readily absorbable and therefore more beneficial. Consuming fortified foods such as 60 g (1200 mg calcium) high-calcium skimmed milk powder daily for 24 months had successfully reduced the percentage of bone loss and increased vitamin D status as assessed using 25-hydroxy vitamin D level among Chinese postmenopausal women (Chee et al., 2003). However, certain segments of Malaysian older persons are not able to consume milk due to various reasons including cultural, affordability and also lactose intolerance. Local foods such as tempeh, can serve as an alternative to milk as it has been reported to have a similar percentage of calcium absorption rate (36.9 ± 10.6%) as compared to milk (34.3 ± 8.6%). However, due to differences in the calcium content of tempeh, four servings (200 g) of this product would be needed to provide the same amount of absorbed calcium as that obtained from a-125 ml glass of milk (Haron et al., 2010). In addition, there is also a need to educate on a healthier way of consuming tempeh, as it is often consumed as deep fried meals.

There are many foods which have adequate amount of calcium to achieve our daily needs of calcium. Green and leafy vegetables including broccoli, mustard greens, kale and spinach can be excellent sources of calcium. Nuts and seeds also provide calcium and are excellent sources of essential fatty acids. Almonds, chickpeas, sesame seeds and walnuts are the type of nuts which are available easily in the market. The nuts can be chopped and added into meals for the older persons. These foods could serve as other sources of calcium which would be more tolerated by individuals with lactose intolerance. Lactose intolerance due to lactase deficiency is a common cause of low calcium intake among older persons (Beto, 2015). Meeting calcium recommendation from food sources are desirable as compared to supplement, as its efficacy on treatment is not clear and further it consumption can cause constipation, bloating, kidney stones and some evidence suggests a small increase in the risk of myocardial infarction (Reid & Bolland, 2019).

Anemia is also a common condition among older persons, particularly among those with advancing age and chronic disease. Anemia is associated with falls, frailty and other negative outcomes, including early mortality (Katsumi et al., 2021). Caloric and protein restriction, iron, vitamin B12, folic deficiency are the causes of nutritional anemia. A minimum of 1700 kcal/day and 1.7 g/kg/day of protein intake are necessary to maintain anabolism in chronic patients to prevent and treat anemia (Bianchi, 2016).

1.3.5 Be involved in the purchasing and preparation of food and ensure food security

A review by Walker-Clarke et al. (2022) reported that psychosocial factors influence eating behaviours in later life. These factors included health awareness and attitudes, food decision making, perceived dietary control, mental health and mood, food emotions and enjoyment, eating arrangements, social facilitation, and social support. Food decision making included self-autonomy in food purchasing, planning and preparation can increase the consumption of food and thus resulted in a decrease food insecurity among older persons. The older persons are encouraged to be involved during food purchasing to enable them to choose foods according to their preferences.

The older persons should also be involved in food preparation including seeking their opinion for menu to stimulate their interest in eating, as 50% of those aged over 65 years old have decreased ability to smell, and 75% of those aged between 80 to 97 years old have an impaired sense of smell (Bernstein & Luggen, 2010). The ability to smell and taste can also be affected by smoking and upper respiratory infections. Certain medications, such as those for blood pressure, may affect taste and smell (DeVere & Calvert, 2011). Family members could also cook the meal in the older persons's home and serve the meal warm. Even if the meal is not cooked there, reheating the food in their home is useful. A long lag time, in contrast, can result in a sense of disconnectedness from the meal, leading to decreased appetite (Brush & Calkins, 2008). Appealing foods may help stimulate appetite, especially in older persons whose senses of taste and smell are not what they used to be. Contrasting colors of foods and an attractive presentation can stimulate appetite. Bright colours on a dark background can make the food appears to be interesting, and a variety of colours, flavours and textures can increase the intake (Wanskink, 2010).

Social support in ensuring food security among older persons is important. Supports such as assistance from neighbours, nearby groceries, mobile groceries or even online ordering could be offered to older persons with mobility problems (Saffel-Shrier et al., 2019). Government plays an important role in securing a safer financial situation for the older persons by providing old-age income security although family support is often seen as an important socio-cultural welfare system in Malaysia and many other Asian countries (Mohd, 2012). Financial assistance should also be provided whenever possible as 34% of Malaysian older persons (65 and above) are categorised as below poverty (Economic Planning Unit, 2013). In Malaysia, food insecurity among older persons is reported at 10.4%, with those from rural areas with no or only primary and secondary education, income less than RM 2000 (USD 477.57), at risk of malnutrition and not receiving very high social support were more likely to be food-insecure (Salleh et al., 2018). Poor socioeconomic status among older persons regardless of settings either urban or rural areas are associated with lack of dietary fibre intake, lesser fresh fruits intake, malnutrition (as assessed using calf circumference), lower functional status and higher disabilty (Shahar et al., 2019). Sustainable and multifacett strategies involving multi stakeholders are needed to ensure adequate support to be given to those in needs.

1.3.6 Eat together with others

Research on health and ageing frequently concern problems related to loneliness (Courtin, 2017) and food intake (Bjornwall, 2021). Loneliness and food-related problems are not independent parts of health since eating, beyond its biological necessity, is an essential portion of social life (Warde, 2016). The practice of sharing a meal has been recognized as one key social influence on eating behavior in later life, which can stimulate more pleasure from food and may improve nutritional status (Vesnaver & Keller, 2011). Older persons were more likely than younger persons to eat with companions if they lived with others. Other study found out that a combination of social and nutritional support would be important for older people who is living alone (Mc Huge et al., 2016). Eating together resulted in 60% increase in total energy intake, compared to eating alone (Bjornwall, 2021). Vesnaver et al. (2016) explored loss of commensality and found that eating alone led to fewer regular meals and less time spent on food. Among the groups of which generate social relationship that occurred at least once a month, social relationship that involved co-engagement in social and daily activities, such as shared meals, conveyed more social support, companionship and positive social (Ashida, 2019).

Older persons are encouraged to eat together with others as part of efforts to provide social facilitation and support leading to food emotions and enjoyment (Walker-Clarke et al., 2022). Spouse and family members play important role in encouraging them to eat meals together. If the older person is living apart from his family, the family should bring a hot meal over to his house or invite him to their house on a regular basis. If necessary, one of the family members may assist during feeding time as the elderly may have difficulties during meal times. Family members can also include the older persons in family gatherings, outings, holidays and celebrations. This might be a good occasion to increase food intake for the older persons with poor appetite or depression. The presence of someone to prepare and serve food may not necessarily increase food intake of older persons. However, the older persons tend to eat more if there is a sense of companionship and interaction during the meal time itself.

If the older persons are forgetful or have memory problems, it is advised to schedule meals at the same time everyday and set reminders about meal times such as notes, short phone messages and alarm clocks. The community also plays an important role in increasing the food intake among the older persons, for example providing food in a tiffin carrier (food container). The community can also utilise existing centres such as place of worship and community centres to provide congregate meals for older persons.

1.3.7 Health supplement (including herbal medicine)

There has been an increasing trend in the use of dietary supplements among the older persons in developing (Dong et al., 2022) and developed countries (Schwab et al., 2014; Gahche et al., 2017) particulary amongst women (Agbabiaka et al., 2018; Cisek-Woźniak et al., 2019). In Malaysia, the prevalence of consumption of dietary supplements including herbal products was reported to be high with one in every three Malaysian older persons uses some form of dietary supplements. Similarly as other reported studies, more Malaysian older women (57.3%) compared to men (42.7%) consume vitamin or mineral supplementations (Vanoh et al., 2021). The most commonly consumer herbal supplements vitamin and mineral supplementations were multivitamin, vitamin B-complex, calcium, vitamin C and fish oil while, most commonly consumer herbal supplements were local herbs such including Tongkat Ali (Eurycoma longifolia), mengkudu (Morinda citrifolia), Misai Kucing (Orthosiphon aristatus), jamun (Syzygium umini), Habbatus sauda (Nigella sativa) and gamat (Holothuroidea) (Vanoh et al., 2021).

Despite the high prevalence of health supplements usage among the older persons, the efficacy of these supplements remains largely unclear (Pitkala et al., 2016). In the meta-analyses published by Jenkins et al., 2018 researchers found that data on the four most commonly used health supplements ie multivitamins, vitamin D, calcium and vitamin C, showed no consistent benefit for the prevention of cardiovascular disease, myocardial infarction or stroke, nor was there a benefit for all-cause mortality. More alarmingly, supplementation of vitamin A, E, and beta-carotene have been associated with an increase in mortality (Bjelakovic et al., 2013; Pitkala et al., 2016) and other toxicity symptoms (Kennedy, 2018). Many older persons consume health supplements with the intention of preventing cognitive decline associated with ageing. However, a longitudinal study conducted amongst Malaysian older persons population reported a contradictory finding. It was reported that dietary supplement intake was not associated with improvement in cognitive function, physical fitness, nutritional status and depressive symptoms (Vanoh et al., 2021).

The contraindications for the concurrent use of prescription and health supplements in the older persons have not been clearly established (Sprouse & Van Breemen, 2016). Nevertheless, the concomitant use of health supplements and conventional medications is very common (Pitkala et al., 2016; Agbabiaka et al., 2018). Potential herb–drug or supplement–drug interactions have been reported and some concomitant use can have hazardous outcome or significant hazard (Agbabiaka et al., 2018; Kahraman et al., 2021; Tan & Lee, 2021) especially with anticoagulant and antiplatelet therapies (Pence et al., 2021).

Despite the high rate of health supplement use among the older persons population, the rate of dietary supplement disclosure to the healthcare professionals is low. With preconceived perception that health supplement as natural and safe, older persons often fail to consider the potential adverse effects of supplement-drug interactions. The concerns regarding health supplement use among the older persons may be due to the adverse effects relating to the continuous use of large amounts of certain nutrients in excess of the upper tolerable intake (UL) e.g.: fat soluble vitamins, minerals including iron and herbal products, the nutrient and nutrient interactions when one supplemental mineral intake exceeds the UL e.g.: excess consumption of zinc is associated with reduced copper status (Kennedy, 2018); the adverse effects relating to the high dosage of herbal products and the concurrent use of dietary supplement with prescription and over-the-counter medicines resulting in altered pharmacodynamics and pharmacokinetics of the drugs (Sprouse & van Breemen, 2016). The reliance on dietary supplements is also being associated with an artificial sense of security about nutrient adequacy or health, thus potentially impairing the adequacy of food intake or delaying the person from seeking effective treatment (ADA, 2005).

1.4 Current Status

1.4.1 Malnutrition and food insecurity

The National Health and Morbidity Survey (NHMS) conducted in 2018 has investigated malnurition and food insecurity among 3977 Malaysian older persons. The prevalence of malnutrition and at-risk of malnutrition, as assessed using Mini Nutritional Assessment Short Form (MNA SF) was 7.3% and 23.5%, respectively. Those in a rural area, low education level, depression, IADL dependency, low QoL, underweight, food insecurity, and inadequate plain water intake were at risk of malnutrition in Malaysia (Ahmad et al., 2020). Malnutrition among Malaysian older persons also been identified as one of key predictor of frailty (Norazman et al., 2020). Malnutrition and food insecurity is interrelated, with the national prevalence of food insecurity among Malaysian older persons is 10.4% (Salleh et al., 2020). A recent review by Sulaiman et al. (2021) indicated that the prevalence of household food insecurity in Malaysia was unexpectedly high, with affected groups including Orang Asli, low-income household/ welfare-recipient households, university students, and the older persons. Consequences of household food insecurity included psychological, dietary (macro- and micronutrient intakes), nutritional status, and health impacts.

1.4.2 Nutrient intake

Data from the National Health Morbidity Survey 2015 (NHMS, 2015) shows only 10.6% of older persons consume adequate intake of vegetables (≥ 3 servings/day), whilst 11.4% of older persons consume adequate intake of fruits (≥ 2 servings/day). With regard to food intake, findings from a relatively larger random sample of 1911 Malaysian older persons from four states (i.e. Johor, Klang Valley, Kelantan and Perak) in a longitudinal study for aging (LRGS TUA) indicated that the mean energy intake for men is 1764 ± 424 kcal/d and women was 1531 ± 375 kcal/d, with more than 80% of the older persons not meeting the recommendation for vitamin D, vitamin E, thiamin, niacin, folate and calcium (Shahar et al., 2016). A subsample analysis of dietary pattern of this study indicated that the healthy dietary patterns for successful ageing include consuming wholemeal and whole grains such as oat, bread and spread, and rich in fibre and micronutrients particularly tropical fruits in the diet (Nik Mohd Fakruddin et al., 2019).

In addition, dietary pattern 'vegetable and healthy cooked dishes' was associated with a lower occurrence of metabolic syndrome among Malaysian older persons (Shahar & Nik Mohd Fakruddin, 2018). In view that food intake data might be over or under reported, other objective measures including biochemical markers should be used. More recently, metabolomic approach has successfully been used to validate dietary pattern of successful aging. Older persons who aged successfully was found to excrete metabolite serotonin and melatonin which were associated with oat intake (Nik Mohd Fakruddin et al., 2020).

Although inadequacy of iron intake has not been reported in the LRGS TUA study, the nationwide study among Malaysian older persons (NHMS Older Adults 2015) found that the prevalence rates of anaemia among older persons is 35.3%. The predictors of developing anaemia included advancing age (aged 80 years old and above), diabetes and limitations or disabilities in walking and vision (Krishnapillai et al., 2022). It should also be noted that the study used a capillary blood sample to determine hemoglobin level using a HemoCue hemoglobinometer, which is not the gold standard for determination of iron deficiency anaemia.

1.4.3 Oral health status and nutrient intake

Poor oral health is prevalent among older persons, with a high level of tooth loss and edentulism (total tooth loss), and high prevalence rates of dental caries, periodontal disease, xerostomia (dry mouth) and oral pre cancer or cancer lesions (Lauritano et al., 2019; Laurito et al., 2019; WHO, 2022). Based on data from the National Oral Health Survey of Adults 2010 (NOHSA) in Malaysia, it was observed that there was a deteriorating oral health status as age increased. Findings had shown that 34.5% of Malaysian aged 60 years old and above were edentulous with the mean number of remaining teeth of only 9.8. There were only 21.7% of Malaysian aged 60 years old and above having a minimum of 20 teeth. As for caries and periodontal disease, high prevalence was observed among the 60 years old and above age group with almost all (99.9%) with caries and only 1.9% was found to have healthy gingivae (NOHSA, 2010).

Numerous studies have provided strong evidence of an association between nutrition and oral health problems among older persons (Toniazzio et al., 2018; Algra et al., 2021; Kotronia et al., 2021). Evidence had also shown that having minimum 20 teeth among older persons played an important role in having a healthy diet rich in fruits and vegetables, a satisfactory nutritional status and BMI (Atanda et al., 2022). Edentulous or those with reduced number of functional teeth were more likely to experience chewing difficulty than dentate individuals, and tend to avoid hard-to-chew natural foods like fruits, vegetables and meat (Zhang et al., 2020). Consequently, this group tends to choose softer, processed foods that are high in fat and cholesterol content and lacking in minerals and vitamins (Nascimento et al., 2022). Thus current dietary recommendations could not possibly be met in the older persons with compromised dentition and no texture modification in food served to them. In Malaysia, a study conducted among older persons aged 60 years old and above living in *'Pondok',* Kelantan demonstrated that the older persons with compromised functional dentition have inadequate calorie intake compared to those with non-compromised functional dentition (Seman et al., 2007).

1.4.4 Other health problems/ geriatric syndromes related to nutrition

There are other health problems and geriatric syndromes commonly occurring among older persons which are related to nutrition. Nutrition plays a role in reducing the risk of health problems or geriatric syndromes; or the occurrence of some of health problems might have effect on nutritional requirements. For the purpose of this guideline, review of the current situation is more focused on prevention rather than treatment.

a. Dementia

The nationwide survey involving 3977 older persons (aged ≥ 60 years old) in the community reported the prevalence of dementia of 8.5% (Sooryanarayana et al., 2020). A more recent scoping review by Anuar et al., (2022) reported the prevalence of dementia among Malaysian older persons is 14.3%. The prevalence of a predemented state, ie. mild cognitive impairment is 16% and is associated with high fasting blood sugar, hyperlipidaemia, disability, lower education level, not regularly involved in technical based activities, limited use of modern technologies, lower intake of fruits and fresh fruit juices and not practicing calorie restriction were among the risk factors of poor cognitive performance in this study (Vanoh et al., 2017). Further, a predementia state involving dual geriatric syndromes, ie. Cognitive prefrailty and cognitive frailty was reported to be affecting 37.4% and 2.2% of a sample of Malaysian older adults, respectively (Rivan et al., 2019). Nutrition (ie. inadequate niacin intake) together with demographic (ie. older age), psychosocial (ie. lack of social support, depression) and functional status have been found as risk factors.

b. Frailty

Frailty and prefrailty are affecting 8.9% and 61.7%, respectively, among a sample of 473 Malaysian older persons in Klang Valley; with being female, abdominal obesity and malnutrition were among the risk factors (Badrasawi et al., 2017). Another study by Sathasivam et al. (2015) involving 789 community dwelling older persons from an urban area in Klang Valley reported the prevalence of frailty and prefrailty of 5.7% and 67.7%, respectively; with physical disability, falls and poor cognition were the risk factors. Study by Norazman et al. (2020) among 301 urban-living community-dwelling older persons in Kuala Lumpur also reported the prevalence of frailty and pre-frail were 15.9% and 72.8%, respectively, in which women appeared to be at a higher risk of frailty. The frailty could be predicted by increasing age, low household income, being at risk of malnutrition, as well as having a low skeletal muscle mass and high serum CRP level. Those who were frail were found to be five times more likely to be at risk of malnutrition (Norazman et al., 2020). Therefore, food intake recommendations are important for frailty prevention including a healthy balanced diet, with more emphasis on protein and vitamin D.

c. Sarcopenia

Sarcopenia, age-related loss of muscle mass and function, is one of the geriatric syndromes affecting the ability of older persons to lead an independent living. A few small-scale cross-sectional studies reported that sarcopenia, as assessed using the Asian Working Group for Sarcopenia criteria, affected at least one third of the older persons. Approximately 33% of community dwelling older persons residing in urban areas of Klang Valley (n=393) (Ranee et al., 2022). The prevalence among community dwelling older persons with type 2 diabetes mellitus (T2DM) (n=506), as recruited in public primary care clinics in Malaysia was 28.5%. Among the risk factors included increasing age, had longer duration of diagnosis with T2DM, not using insulin sensitizers, using less than 5 medications, low body mass index, engaging in low and moderate physical activities were among the risk factors (Shariff et al., 2020). Nutrition and physical activity have been recognised as modifiable risk factors to be targeted in prevention of sarcopenia.

d. Psychosocial

Data from a nation-wide community-based cross-sectional study among 3772 older persons indicated that the prevalence of depression was 11.5% (95% confidence interval [CI] 9.4, 13.4). Poor functional status as assessed using activities of daily living and instrumental activities of daily living, urinary incontinence, chronic medical illness, current smoker, poor social support, do not have partner, ethnic minorities and low individual monthly income were the risk factors (Ahmad et al., 2020). Psychosocial distress among Malaysian older persons has to be addressed as it is inter related with poor nutrient intake and food insecurity, especially during crisis such as covid 19 pandemic (Malek Rivan et al., 2021). It is acknowledged that approximately 10% of Malaysian older persons living alone that could be associated with loneliness and psychosocial distress. However, findings from the LRGS TUA indicated that the effects of social network may exceed the impact of living arrangements. It is recommended that health professionals pay more attention to the social networks of older persons Malaysians to harness its benefits in improving their mental health status (Hamid et al., 2021).

1.5 Key Recommendations

Key Recommendation 1

Choose your daily foods from each food group

KM1

Eat a variety of wholesome foods within your recommended servings

How To Achieve:

1. Enjoy a variety of wholesome foods from each food group according to the Malaysian Food Pyramid 2020 (Figure 1.1).

2. Determine the number of servings recommended based on calorie needs (Refer Tables 1.1, 1.2, 1.3 and 1.4 for menu examples). For household measurements please refer to Appendix 1.1.

3. Eat three to five servings of rice, noodle, cereal products (preferably whole grain) and tubers per day.

4. Consume adequate protein including animals and plant sources such as fish, chicken, meat, egg, legumes, and nuts.

5. Choose wholesome food as compared to processed and ultra-processed foods. For example; chicken meat as compared to chicken nuggets; fresh vegetables/ fruits as compared to canned vegetables/ fruits.

Figure 1.1 Malaysian Food Pyramid 2020

Notes:
- The number of servings is calculated based on 1500 to 2300 kcal.
- This pyramid is meant for children aged 7 years and older; for younger children, refer to the Malaysian Dietary Guidelines (MDG) for Children and Adolescents.
- For adolescents aged 13 to 15 years, the recommendation for fruits is 2-3 servings and for milk and milk products 2-3 servings.
- For adolescents aged 16 to < 18 years, the recommendation for fruits is 2-3 servings, milk and milk products 2-3 servings and for rice, other cereals, whole grain cereal-based products and tubers 3-6 servings.

* This includes ultra-processed foods which contain artificial substances such as colours, sweeteners, flavours, preservatives, and other additives.

Table 1.1 Recommended number of servings for each food group based on 1500 kcal, 1800 kcal and 2000 kcal per day

Food group	1500 kcal*	1800 kcal*	2000 kcal*
Vegetables	≥ 3	≥ 3	≥ 3
Fruits[1]	2	2	2
Rice, other cereals, wholegrain cereal-based products and tubers[2]	3	4	5
Poultry/ Meat/ Egg[3]	1	1	2
Fish[4]	1	1	1
Legumes (combine bean, lentil and soy)[5]	1	1	1
Milk & milk products[6]	2	2	2
Fats/ oils (including 1 serving from nuts and seeds)[7]	6	8	9
Sugar[8]	1	1	2

[1]Based on 15 g carbohydrate and 60 kcal per serving
[2]Based on 30 g carbohydrate, 4 g protein, 1 g fat and 150 kcal per serving
[3]Based on 14 g protein, 8 g fat and 130 kcal per serving
[4]Based on 14 g protein, 2 g fat and 70 kcal per serving
[5]Based on 40 g carbohydrate, 14 g protein, 0.5 g fat and 220 kcal per serving
[6]Based on 15 g carbohydrate, 8 g protein, 1 g fat and 90 kcal per serving
[7]Based on 5 g fat and 45 kcal (including 1 serving of nuts & seeds = 5 g of fat and 65 kcal)
[8]Based on 15 g CHO and 60 kcal. 1 serving of sugar = 3 teaspoons; 1 teaspoon = 5 g of CHO and 20 kcal.
Sources: Suzana et al., (2015), NMHS (2018)

One serving of cereals and cereal products and tubers (30 g carbohydrate per serving)	**Household measurement**
Bihun, soaked	1 ½ cups
Biscuits, cream crackers	6 pieces
Bread, white	2 slices
Bread, wholemeal	2 slices
Laksa, soaked	1 ½ cups
Mee or kuey-teow, wet	1 cup
Potato	2 wholes
Putu mayam	2 pieces
Rice, white, cooked	2 scoops/ 1 cup
Rice porridge, plain	2 cups
Sweet potato or yam or tapioca	1 cup

One serving of fruit (15 g carbohydrate per serving)	**Household measurement**
Apple, chinese pear, mango, ciku	1 whole
Mango	½ whole
Banana *(Berangan)* (medium size)	1 whole
Banana *(Emas)*	2 wholes
Durian	3 ulas
Grapes	8 small
Guava/ pear	½ whole
Mandarin orange (small to medium)	1 whole
Papaya, pineapple, watermelon	1 slice
Prunes	4 small
Raisins	1 dessert spoon

One serving of vegetables	**Household measurement**
Dark green leafy-vegetables with edible stem, cooked	½ cup
Fruit vegetables, cooked	½ cup
Ulam, raw	1 cup

One serving of fish, poultry and meat (14 g protein per serving)	**Household measurement**
Anchovies (head removed)	⅔ cup
Beef, lean (7.5 cm × 9 cm × 0.5 cm)	2 pieces
Chicken, drumstick	1 piece
Cockles without shelves	1 cup
Eggs	2 whole
Ikan kembong	1 medium
Ikan selar	1 medium
Ikan tenggiri (14 cm × 8 cm × 1 cm)	1 piece
Chicken liver	2 pieces
Telur puyuh	12 whole
Squid	2 medium

One serving of legumes (7 g protein per serving)	**Household measurement**
Chickpea and yellow dhal (cooked)	½ cup
Green/ Mung bean (cooked) and canned baked bean	¾ cup
Soyabean fermented cake *(Tempe)*	1 piece
Soyabean cake *(Tau-kua)*	1piece
Soybean curd *(Tau-hoo-fah)*	1 tub
Unsweetened soya bean milk	1 glass

(Source: Shahar et al., (2015)

One serving of milk and dairy products (8 g protein per serving)	**Household measurement**
Cheese, processed, cheddar	2 thin slices
Cow's milk fresh, full cream	1 glass
UHT milk, low fat	½ glass
Milk, evaporated	2/3 cup
Milk, powdered (heaped)	4 dessert spoon
Yoghurt, low fat, plain	¾ cup

(Source: Shahar et al., (2015)

Table 1.2 Example of one day menu and *serving size of food group for 1500 kcal

Meal time	Example of one day menu
Breakfast	2 slices (60 g) of wholegrain bread
	1 glass of milk
Morning tea break	1-2 pots of plain yoghurt
Lunch	2 scoops of brown rice
	1 scoop (60 g) of stir-fried spinach with tauhu
	1 scoop (30 g) of string beans + brinjal
	1 piece of fried fish with chilli (80 g)
	1 whole medium banana
	1 glass of plain water
Afternoon tea break	1 glass of unsweetened soya milk
Dinner	2 scoops of brown rice
	1 scoop (60 g) of stir-fried *kangkung*
	1 bowl (250 g) of chicken soup with carrot
	1 glass of plain water
Supper	1 slice (159 g) of papaya medium size

Food group	*Total number of servings in a day
Vegetables	3 servings
Fruits	2 servings
Rice, other cereals, wholegrain cereal-based products and tubers	3 servings
Poultry/ Meat/ Egg	1 serving
Fish	1 serving
Legumes	1 serving
Milk & milk products	2 servings
Fats/ oills	6 servings
Sugar	1 serving

Table 1.3 Example of one day menu and *serving size of food group for 1800 kcal

Meal time	Example of one day menu
Breakfast	2 slices (60 g) of wholegrain bread
	1 slice of cheese
	1 glass of milk
Morning tea break	1-2 pots of plain yoghurt
Lunch	2 scoops of brown rice
	1 scoop (60 g) of stir-fried spinach with *tauhu*
	1 scoop (30 g) of string beans + brinjal
	1 piece of fried fish with chilli (80 g)
	1 whole medium banana
	1 glass of plain water
Afternoon tea break	4 pieces of wholegrain crackers (20 g)
	½ cup of boiled chickpeas (80 g)
Dinner	2 scoops of brown rice
	1 scoop (60 g) of stir-fried *kangkung*
	1 bowl (250 g) of chicken soup with carrot
	1 glass of plain water
Supper	1 slice (159 g) of papaya medium size

Food group	*Total number of servings in a day
Vegetables	3 servings
Fruits	2 servings
Rice, other cereals, wholegrain cereal-based products and tubers	4 servings
Poultry/ Meat/ Egg	1 serving
Fish	1 serving
Legumes	1 serving
Milk & milk products	2 servings
Fats/ oills	8 servings
Sugar	1 serving

Table 1.4 Example of one day menu and *serving size of food group for 2000 kcal

Meal time	Example of one day menu
Breakfast	2 slices (60 g) of wholegrain bread
	1 slice of cheese
	1 omelette
	1 glass of milk
Morning tea break	1 pot of plain yoghurt
	1 cup whole grain wheat breakfast cereal without added sugar
Lunch	2 scoops of brown rice
	1 scoop (60 g) of stir-fried spinach with *tauhu*
	1 scoop (30 g) of string beans + brinjal
	1 piece of fried fish with chilli (80 g)
	1 whole medium banana
	1 glass of plain water
Afternoon tea break	4 pieces of wholegrain crackers (20 g)
	1½ pieces (158 g) of stuffed *tauhu*
Dinner	2 scoops of brown rice
	1 scoop (60 g) of stir-fried *kangkung*
	1 bowl (250 g) of chicken soup with carrot
	1 glass of plain water
Supper	1 slice (159 g) of papaya medium size

Food group	*Total number of servings in a day
Vegetables	3 servings
Fruits	2 servings
Rice, other cereal, wholegrain cereal-based products and tubers	5 servings
Poultry/ Meat/ Egg	2 serving
Fish	1 serving
Legumes	1 serving
Milk & milk products	2 servings
Fats/ oills	9 servings
Sugar	1 serving

Key Recommendation 2

Eat at least three main meals every day according to the Malaysian Healthy Plate (Figure 1.2); with additional 1 to 2 times nutritious snacks.

How to Achieve:

1. Eat a minimum of three main meals each day i.e. breakfast, lunch and dinner. Consume heavier breakfast/lunch compared to dinner.
2. Consume nutritious snacks (e.g. wholemeal and whole grains food such as wholemeal bread and crackers, oat, cereals; fruits, yogurt, milk, tau-foo-fah, boiled chick pea) between main meals if necessary, e.g. morning tea, afternoon tea and supper.
3. Take small, frequent meals, containing a variety of foods that are easy to prepare, appetizing, and easy to chew or swallow.
4. Choose nutrient-dense foods (high nutrient contents) such as mixed porridge (with added fish/ chicken/ meat & vegetable), egg omelet with chopped vegetables, *tauhu sumbat, begedil,* cut mixed fruits and milk drinks.
5. Take a right portion of food for daily meals based on Malaysian Healthy Plate and recommended food intake for the older persons.
6. Consume the last meal (dinner/ supper) of the day at least 2 to 3 hours before bedtime.

Key Recommendation 3

Eat more vegetables (including *ulam*), fruits and adequate amount of legumes. Increase the consumption gradually.

How To Achieve:

Vegetable intake guidelines:

1. Eat at least 3 servings of vegetables (including *ulam*) every day. Eat at least 1 serving of vegetables during your main meals.
2. Modify vegetable texture (e.g. cut vegetables into small pieces, boil or blanch to soften and make it chewable) if older persons have chewing difficulty.
3. Choose fresh dark green, orange, yellow or purple vegetables (including *ulam*) because they are rich in vitamins, minerals, fibres and phytonutrients.
4. Add more vegetables in dishes such as fried rice, soup, porridge and noodles.

Fruit intake guidelines:

1. Eat 2 servings of fruits every day. It is best to consume fresh fruits.
2. Increase intake of fruits by:
   i. Drink fresh fruit juices including pulp or make it to smoothies (1 serving per day).

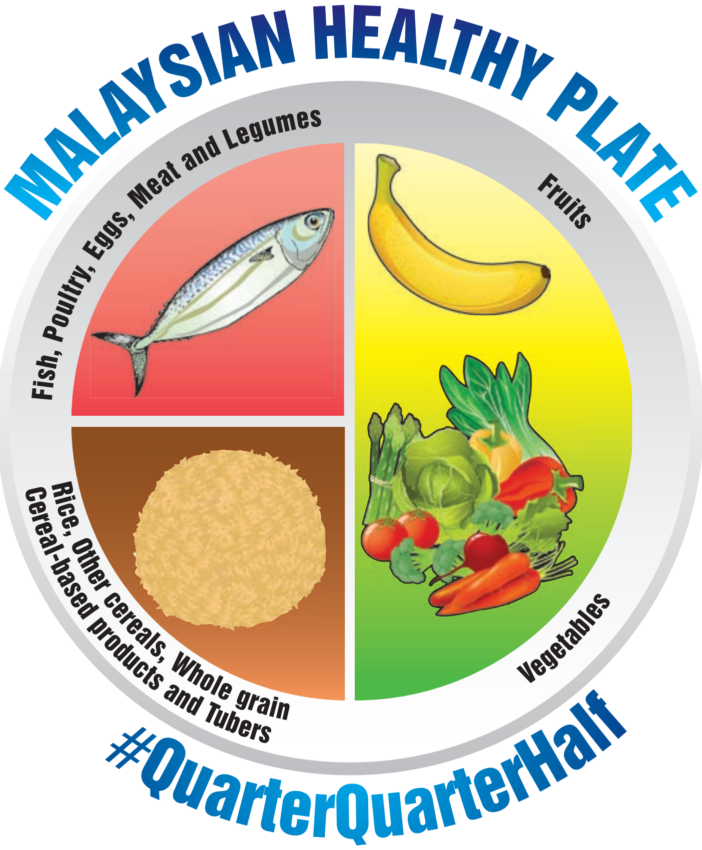


Figure 1.2 Malaysian Healthy Plate

ii. Using or adding fresh fruits as natural sweeteners in dessert or *kuih-muih* such as *pengat pisang, bubur* etc.

3. Choose fresh fruits as snack.

Additional notes:

- If you are facing discomfort or have specific food belief such as cabbage or jackfruit choose other types of fruits or vegetables.

- Include wholegrain products (e.g. cereals, oat, wholemeal bread, barley, millet) in your daily meal to achieve adequate fibre intake in addition to fruits and vegetables.

Legumes intake guidelines:

1. Choose a variety of legumes including dhal and legume products such as tempeh, chickpea and tofu to prepare meals.

2. Add legumes (peas, beans or lentils) to soups and dishes.

Key Recommendation 4

Consume adequate protein, calcium, vitamin D and iron rich foods.

How To Achieve:

1. Consume adequate protein from both animal and plant sources, at least 1g per kg body weight per day.

2. Drink 2 glasses of milk and milk products or plant-based milk a day as one source of protein and to achieve your calcium requirements.

3. Add fresh milk or powdered milk in beverages such as tea and coffee.

4. Add milk into your food during cooking, e.g. egg omelette, pudding, gravies and porridge.

5. Replace partially coconut milk *(santan)* with milk in your cooking.

6. Add calcium-rich foods such as anchovies, tempeh, fucuk or tau-hoo in your cooking for those who does not drink milk. These foods are also rich source of protein and calcium.

7. Ensure obtain adequate sunlight exposure for vitamin D to assist calcium absorption.

8. Include vitamin D-rich foods such as fish (particularly oily fish, e.g. mackerel and sardines), egg yolks, red meat, fortified breakfast cereals and dairy products in the diet.

9. Include iron rich foods such as red lean meats, poultry, fish, eggs, lentils, beans, tofu, green leafy vegetables, iron fortified cereals and bread.

Additional notes:

- 1 glass of milk can be exchanged with other foods high in calcium [300 mg]. (Refer Table 1.5)
- If you do not like milk, you can consume the foods stated in Table 1.5 to achieve your calcium requirement.
- Sweetened condensed milk or sweetened creamer is not the recommended because it is high in sugar content and very low in calcium and other nutrients.

Key Recommendation 5

Be involved in the purchasing and preparation of food and ensure food security.

How To Achieve:

Purchasing of food:

1. Involve older persons when purchasing food. This would enable them to choose foods according to their preferences.

2. Seek assistance from neighbours or nearby groceries for delivery if the elderly has mobility problems.

3. Assist in online food ordering and delivery if the elderly has mobility problems.

4. Provide financial support if the elderly does not have enough money to buy food.

5. Keep extra food at the home of the older persons such as ready-to-eat, canned, frozen and non-perishable foods to ensure continuous supply of food. However, limit intake of processed foods (e.g. nugget, sausage, fries, ham) and avoid ultra processed foods (e.g. soft drinks, sweetened breakfast cereals, salty fatty packaged snacks and instant noodles) which are nutritionally unbalanced.

Table 1.5 Calcium exchanges for 1 glass of milk

Food	Portion	Calcium content (mg)
Sardines	1 small tin (with bones) (155g)	495
Cheese, processed, cheddar	4 thin slices	313
Tau-kua/ hard tau-hoo (soya bean cake)	3 pieces	315
Tempeh (soya bean fermented cake)	10 pieces (12 cm × 9 cm × 0.5 cm)	304
Anchovies	1½ cups	300
Green leafy vegetables (raw e.g. *kailan*)	3 cups	336
Ulam (ulam raja)	3 cups	270
Homemade soya bean milk without sugar (250ml)	6 glasses	300
Calcium fortified soya bean milk (250ml)	1 glass (200ml)	400
Tau-hoo-fah, soya bean curd without sugar (368g)	1 tub	243
Yoghurt (135g/ tub)	2 tub	324
Plant-based milk (eg. Almond, pea and flaxseed milk)	250ml	188

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Eat a variety of wholesome foods within your recommended servings

6. Add-on or provide nourishing formula with complete nutrients in order to help the elderly to meet nutrient requirement, if they are having loss of appetite or chewing difficulties.

Preparation of food:

1. Involve the older persons in meal preparation. Alternatively, seek their opinion for menus to stimulate their interest in eating.
2. Take into consideration their meal preferences.
3. Cook the meal in the older persons's home and serve the meal warm. Even if the meal is not cooked there, reheat the food in his home.
4. Contrasting colours of foods and an attractive presentation can stimulate appetite. Bright colours on a dark background make food looks interesting, and a variety of colours can increase intake.
5. Appealing foods may help stimulate appetite, especially in someone whose senses of taste and smell are not what they used to be. Flavours can be intensified using herbs and spices.
6. Soupy dishes can be served to increase appetite.

Key Recommendation 6

Eat together with others.

How To Achieve:

Tips for elderly:

1. Eat together with the family members at least one meal per day.
2. Bring food and eat together with friends at local organization (e.g. mosque, place of worship, multipurpose hall) or event (e.g. *kenduri*, party, small occasion) to minimize loneliness and isolation condition.
3. Visiting friends and other related family members during festival ceremony (e.g. Hari Raya, Chinese New Year, Deepavali) to nurture relationship with others.
4. Loss of appetite can be dealt with by living an active life to overcome loneliness.

Tips for family:

1. Eat together with the older persons at least one meal per day.
2. Bring a hot meal over to their house, or invite them to your house on a regular basis if the older persons individual is living apart from their family.
3. Assist the older persons during feeding time if he has difficulties during meal times.
4. Include the older persons in family gatherings, outings, holidays and celebrations. This might be a good occasion to increase food intake for individuals with poor appetite or depression.
5. Schedule meals at the same time each day and give reminders about meal times such as notes, short phone messages and alarm clocks, if elderly individuals are forgetful or have memory problems.

Tips for community:

1. Share food with your older persons neighbours as community initiative.
2. Utilise existing centres for older persons to eat together (e.g. place of worship, community centre).
3. Organise "pot luck" during gathering and *kenduri* involving the older persons.

1.5.1. Additional Recommendations:

1.5.1.1 Maintain a healthy dentition

It is important to maintain healthy natural teeth until old age. Healthy teeth contribute to good appearance and effective mastication. In order to achieve these conditions, an older persons should practice good oral hygiene such as brushing teeth twice a day with fluoride toothpaste and flossing daily. Limit the intake of foods and drinks that contain high sugar content. These practices can prevent tooth loss due to tooth decay, dental caries and gum disease. If there are missing teeth, replace them with dental prostheses such as dentures or dental bridge. With this, chewing and appearance can be improved. Get advice on selection of appropriate dental treatment in a dental clinic nearby. If the older persons experience chewing problems, modify the texture of the food, such as soft and finely chopped foods.

1.5.1.2 Consumption of health supplement

Eating a variety of foods daily based on the Malaysian Food Pyramid 2020 should provide all the nutrients needed by the body. Therefore, supplements are not necessary for older persons who are able to meet the nutrient recommendations. Furthermore, supplements do not supply all the essential nutrients needed by the human body. Supplements cannot be used as a substitute for proper food choices and regular consumption of supplements in high dosages can be harmful. However, supplements may be needed for older persons who could not meet the nutrient recommendations from diet alone. Health supplement should only be taken on the advice of healthcare professionals.

Appendix 1.1

Standard Household Measurement

LIQUIDS VOLUME MEASUREMENTS

5 ml	=	1/6 oz	=	1 tea spoon
10 ml	=	⅓ oz	=	1 dessert spoon
15 ml	=	½ oz	=	1 table spoon (3 tea spoon)
60 ml	=	2 oz	=	¼ cup (4 table spoon)
85 ml	=	2 ½ oz	=	1/3 cup
90 ml	=	3 oz	=	3/8 cup (6 table spoon)
125 ml	=	4 oz	=	½ cup
180 ml	=	6 oz	=	¾ cup
250 ml	=	8 oz	=	1 cup (16 table spoon)
1 liter	=	32 oz	=	4 cups

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Key Message 2

Maintain an optimal body weight and body composition for healthy ageing

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Maintain an optimal body weight and body composition for healthy ageing

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2.1 Terminology

Abdominal obesity (central or android obesity)
Abdominal obesity refers to a waist circumference of more than 90 cm and 80 cm for men and women, respectively. Clinically, excessive abdominal fat around the stomach and abdomen has negative impacts on health.

Body composition
Body composition is a general term that encompasses total body weight components consisting of relative proportions of fat mass and fat-free mass whereby the latter comprises muscles, bones, organs ligaments, tendons and water. Many techniques are available to allow measurement of body composition including densitometry, bioelectrical impedance analysis, absorptiometry, circumferences and skinfolds. Each has benefits and drawbacks.

Body Mass Index (BMI)
Body mass index (BMI) is calculated as body weight in kilograms divided by height squared in meters. This is a quick method to assess body weight status with regard to height. In Malaysia, two recommendations or cut-offs of BMI for adults are available, namely the World Health Organization (WHO) (1998) and Clinical Practice Guidelines on Management of Obesity (Malaysia) (2023). There is growing evidence that the existing cut-offs or thresholds developed using data from young adults may not applicable for older persons.

Normal weight obesity

A condition in which a person has a normal BMI but an excessive body fat percentage (De Lorenzo et al., 2006). Normal weight obesity is commonly associated with a significantly higher risk of developing metabolic syndrome, cardiometabolic dysfunction and with a higher risk of mortality.

Sarcopenia

Sarcopenia is defined as age-related loss of muscle mass, low muscle strength, and/or low physical performance (Chen et al., 2020). It is independent of the diagnosis of frailty and cachexia although they may have overlapping signs and symptoms (Dent et al., 2021).

Sarcopenic obesity

Sarcopenic obesity is defined as the presence of sarcopenia and obesity in the same individual (Donini et al., 2022). It is a condition where loss of skeletal muscle mass and function is paralleled by relative or absolute body fat status.

Unintentional weight loss

Unintentional weight loss is the involuntary decline in total body weight over time that is not related to the consequences due to treatment or illness (Wong, 2014). It is often defined as weight loss of more than 5% of the usual body weight within the past 6 months, or more than 10% beyond 6 months (Cederholm et al., 2019). Unintentional weight loss may reflect disease severity (e.g., in patients with heart failure, advanced lung disease or malignant disease) or to be consequences of an undiagnosed disease itself.

2.2 Introduction

Globally, older persons are at risk of different forms of malnutrition that include undernutrition (wasting, stunting, underweight) inadequate vitamins or minerals, overweight and obesity. Either underweight or overweight or obesity condition increases potential health risks to older persons. Obesity increases the risk of chronic diseases leading to serious medical complications affecting the quality of life of older persons. On the other hand, underweight and chronic energy deficiency among older persons are associated with a risk of malnutrition, compromised immune function, sarcopenia, fragility fractures, increased susceptibility to infectious illnesses, and reduced survival rates in the older age groups. Therefore, maintaining a healthy body weight is important in the later years to ensure optimal health and good quality of life.

Other than weight changes, the ageing process is accompanied by changes in body composition. There will be an inevitable decline in lean body mass and a relative increase in body fat. From as early as age 25 years old, an average loss of muscle mass by the time adults hit old age is about 11.3% and 15.4% in normally healthy men and women (Strugnell et al., 2014). The muscle loss is further accelerated as ageing progresses after age 60 years old. This unintentional loss of muscle mass, strength, and function is a fundamental cause and contributor to disability in older adults (Volpi, Nazemi & Fujita, 2004). The age-related loss in skeletal muscle mass is associated with a decline in physical functioning, and impaired mobility and balance (Dent et al., 2021). Fat mass on the other hand has a curvilinear relationship as we age with fat mass peaking near age 60 years old and then plateaus or declines from there (Chumlea et al., 2002; Strugnell et al., 2014). However, in European cohorts, increases in body fat mass continue after that age (Böhm & Heitmann, 2013). More worrisome is the change in the location of fat deposits where abdominal adiposity becomes more prominent in ageing (Al-Sofiani et al., 2019). Older persons with sarcopenic obesity have worse health outcomes than those who are without sarcopenia or obesity (Dent et al., 2021).

2.3 Scientific Basis

In epidemiological studies, body mass index (BMI), an indicator of relative weight for height (weight in kilograms divided by the square of height in meters) and waist circumference are frequently used as surrogates for the assessment of excess body fat and obesity. There is increasing evidence that abdominal obesity may be more important than the overall BMI (Allison et al., 1997) in predicting mortality risk in older persons. Waist circumference is an effective surrogate measure of abdominal obesity, which demonstrates a strong relationship with increased risk for numerous health outcomes as well as mortality (Coutinho et al., 2011). In the local context, waist circumference cut-off points have low sensitivity to identifying overweight and obesity, especially among men, hence may result in the underclassification of a large proportion of overweight and obese individuals (Kee et al., 2011).

Obesity in older persons is one of the increasingly important issues for today in light of the magnitude of obesity among this population. Globally, the prevalence of obesity in older persons, classified using BMI, continues to increase over time. It has increased tremendously with 30%-40% of older persons in developed countries being obese (Castle 2015; Flegal et al., 2016). These estimates have been replicated and increasing in other developing countries (Seo et al., 2018; Tian & Wang, 2022) including Malaysia (IPH, 2019).

There is growing literature on the impact of body weight on the physical and mental health of older persons. Excessive body weight (overweight and obese) is associated with functional limitations such as osteoarthritis (Zheng & Chen, 2015), metabolic syndrome, non-communicable diseases (type 2 diabetes, lipid disorders, hypertension, coronary heart disease and certain cancers) and increased mortality (Villareal et al., 2005; Flegal et al., 2013; Chan et al., 2017; Omar et al., 2017; Keramat et al., 2021). Despite the widely known deleterious effects of obesity on overall health, there is growing evidence that underweight is related to better mobility, but increased frailty, sarcopenia, disability, poor mental health, lower health-related quality of life and increased mortality among older persons (Flegal et al., 2013; Selvamani & Singh, 2018; Chen et al., 2020; Gao et al., 2021a).

2.3.1 Physiological changes associated with ageing

Ageing is associated with several physiological changes including neurological, musculoskeletal, changes in five senses and body composition. The sense of taste and smell deteriorates with age. As we get older, our olfactory function declines. Not only do we lose our sense of smell, but also the discriminative ability to differentiate smells. It has been reported that more than 75% of people over the age of 80 years old have a major olfactory impairment, and that olfaction declines considerably after age 70 years old (Doty et al., 1984). In the same study, the declines in sense of smell and taste were higher when we aged. The decline in sense of smell decreases food intake in older people and can influence the type of food eaten. It has been shown that a reduced sense of smell was associated with reduced interest in foods intake. (Ahmed & Haboubi, 2010). Other than smell dysfunction, the most common causes of taste dysfunction are upper respiratory infection, head injury and drugs (such as Parkinson's medications and antidepressants) and idiopathic causes (Miller, 1995; Mathey et al., 2001; Bromley & Doty, 2002; Jeon et al., 2021). Besides that, reduced number of taste buds, chewing problems associated with tooth loss and dentures can also interfere with taste sensations, along with xerostomia (dry mouth) (Boyce & Shone, 2005; Watson, Leslie & Hankey, 2006; Jeon et al., 2021).

Older persons commonly complain of increased fullness and early satiation during a meal which may be caused by changes in gastrointestinal sensory function. Ageing is associated with impairment of receptive relaxation of the gastric fundus, causing rapid antral filling and distension and earlier satiety (Morley, 2001). Ageing is also accompanied by a decrease in the production of saliva. On average, an approximately 40% reduction in salivary flow was reported in older adults compared with young adults (Vandenberghe-Descamps et al., 2016). On the other hand, approximately one-third of older persons above the age of 65 years old experienced reduced saliva production (Pilgrim et al., 2015) which slows down peristalsis and delays oesophageal emptying, contributing to early satiety (Amarya, Singh & Sabharwal, 2015). Early satiety is a substantial contributor to reduce food intake in the older persons population.

2.3.2 Changes in body weight and body composition

As the number of older persons increases rapidly, the epidemic of obesity is also affecting the aging population. The ageing process is associated with significant changes in body weight and body composition, including quantitative and qualitative progressive loss of skeletal muscle mass and body fat redistribution. In Australia, the prevalence of obesity increased from 11% to 23% in older persons between the 1980's and early 2000's (Bennet, Magnus & Gibson, 2004). In Malaysia, the National Health and Morbidity Survey 2018 found that more than half, about 54.6% of Malaysian older persons were either overweight or obese (IPH, 2019), which demonstrates a consistent increasing in the trend compared to previous National Health and Morbidity Surveys (IPH, 1996; Kee et al., 2008, IPH, 2011; IPH, 2015).

Besides substantial weight gain, advancing in age is often associated with changes in body composition namely loss of lean muscle mass and the gain of fat mass. As we aged, lean muscle mass or fat-free mass loss at a faster rate due to reduced muscle fibre regeneration, lower protein turnover from dietary sources and increases in protein breakdown (Tieland et al., 2018). These changes can lead to normal weight obesity, which is defined as a condition in which a person has a normal BMI but an increased body fat percentage and a greater risk of developing non-communicable diseases (Franco, Morais & Cominetti, 2016). As individual with normal weight obesity often has metabolic disorders of lesser magnitude and increased visceral fat that is not easily observed (De Lorenzo et al., 2007), individuals with normal weight obesity are possibly under-diagnosed and unaware of the risks to which they are exposed (Franco, Morais & Cominetti, 2016). This may be reflected in the decreased accuracy of the BMI and waist circumference and other crude indicators in older persons (Ji et al., 2020). It is noteworthy that in a large prospective cohort study of 156,624 postmenopausal US women, normal-weight abdominal obesity was associated with higher risk of all-cause, cardiovascular disease and cancer mortality compared with normal weight without abdominal obesity individuals (Sun et al., 2019).

In addition to gain body fat mass and a concomitant decrease in lean mass, adipose inflammation leads to the redistribution of body adiposity. This results in the increase of abdominal fat and fat infiltration in organs such as skeletal muscle and liver while accompanied by decreases in subcutaneous fat (Li et al., 2022). Such changes in body composition, which are independent from the general fluctuations in body weight and BMI, are the risk factors for diseases such as cardiovascular diseases, osteoporosis, sarcopenia and decrease in overall strength and functionality (Ponte et al., 2020; Li et al., 2022). Thus, achieving and maintaining an optimal body composition that comprises fat, muscle and bone components is crucial to preserve metabolic homeostasis and contributing to healthy aging.

Loss of skeletal muscle mass among older persons increases the risk of sarcopenia. Sarcopenia is an emerging public health problem among older persons and predisposes them to higher risks of falls, fractures (Yeung et al., 2019) and mortality (Xu et al., 2021) than non-sarcopenic individuals. The risk of gaining additional weight due to ageing-related reduced metabolic rates alongside muscle loss, can further accelerate physical frailty and compounded by the reduced capacity to perform daily physical activities (Bales & Porter Starr, 2018). This would further worsen the adverse outcomes of obesity in older persons, resulting in sarcopenic obesity (Batsis & Villareal, 2018). Global estimates show that 1 in 10 older persons is affected by sarcopenic obesity (Gao et al., 2021b), which can have a detrimental impact on daily functional and clinical outcomes of older persons (Roh & Choi, 2020) including metabolic and cardiovascular dysfunctions (Hunter et al., 2019). Older persons with sarcopenic obesity have twice the risk of frailty and disabilities compared to older persons with only obesity or sarcopenia (Hirani et al., 2017). Similar findings were replicated among Japanese-American men whereby sarcopenic and obese men had significantly greater risk of all-cause mortality as compared to men who were obese or sarcopenic alone (Sanada et al., 2018).

The subject of weight loss interventions in older persons remains controversial. Despite the association of weight loss with improved clinical outcomes in younger adults, weight loss has been discouraged in older persons, partly because of health concerns over inadvertent reductions in muscle and bone mass, which is known to accompany overall weight loss (Kammire et al., 2019; McKee & Morley, 2021). Weight loss in older persons, whether intentional or not, occurs at the expense of muscle mass. The common weight regains after a weight-reducing diet is predominantly a regain in fat mass and not in lean mass (Parr, Coffey & Hawley, 2013), which might contribute to the development of sarcopenic obesity. Hence, the golden years are definitely not the time for extreme diets or drastic weight loss. Fad diets frequently eliminate certain or entire food groups, which can lead to serious nutrient deficiencies such as calcium, vitamin B and protein which are essential nutrients for health maintenance in older persons. Weight reduction in older persons is further complicated by other physiological changes such as a decline in muscle mass and function leading to sarcopenia.

Weight loss therapy that minimizes muscle and bone loss is beneficial for older persons who are obese and have functional impairment or metabolic complications. It is recommended that a comprehensive medical history physical examination, and an assessment of readiness to change are essential prior to the implementation of a weight loss programme (Villareal et al., 2005). However, rapid weight loss that often leads to a loss of lean body mass should be avoided by older persons. Thus, for weight management in older persons, assessment of body composition is critical in relation to health and well-being. In the absence of medications and treatment to increase muscle mass or reduce fat mass, appropriate nutrition and exercise are keys to having good weight management in older persons. Studies found that weight loss programs embedded with physical activity and caloric-reduction interventions were associated with improvements in clinical parameters of cardiometabolic risk and mobility (Beavers et al., 2013) as well as physical functions (Villareal et al., 2011). These interventions also help temper some of the safety concerns regarding the recommendation of intentional weight loss strategies for older persons. Practical strategies for dietary weight loss sparing lean and bone mass as well as bone structure are highly recommended for older persons with obesity-related health problems who are not able (and/or sometimes not willing) to exercise (Goisser et al., 2020).

2.3.3 Unintentional weight loss

The definition of unintentional weight loss is varied without any formal consensus. Generally, unintentional weight loss is defined as a loss of more than 5% of the usual body weight within six months or more than 10% beyond six months (Cederholm et al., 2019). Unintentional weight loss may be poorly recognized in primary care settings as it represents a striking contrast to the epidemics of overweight and obesity in many countries and may be considered as attempt at intentional weight reduction (El-Kareh et al., 2015).

Undeniably, unintentional weight loss is prevalent among older persons. In population-based cohort studies, the prevalence of unintentional weight loss averaged 33.3% from a combined 10,307 older persons aged 65 years old and above (De Stefani et al., 2018). Significant unintentional weight loss was also recorded in about 30% and 20% of older persons in the population-based cohorts of Health and Retirement Study US and English Longitudinal Study of Ageing UK (Jackson, Beeken & Wardle, 2014), respectively. In general, unintentional weight loss in older persons showed a higher all-cause mortality in older persons at a relative risk of 1.8 times (De Stefani et al., 2018).

The etiology of unintentional weight loss among older persons can be complex and is multifaceted. Major factors associated with greater risks of unintentional weight loss including poor appetite (Gill et al., 2015; Gaddey & Holder, 2021), presence of diseases (such as malignant gastrointestinal and respiratory diseases), psychological factors (e.g., depressed mood, bereavement) (Potter, McQuoid & Steffens, 2015), pharmacological treatment and history of hospitalization (Sorbye et al., 2008). Normal age-related physiological changes or physiological anorexia of aging can also predispose an older persons to unintentional weight loss, including declined smell and taste, slowed gastric emptying, reduced efficiency of chewing and decreased saliva production (Wysokiński et al., 2015; Xu et al., 2019).

A critical aspect of weight loss is protein-energy malnutrition, which is associated with impaired physiologic function such as impaired cell-mediated, mental well-being, immune function and healing (Mathewson et al., 2021). Weight loss may also be characterized by the decrease in lean body mass relative to body fat, results in skeletal and cardiac muscle wasting and loss of visceral protein. In the English Longitudinal Study of Ageing UK, 6,864 community-dwelling men and women showed significantly higher risk of all-cause mortality in those who experienced weight loss over the past 4 years compared to those with maintained body weight (Hamer & O'Donovan, 2017). Given the prevalence of unintentional weight loss and the associated clinical importance of weight loss and low body weight, it is crucial to ascertain factors such as the presence of any underlying serious illness, clinical implications of the weight loss and other probable factors in older persons.

2.3.4 BMI and health outcomes in older persons

BMI classifications were developed based on associations between BMI and chronic disease and mortality risk in younger populations (WHO, 2000).

The assessment is inexpensive, easy and quick to perform, thus, making it a practical measure to use in clinical settings. The cut-offs or thresholds developed using data from adults aged 18 years old and older and generalized to the entire adult population, including older persons. In the younger population, a plot of the relative risks of mortality and morbidity against BMI always presented with a U-shaped or J-shaped curve with the minimum mortality close to a BMI of 25 kg/m^2. Hence, the risk of mortality increases as BMI increases above 25 kg/m^2 and decreases as BMI falls below 25 kg/m^2.

In the past decade, there are increasing evidence that patients, especially older persons, with several chronic diseases and elevated BMI may demonstrate lower all-cause and cardiovascular mortality compared with patients of normal weight, highlighting a paradoxical condition known as the "obesity paradox". This paradox suggests that fat mass becomes less harmful or even protective with increasing age (Bosy-Westphal & Müller, 2021). Many adverse outcomes in older persons are directly linked to different nutritional status and are not present in middle-aged adults, and their ideal nutritional status may differ. Overall, older persons with low BMI were at higher risk of health problems such as mortality and difficulties in physical functioning as compared with middle-aged adults. Older persons who are overweight may be less at risk of health problems as compared to those with normal weight (Kuzuya, 2020). On the other hand, obesity in older age may compromise health, well-being and functional status (DiMilia, Mittman & Batsis et al., 2019). Hence, it is timely to delineate the impact of weight on older persons' health.

Evidence is growing on the relationship between body weight and risk of mortality among older persons. There is an increase in mortality with increasing BMI in younger adults, but among the older persons, the relative risk is lower than that of young and middle-aged adults, where the lowest mortality is correlated with an average BMI of between 23.5 kg/m^2 and 27.5 kg/m^2, higher than that in young and middle-aged adults (Donini et al., 2012). More recent studies also supported that existing BMI thresholds may not accurately reflect risk in older persons as being overweight or obese is not always associated with adverse health outcomes (Javed et al., 2020) or increased mortality risk (Winter et al., 2014).

In a meta-analysis of 32 cohort studies (level II and III, n=197,940) in community-dwelling older people aged ≥ 65 years old, a U-shape association with a broad base was found between all-cause mortality and BMI, with mortality risk being lowest at BMI 24 kg/m^2 – 31 kg/m^2 (Winter et al., 2014). The nadir of the curve for BMI and mortality was between BMI 24.0 kg/m^2 and 30.9 kg/m^2, with the lowest risk being between BMI ranges of 27.0 kg/m^2 to 27.9 kg/m^2. The overweight BMI range (BMI of 25 kg/m^2–29.9 kg/m^2) was not associated with an increased all-cause mortality risk. There was a 4%–10% lower mortality risk in overweight older adults for that BMI range (Winter et al., 2014). While the overweight BMI range may seem to be protective, obese older persons were associated with higher mortality risk, with a 21% increase in mortality risk for the BMI range of 35.0 kg/m^2–35.9 kg/m^2. In an earlier systematic review, a BMI of 27 kg/m^2 was a significant prognostic factor for increased mortality risk for older persons that was attenuated with age and usually disappeared after the age of 75 years old (Heiat, Vaccarino & Krumholz, 2001). On the other hand, all-cause mortality risk also started to increase at a BMI of <23.0 kg/m^2, which falls within the WHO healthy weight range for adults (BMI: 18.5 kg/m^2–24.9 kg/m^2). Mortality risk increased at 12% and 28% for older persons at BMIs of 21.0 kg/m^2–21.9 kg/m^2 and < 20 kg/m^2, respectively, compared to those with a BMI between 23.0 kg/m^2 and 23.9 kg/m^2 (Winter et al., 2014). Hence, both obesity and thinness appear to have a greater risk for mortality in older persons, especially when older persons experience weight loss that leads to thinness particularly in conditions of involuntary weight loss compared to overweight older persons.

While most evidence on the protective effect of higher BMI among older persons was initially from Caucasian-based studies (Carr et al., 2023; Flegal et al., 2013; Winter et al., 2014), evidence from Asian cohorts are depicting similar patterns. A scoping review comprised of studies among Asians, Caucasians and Europeans showed the current WHO BMI classification of overweight (25-29.9 kg/m^2), and mild obesity (class 1), demonstrated a reduced all-cause mortality risk for older persons (Javed et al., 2020). In Taiwan, a J-shaped relation was found between BMI with the lowest all-cause mortality risk was documented in older persons with a BMI between 27 kg/m^2 to 28 kg/m^2 (Lin et al., 2021). In addition, to Lin et. al (2021) found that BMI ranges between 27-28 kg/m2, 28-29 kg/m^2, 29-30 kg/m^2 and 31-32 kg/m^2 were associated with the lowest infection, circulation, respiratory and cancer mortalities, respectively. Another observational study in Taiwan found U-shaped curves between BMI with all-cause mortality and expanded cardiovascular diseases (CVD) mortality in older persons. Overweight and grade 1 obesity were associated with lower mortality risks with risk elevated at ≥ BMI 35 kg/m^2 (Wu et al., 2014). Similarly, an earlier longitudinal study among Chinese older persons from 2004 to 2013 showed that older adults with a BMI of 24 to 28 kg/m^2 had lower all-cause mortality, infection-related mortality, cardiovascular-related mortality and all-cause hospitalization (Chan et al., 2015). A recent prospective cohort study among older persons in China demonstrated similar findings, with increased BMI associated with lower all-cause and non-cardiovascular disease mortality (Lv et al., 2022). The prospective cohort study among older persons, followed up from 2003 to 2011 in Singapore Longitudinal Ageing Studies found that BMI showed a U-shaped relationship with mortality (Ng et al., 2017). While overweight or obese was not associated with excess all-cause mortality, underweight was associated with increased all-cause mortality, including CVD and stroke mortality in older persons. To date, the available evidence among Malaysian adults aged 18 years old and older from the National Health and Morbidity Survey found that being underweight was associated with an increased risk of all-cause mortality while obesity increased the risk of CVD mortality (Kee et al., 2017). With the increasing evidence of BMI in relation to mortality risk in Asian older persons, it warrants serious reconsideration of how we interpret and advise on BMI in this specific population.

Besides mortality risks, there is increasing research and clinical interest on the relationship between body weight and other health outcomes among older persons, including muscle strength, functional capacity, and cognitive function. Recently, older adults with BMI < 25 kg/m^2 and > 35 kg/m^2 were at higher risk of poor functional capacity and fall (Kiskac et al., 2022). Evidence is also consistent regarding low body weight or BMI and increased risk of sarcopenia among different populations (Yu et al., 2014; Sato, Ferreira & Rosado, 2020; Choi et al., 2022; Lim & Kong, 2022; Murawiak et al., 2022). Besides weight and BMI, it is noteworthy that body composition plays a critical role in determining the risk of sarcopenia. Recently, a decrease of 1% in body fat accompanied by a 1 unit increase in muscle mass increased muscle strength substantially among older males and females (Hiol et al., 2021). This indicated the necessity and importance of appropriate body composition to preserve muscle strength in older persons.

Evidence associating BMI in midlife with poor cognitive outcomes in late life were consistent (Fitzpatrick et al., 2009; Smith et al., 2011; Li et al., 2021), with earlier study showing obese individual in midlife had 2 to 3 fold higher dementia risk in old age compared to non-obese individuals (Wotton & Goldacre, 2014). In the recent 38 years of follow-up of the Framingham study, obesity at ages 40–49 years old was associated with a higher risk of dementia, with each 1-unit increase in BMI at ages 40–49 years old was associated with higher risk of dementia (Li et al., 2021). While the exact biological mechanisms explaining the risk of dementia associated with obesity at midlife are poorly understood, it was hypothesized obesity at midlife was associated with greater brain atrophy (Debette et al., 2010; Ronan et al., 2016).

The relationship between BMI and dementia risk was heterogeneous across the adult age range (Li et al., 2021). While findings on BMI in midlife with poor cognitive outcomes in late life were consistent, findings on BMI, WC and late-life cognitive decline among older adults are mixed. Apparently, higher BMI and WC have both detrimental and protective effects on the decline in cognitive function. Many studies showed higher BMI were associated with the decline of cognitive function (Cournot et al., 2006, Debette et al., 2011), as well as increased risk of dementia and Alzheimer's Disease (Whitmer et al., 2005; Nepal, Brown & Ranmuthugala, 2010; Xu et al., 2011; Anjum et al., 2018), several studies contradicted these findings. Older persons who are overweight or obese had a better cognitive status or lower risk of cognitive impairment (Qizilbash et al., 2015; Skinner et al., 2017; Hou et al., 2019; Kronschnabl et al., 2021; Liang et al., 2022). In addition, there is growing evidence that higher later-life BMI is related to a slower rate of cognitive decline (Kim et al., 2016; Rodriguez-Fernandez et al., 2017; Arvanitakis et al., 2018; Aiken-Morgan, 2020). These findings were corroborated in the model trajectories of nearly three decades where overweight BMI at age 70 years old among British men and women had a lower risk of mortality (Singh-Manoux et al., 2018).

On the other hand, maintaining a stable body weight is important for older persons. Substantial changes in weight among older persons were associated with poorer cognition or a higher risk of dementia (Roh et al., 2020; Eymundsdottir et al., 2021; Kang et al., 2021). In the latest evidence, greater increases decreases, or variability in BMI over time were associated with an accelerated rate of cognitive decline, irrespective of the initial body weight status, whether a person has normal, overweight, or obese BMI (Beeri et al., 2022). In the Helsinki Businessmen Study which examined the effect of weight trajectories, researchers compared those with constantly normal weight with weight-loss group and found a 2-fold and approximately 4-fold higher risk of disability and frailty in late life, respectively, for the latter group (Strandberg et al., 2013). Noting the importance of tracking BMI change, it may be critical to avoid flatuations in body weight that may predict poorer cognitive trajectories and other clinical outcomes in older persons (Beeri et al., 2022).

It is increasingly recognised that current WHO's BMI cut-offs are less sensitive in predicting mortality risk and other health outcomes in older persons. In contrast, there is growing evidence of the use of body composition namely lean mass, muscle mass and body fat percentage in the prediction of health outcomes among older persons. Lower body fat for example was associated with better global cognition, memory and attention, and information processing, as well as verbal learning and memory in a cohort of older Malaysian men and women (Won et al., 2017b).

While there are insufficient data to propose or generate appropriate BMI cut-offs for older persons in Malaysia, collectively, available evidence suggests that being moderately overweight is a marker of a healthy aging process and it might protect, at least in part, against comorbidity (Pes et al., 2019), with many studies showing a 'U-shaped' curve for the relationship of BMI and mortality, sarcopenia, frailty, and cognitive function in older adults. Recognising the strongest evidence from meta-analysis, the current working group proposed to adopt the desirable BMI cut-off range proposed by Winter et al. (2014). This is further adjusted by taking into consideration the available evidence in Asian populations (Wu et al., 2014; Chan et al., 2015; Ng et al., 2017; Won et al., 2017a; Javed et al., 2020; Lin et al., 2021; Lv et al., 2022). Hence the working group proposes that the desirable BMI range for older adults is 24 kg/m^2 – 27 kg/m^2.

Besides BMI, there is growing evidence on the relationship between abdominal obesity and health risk among older persons. Abdominal obesity, commonly measured by waist circumference or waist-to-hip ratio, has been linked with metabolic dysfunction and structural abnormalities in the brain, which are the renowned risk factors for cognitive impairment and dementia (Tang et al., 2021). Abdominal obesity represented by elevated waist circumference is also a strong risk factor for non-communicable diseases including diabetes mellitus among older persons (Wu et al., 2021; Bai et al., 2022). Besides waist circumference, waist-to-hip ratio (WHR) has also been used for assessing abdominal obesity, visceral fat and risk fo chronic diseases. These two measures are recommended by the National Institute for Health and Clinical Excellence (NICE) (2006) and the National Institute of Health (NIH) (2000) as practical tools to assess risk factors for diseases like diabetes and hypertension, especially in persons with a BMI range under 35 kg/m^2. As Asians tend to have an increased metabolic risk at lower waist circumference and waist-hip ratio, hence a lower waist circumference (85 cm and 80 cm for men and women, respectively) and waist-hip ratio (0.90 and 0.80 for men and women, respectively) are recommended (WHO, 2008). In addition to waist circumference and waist-hip ratio, the waist-to-height ratio (WHtR), which is calculated by dividing the waist circumference by height, has increasing gained attention as an anthropometric index for abdominal adiposity. With a cut-off of 0.5, WHtR is an easy-to-use and less age-dependent index to identify individuals with increased cardiometabolic risk such as hypertension (Kawamoto et al., 2020). A systematic review highlights WHtR as a valid anthropometric index that is very useful in assessing adiposity in elderly and to predict non-communicable diseases, supercede other anthropometric parameters including waist circumference (Corrêa et al., 2016). One of the major applicabilities of WHtR is due to its easy understanding by the general population. The message: "Keep your waist circumference less than half your height" translates the simplicity of WHtR to allow individuals to aware of their health risks and portrays the potential benefits for public health.

In addition to non-communicable diseases, increase in age is often associated with involuntary loss of skeletal muscle mass and muscle strength which predisposes an older person to a higher risk of sarcopenia and poor physical performance. Anthropometric measurements can be useful tools for the clinical assessment of sarcopenia in older persons. According to the Asian Working Group on Sarcopenia (AWGS, 2019), bioelectrical impedance analysis (BIA) and dual-energy X-ray absorptiometry (DXA) are tools for measuring body composition and sarcopenia assessment. However, the applicability and availability of BIA and DXA among older persons may be limited as both methods are costly. In addition, BIA is not recommended for those with an implantable pacemaker while DXA is not portable, which limits their use in community settings, signifying the need of other measurements. Among all, calf circumference was positively correlated with appendicular skeletal muscle mass and skeletal muscle index, and is suggested as a surrogate marker of muscle mass for diagnosing sarcopenia (Kawakami et al., 2014; Hwang et al., 2018; Kim et al., 2018; Kawakami et al., 2020). According to AWGS (2019), the suggested cut-off values of calf circumference represented the discriminant criterion for predicting low muscle mass are < 34 cm in men and < 33 cm in women (Chen et al., 2020). In addition to calf circumference, there is a suggestion to use mid-upper arm circumference (MUAC) to reflect the amount of muscle mass and subcutaneous fat among older persons (Hu et al., 2021). In the above study in China, the recommended optimal MUAC cut-off values for predicting low muscle mass were ≤ 28.6 cm for men and ≤ 27.5 cm for women (Hu et al., 2021). Nevertheless, compared to calf circumference, evidence on the suitability to support using MUAC as a simple proxy measure for muscle mass is scarce.

In summary, as aging may be characterized by loss of lean mass and an increase in adipose tissue, possibly without weight gain (normal weight obesity), traditional adiposity measures such as BMI measurement are not that helpful. Hence, alternative anthropometric tools including body composition analysis would be useful to provide a better outlook in relation to healthy aging. Therefore, the introduction of other body composition measurements in the routine nutritional screenings for older persons including waist circumference, hip circumference, calf circumference, waist-to-hip ratio, waist-to-height ratio and other indicators besides BMI are warranted to avoid misclassification of individuals with high health risk due to high body fat depositions despite having a "normal" weight by BMI.

2.4 Current Status

Overweight and obesity

Obesity impacts not only morbidity and mortality, but also impacted the functional status and quality of life in older adults (Malenfant & Batsis, 2019). The condition can result in limiting daily life activities and increasing functional disabilities (Chen & Guo, 2018). Findings from the National Health and Morbidity Survey (2018) reported that the prevalence of obese, abdominal obesity and muscle wasting in Malaysian older persons were 16.7%, 59.6% and 2.9%, respectively (Singh et al., 2021). In comparison, nationwide data from South Korea reported that 32% of the older persons aged 65 years old and above were classified as overweight or obese (Park & Lee, 2017). In addition, there was a high percentage, about 63%, of normal BMI in this population. Similar findings were observed when comparisons were made to our neighbouring country, Singapore in the Well-Being of the Singapore Elderly (WiSE) study (Fauziana et al., 2016). Prevalence of normal weight of Malaysian older persons is noticeably lower compared to older persons in South Korea and Singapore.

Figure 2.1 shows the nutritional status of older persons in Malaysia according to the findings from the National Health Morbidity Survey (NHMS) series. Generally, from five survey cohorts, there is an increasing trend of overweight and obese among older persons. On the contrary, the prevalence of underweight and normal showed decreasing trends. Specifically, the prevalence of overweight and obesity increased for more than two dan five folds, in older persons from 1996 to 2018, respectively. These findings have raised concern because obesity among older persons involves interaction between various factors that are not only due to the aging process that increases fat adiposity (Zainuddin et al., 2017). Other factors also contribute to obesity in older persons include endocrine, immunology, lifestyle, environment, genetics, and epigenetics (Kim, 2018). Currently, a nationwide study conducted among adults aged 40 years old and above has reported the prevalence of obesity as 34.5% and 23.7% for older persons aged 60 to 69 years old and above 70 years old, respectively (Apalasamy et al., 2021). In this study, a higher percentage of women were obese (42.8%) as compared to men (30.9%).

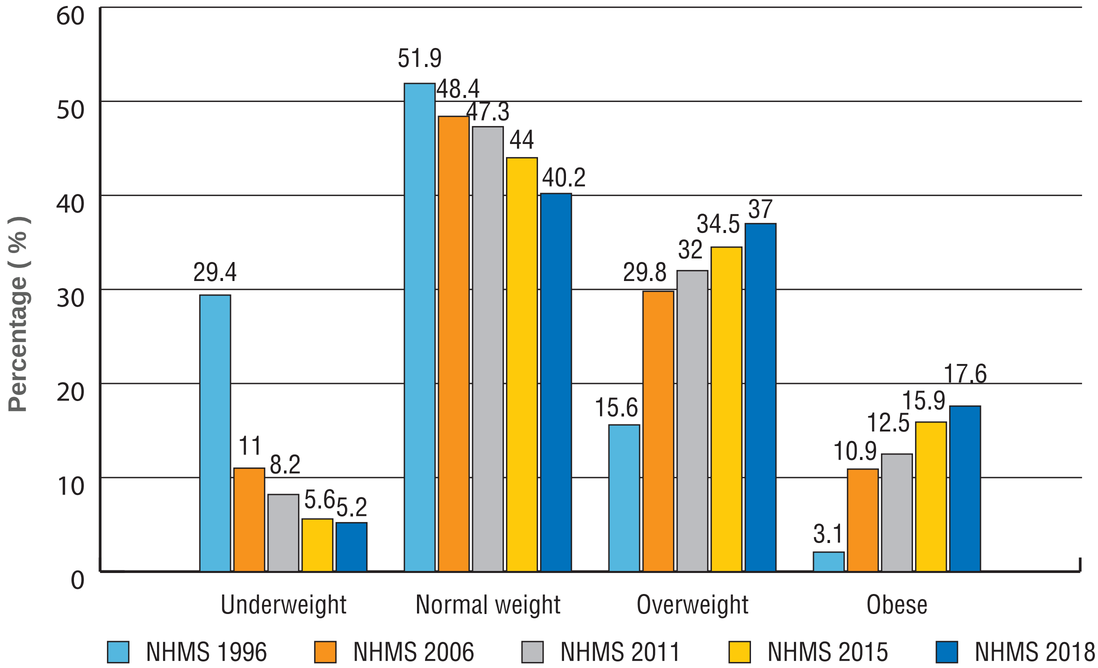

Figure 2.1 Prevalence of underweight, normal weight, overweight and obesity among older persons (60 years old and older) based on NHMS (1996), NHMS (2006), NHMS (2011), NHMS (2015) and NHMS (2018).

Abdominal obesity

Abdominal obesity refers to abdominal fat accumulation measured by the waist circumference (NIH, 1998). It may be a more appropriate measure of mortality risk for older adults compared to BMI (Alharbi et al., 2022). Figure 2.2 shows the prevalence of abdominal obesity among older persons according to gender. Women have a higher prevalence of abdominal obesity than men in all four NHMS cohorts. Furthermore, the trend for abdominal obesity prevalence is increasing by nearly eight times for men and doubled for women between 2006 and 2018. Abdominal obesity was significantly higher among older persons with chronic diseases such as hypertension, hypercholesterolemia and diabetes (Baharudin Shaharuddin et al., 2020). A study conducted in India among urban and rural adult populations found similar trends of significant association between abdominal obesity and chronic diseases including hypertension and diabetes (Pradeepa et al., 2015). In addition, hypertension, diabetes and hyperlipidemia were significantly associated with abdominal obesity as reported in a study conducted among adults in Northeast China (Zhang et al., 2016). Abdominal obesity in older persons was associated with a 49% increased mortality risk compared to non-abdominal obesity as reported in a study that involved community-dwelling older adults in France (Alharbi et al., 2022).

Findings from the Malaysia Ageing and Retirement Survey (MARS) reported the prevalence of abdominal obesity as 63.7% and 60.9% for older persons aged 60 to 69 years old and above 70 years old, respectively (Apalasamy et al., 2021). In this study, a higher percentage of men were abdominally obese (40.8 %) as compared to women (34.1%). Similar findings were reported in a study conducted in Purworejo District, Central Java, Indonesia where abdominal obesity was five-fold more prevalent in women than men (Pujilestari et al., 2017). The higher prevalence of abdominal obesity in women may be explained by the larger abdominal subcutaneous adipose tissue as compared to men (Suzana et al., 2012; Norafidah et al., 2013).

The prevalence of abdominal obesity in older persons from different age groups is depicted in Figure 2.3. Although there are no specific trends in terms of prevalence based on age groups for all four NHMS cohorts, overall, the prevalence of abdominal obesity is lower in older adults aged above 75 years old. The same trend was reported in a study conducted in Indonesia where the prevalence of abdominal obesity was higher in older adults aged 60 to 69 years, compared to those aged above 80 years old (Pujilestari et al., 2017).

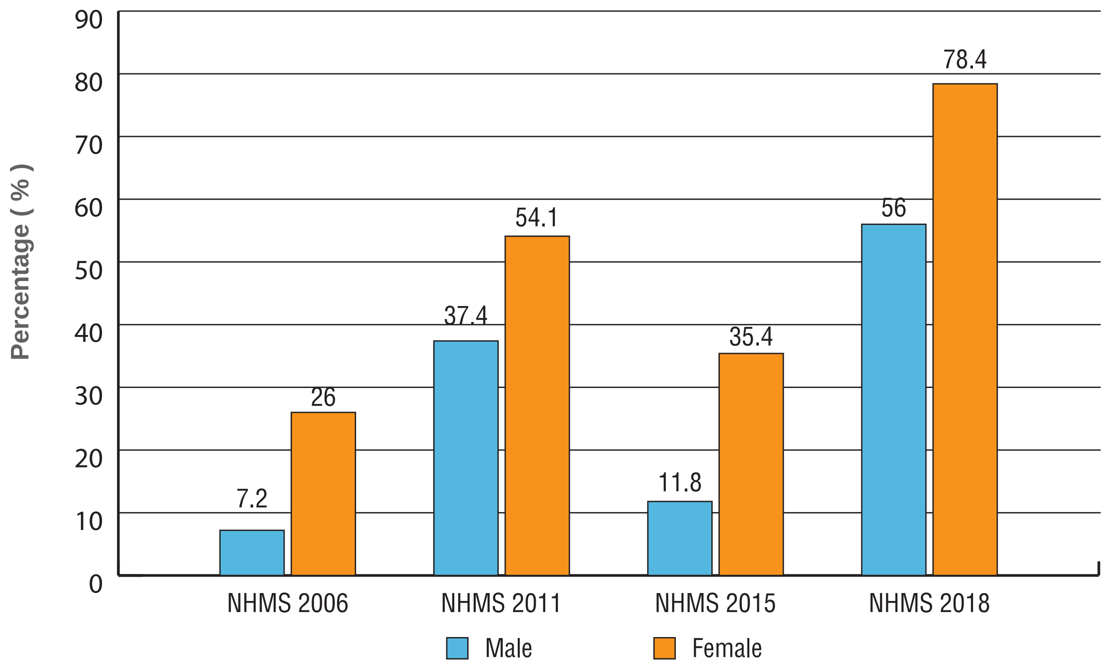

Figure 2.2 Prevalence of abdominal obesity among older persons according to gender

Malnutrition and at-risk of malnutrition

Malaysia is projected to be an aged nation in the year of 2035 with 15% of total population will consist of older adults (Daim, 2019). The increase in the older adult population has raised concern on malnutrition that may occur in this age group. Malnutrition may lead to physically disability, and increase the morbidity and mortality, thus, can impact the quality of life and activity of daily living of the older adults (Norman et al., 2021). Based on results from the National Health and Morbidity Survey (2018), 7.3%

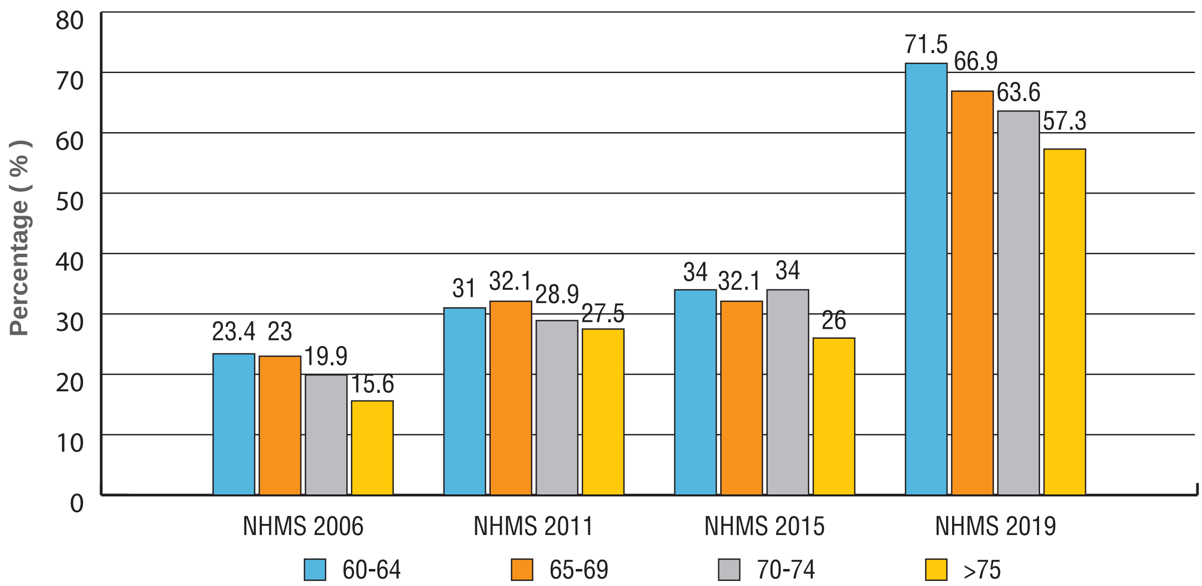

Figure 2.3 Prevalence of abdominal obesity among older persons according to age groups

and 23.5% of the older persons in Malaysia were malnourished and at-risk of malnutrition, respectively (Ahmad et al., 2021). Various factors had increased risk of malnutrition in Malaysian older persons including those living in a rural area, low education level, depression, low functional capabilities, low quality of life, underweight, experienced food insecurity and inadequate intake of plain water (Ahmad et al., 2021).

Muscle wasting has various impacts on the health and well-being of older adults. These include inability to perform activities of daily living which can lead to frailty and increase the risks of falls, dependent living, depression and social isolation, sedentary lifestyle, increase risk of chronic diseases and all-cause of mortality (Arango-Lopera et al., 2013). In addition, declines in strength and muscle function with aging process may lead to hospitalization and reduced the mobility of older adults (Oikawa et al., 2019). Peripheral muscle wasting is prevalent among older persons in Malaysia, with 10.5% of the older persons was documented as at risk of muscle wasting (IPH, 2019). Despite the lower prevalence than the earlier national surveillance cohort, NHMS 2015 (Chan et al., 2017), it is noticeable that many Malaysian older adults are at higher risk of muscle wasting compared to their similar age counterparts in Netherlands (Kruizenga, 2003) and the United Kingdom (Edington et al., 1996). On the other hand, prevalence of muscle wasting tends to increase with age. According to NHMS 2011 (IPH, 2011) and NHMS 2015 (Chan et al., 2017), approximately one in four and one in three older persons in the age group of 70-74 years old and more than 75 years old, experienced muscle wasting, respectively. A study in China assessed older adults' muscle mass at two time points within a 4-year period showed an increment in the prevalence of low muscle mass from 32.1% at baseline to 36.7% after four years (Zhang et al., 2020).

The magnitude of muscle wasting vary depending on the condition (i.e., muscle disuse or aging) and may by influenced by gender (McMillin et al., 2022), however findings are inconsistent. A recent study found that 59.9% of Singaporean older adults have low appendicular skeletal muscle mass index (Tey et al., 2021), where the prevalence was higher among women (61.8%) compared to men (57%). The factors associated with poor muscle health in older persons in Singapore on the other hand include low socio-economic status, presence of chronic diseases, poor nutritional intake, and inapproprite lifestyle (Tey et al., 2021). In Malaysia, the prevalence of muscle wasting according to gender among older persons in Malaysia is displayed in Figure 2.4. Generally, the prevalence of muscle wasting was higher among men as compared to women. In addition, the decline in trends can be seen from 2006 to 2018 in the percentage of muscle wasting based on gender. In the Malaysian context, higher prevalence of those at risk muscle wasting was reported in older persons who are from rural areas, with no formal education, unemployed and low monthly income (IPH, 2019).

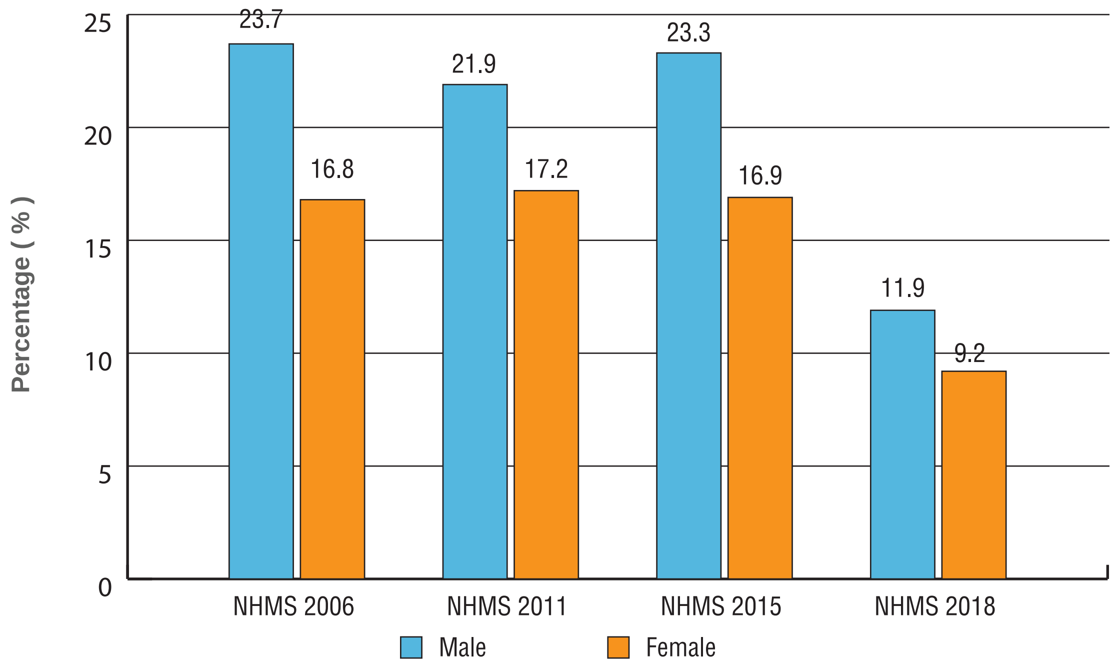

Figure 2.4 Prevalence of muscle wasting according to gender

2.5 Key Recommendations

Key Recommendation 1

Know your weight status regularly.

How to achieve:

1. Weigh yourself at least once a month.
2. Maintain desirable body weight range (BMI= 24 – 27 kg/m^2) and avoid unnecessary weight changes.
3. Seek help from health care professionals (nutritionists or dietitians) to determine your weight status.

Key Recommendation 2

When weight loss is required, ensure lean or muscle mass is not compromised.

How to achieve:

1. Avoid drastic weight loss. Set a realistic weight-loss goal of 0.5 – 1 kg per week. Total weight loss to achieve is 5 – 10% of initial body weight after 6 months.
2. Aim for a steady weight loss by decreasing calorie intake while maintaining an adequate nutrient intake and increasing physical activity.
3. Consume an adequate amount of protein. Older persons are advised to consume at least 1g protein per kg body weight per day.
4. Take smaller portion sizes for foods and limit intake of high calorie snacks (e.g. fried banana fritters, curry puff) and beverages (e.g. *teh tarik*).
5. Perform appropriate exercises to lose weight healthily and preserve muscle mass (Refer to Key Message 3 for details on exercise recommendations).
6. Perform regular physical activity, household chores, and gardening.
7. Reduce sedentary activities such as watching television, sitting and lying.
8. Consult nutritionists or dietitians on how to lose weight safely before starting a weight-reduction programme.

Key Recommendation 3

When weight gain is required, it should be done with caution to ensure the additional gain will contribute to lean or muscle mass, but not fat mass.

How to achieve:

1. Eat an adequate amount of energy and protein daily.
    - i. Consume enough energy of at least 30 kcal per kg body weight per day.
    - ii. Consume enough protein of at least 1g per kg body weight per day.
    - iii. Adjust the calorie and protein requirement according to individual nutritional status, physical activity level, disease status and tolerance.
2. Consume 3 main meals and 1-2 snacks a day. Prepare additional healthy snacks to promote food intake of older persons (Refer to Key Message 1 for more details).
3. Consume energy-dense foods such as rice, porridge, noodles or oatmeal that are added with chicken, fish, meat, vegetable, oil, full cream milk or unsalted butter (Refer to Key Message 1 for more details).
4. Consume full cream milk, creamy soups, soupy dishes, fruit juices, milk shakes or smoothies and others if you have poor appetite. Eat together with the family members at least one meal per day.
5. Perform appropriate exercise and be physically active to maintain or improve muscle mass and function.
6. Consult nutritionists or dietitians on weight gain strategies.

Key Recommendation 4

Be wary of any unintentional weight loss.

How to achieve:

1. Weigh yourself at least once a month.
2. Maintain a good eating pattern (Refer to Key Message 1 for more details).
3. Consult nutritionists or dietitians if older persons experienced significant weight loss (≥ 5% in 6 months or ≥10% beyond 6 months).

Key Recommendation 5

Achieve appropriate body composition.

How to achieve:

1. Measure your waist, hip and calf circumferences, waist-to-height ratio at least once a month.
2. Practice healthy eating and appropriate exercise regularly with safety measures (Refer to Key Message 1 for details on good dietary practices and Key Message 3 for exercise recommendations).
3. Consult nutritionists or dietitians to achieve and maintain optimal body composition (waist, hip, mid-upper arm and calf circumferences, waist-to-height ratio, lean mass, fat mass and body fat percentage)

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DANCING

SPORT ACTIVITIES

CYCLING

GARDENING

Key Message 3

Be physically active for optimal health

Key Message 3

Be physically active for optimal health

Assoc. Prof. Dr. Arimi Fitri Mat Ludin, Dr. Syahrul Ridhwan Morazuki, Mr. Mohd Adha Nawawi, Mr. Hairul Hapizi Samaon and Ms. Faridah Omar

3.1 Terminology

Aerobic exercise
Aerobic exercise refers to the exercise which involves repeated contraction of muscle groups without any added resistance. This type of exercise increases heart rate for an extended period, and raises core body temperature (e.g.: walking, dancing and swimming).

Balance exercise
Exercise that helps maintain stability during daily activities and other exercises, preventing falls. It can be static (e.g.: stand on one leg) or dynamic (e.g.: walk on a straight line), with hand support as needed.

Duration
The amount of time spent participating in a physical activity session.

Flexibility or stretching exercise
This exercise lengthens muscles to improve the range of motion of a joint.

Flexibility
Flexibility refers to the ability to move a joint through a series of articulations in a full non-restricted, pain-free range of motion (ROM).

Frequency
This refers to the number of exercises in a given period of time. Normally it tells how often one needs to do an exercise in a week.

Lifestyle modification
An approach in altering the existing habits (typically eating and physical activity) and maintaining the new behaviors for months or years.

Moderate intensity
Moderate intensity requires moderate amount of effort and noticeably accelerates the heart and respiratory rate but still be able to talk (e.g.: brisk walking, gardening, housework & domestic chores)

Muscular power
Ability to exert maximal force as quickly as possible.

Physical activity
Any bodily movement produced by the contraction of skeletal muscles resulted in an increase of energy expenditure.

Physical exercise
Physical exercise is a structured, planned, repetitive and purposive physical activity with the intent of improving or maintaining physical fitness.

Physical fitness
Physical fitness is a summation of four factors: cardiorespiratory fitness, muscular fitness, flexibility, and body composition.

Resistance or strength exercise
This type of exercise requires muscles to generate force to move or resist weight or any kind or resistance (e.g.: weight training).

Repetition
Repetition defines the number of times to perform an exercise.

Set
Set is the number of cycles of repetition.

Vigorous or high intensity
This requires a large amount of effort and causes rapid breathing & a substantial increase in heart rate (e.g.: running, fast cycling, walking briskly up a hill, fast swimming).

3.2 Introduction

Physical activity, physical exercise and physical fitness are often used interchangeably though each term has its own meaning. Physical activity is defined as any body movement produced by skeletal muscles that requires substantial energy expenditure above and beyond resting energy expenditure and produces progressive health benefits (Winzer et al., 2018; Nikitas et al., 2022). Physical activity is frequently measured in populations as occupational, household, leisure time, recreational and activities of daily living.

Physical exercise is a subset of physical activity and is defined as planned, structured, repetitive activities that improve or maintain the components of physical fitness (Bouca-Machado et al., 2019). On the other hand, physical fitness is conceptualized as the ability to carry out daily tasks, including exercise with vigor, alertness and without undue fatigue (Bouca-Machado et al., 2019).

The goals of an exercise are to improve oxygen delivery and metabolic process, build strength and endurance, decrease body fat and improve movement in joints and muscles. All those benefits are essential for health. Exercise should not be used or applied to a specific group of people and should be used as a preventive rather than corrective tool. Even the cancer patients and survivors are encouraged to exercise to improve their physical functioning and quality of life (Lee, 2020).

There are few types of exercise or physical activities that can be applied to obtain different benefits. They are divided into three general categories, namely aerobic, strength and flexibility exercises. It is also recommended for a well-balanced exercise to contain all these three categories.

Jogging, swimming, cycling, stair climbing, and aerobic dancing are a few examples of aerobic exercise. Aerobic exercises build endurance and keep the heart pumping at a steady but elevated rate for a longer period. On the other hand, based on the physical activity pyramid (PAP), moderate intensity aerobic exercise is recommended for 30 to 60 minutes, five to seven times per week to achieve the desired physiological effect on an individual body system (MDG, 2020). For those who are accustomed with the moderate intensity, can choose to engage at least 75 to 150 minutes of vigorous or high intensity aerobic physical exercise for at least 10 minutes per day is recommended (WHO, 2020).

The purpose of strength or resistance training is to develop muscle strength. With reference to PAP, strength training is recommended two or three times a week, one to three set with eight to twelve repetitions per set. The American College of Sports Medicine and WHO (2020) has recommended for strength training to be performed at least two times a week. The resistance training should involve all major muscle groups.

Flexibility training uses stretching exercise and if done properly it can prevent muscle cramps, stiffness and injuries as well as increase joint flexibility (Iwata et al., 2019). It also improves the range of motion of a joint. Older persons are recommended to perform flexibility exercises for 10 to 12 minutes at least two to three times per week (NIH, 2015). For instance, yoga could be one of the exercises that suitable for the elderly as it is a low-impact activity. It can improve muscle strength, body balance, flexibility, cardiovascular endurance, decrease blood pressure levels, reduce blood glucose level and improve bone health in the elderly (Buttichak et al., 2022).

Older persons who are unable to perform the recommended amounts of exercises due to personal health conditions, they should be as physically active as possible according to their abilities and health conditions (WHO, 2020).

3.3 Scientific Basis

3.3.1 Exercise, cardiovascular diseases, type 2 diabetes mellitus and other chronic diseases

Physical inactivity is a key risk factor for non-communicable diseases which negatively impacted individual life in general. This trend is still rising worldwide (Guthold et al., 2018). Prolonged physical inactivity including older persons aged 65 years old and above leads to the increase in the prevalence of cardiovascular disease, type 2 diabetes mellitus and cancer (Biswas et al., 2015; WHO, 2020). Conversely, older persons participate in 150 minutes of moderate physical activity a week, or its equivalent is estimated to reduce the risk of ischemic heart disease by approximately 30%, the risk of diabetes by 27%, and the risk of breast and colon cancer by 21%-25% (WHO, 2009). Additionally, regular physical activity lowers the risk of stroke, hypertension, and depression. It is a key determinant of energy expenditure and thus fundamental to energy balance and weight control (WHO, 2011).

The health benefits of physical activity in older people have been well-established (Musich et al., 2017). Winzer et al. (2018) indicated that being physically active minimizes the risk of developing cardiovascular. A review by Ciumărnean et al. (2022) indicated that physically active people even at a modest level had a lower risk of suffering from cardiovascular disease as compared to those inactive. For those currently inactive, even at a small increase in physical activity would provide protective benefits against the cardiovascular disease (Garcia et al., 2023). Moreover, a recent study by Li (2023) indicated that 20 minutes of resistance training using elastic band for 14-week, 4 times a week, improved the cardiovascular health among the older persons.

Evidence demonstrated that, compared to less active individuals, older persons who are more active and have higher level of physical activity coincides with the reduced morbidity and mortality from various chronic diseases, postpone disability and prolong independent living (Hupin et al., 2015; Warburton & Bredin, 2017).

Increased risk of mortality in relation to increased blood pressure is reported in both men and women (Wei et al., 2017) and it is even higher among those with elevated morning blood pressure (Xie et al., 2021). To date, high blood pressure remains as one of the common health problems among the elderly and healthy lifestyle is the most recommended therapy (de Andrade et al., 2023). Older persons who are actively exercising benefits from the lowering effect of the resting blood pressure (Igarashi et al., 2021). This notion has been further strengthened by the meta-analysis which concluded that physical exercise is effectively lowered blood pressure when prescribed as adjuvant treatment (Saco-Ledo et al, 2020).

Dyslipidaemia is another risk factor for cardiovascular disease, and it is described by a disorder of lipoprotein metabolism causing elevated blood cholesterol and triglyceride. Recent meta-analysis reported exercise interventions were found to improve lipid profile in the study subjects (Yun et al., 2023).

Steven et al. (2015) concluded in a meta-analysis lifestyle, including exercise and dietary modification interventions have been shown to slow down the progression impaired glucose tolerance to type 2 diabetes mellitus. A linear meta-analysis of the association between physical activity and mortality rate the result showed each 10 METs task of physical activity performed in a week by older adults reduced their mortality rate due to diabetes mellitus type 2 by 4% (Geidl et al., 2020). Similarly, Rietz et al. (2022) concluded that physical activity is associated with lower diabetes complications.

3.3.2 Exercise and musculoskeletal function (sarcopenia, balance and fall)

Musculoskeletal issues substantially reduced the health condition especially among older adults. Sarcopenia, a condition of loss of muscle mass and strength is one of them (Minetto et al., 2020). Being old age is one of the risk factors for the loss of muscle mass (Larson et al., 2019). The process of losing muscle mass and strength is rather slow but resulted in quantitative and qualitative changes in skeletal muscle structure and function. Siervo et al. (2012) reported that the prevalence of older women with low muscle mass was much higher (27.4%) as compared to the prevalence for the total population which was only 9.2%. A study by Badrasawi et al. (2016) showed frail older persons had a significantly lower physical performance as compared to pre-frail older persons.

Falls are more prevalent in older persons and good balance could minimize the risk of falling (Kushkestani et al., 2023). Muscle strength is strongly correlated to the ability to maintain physical balance. The ability to maintaining balance is important for an independent life especially in older persons. This ability enables them to complete their activities of daily living (ADL) and thus for the avoidance of falls. This ability however is affected by the aging process. A cross sectional study observation among Malaysian older persons found muscle strength significantly reduced with advancing age and physical inactivity in both male and females (Shamsul Azhar Shah et al., 2022). The loss of muscle strength can be restored by performing resistance training. This has been shown after a 12-week resistance training for twice a week session using a resistance bands (Motalebi et al. 2016). As reported by (Mat et al. 2018), a homed-based resistance training improved postural control among older people with osteoarthritis. The same study also reported a reduction in fear of falling among the studied participants. The usage of elastic band at home could be useful for the older persons to do resistance training at home (Li, 2023).

3.3.3 Exercise, cognitive function and psychological wellbeing

Brain aging involve structural changes and depletion in neurochemical availability. These processes will then lead to the alteration of cognitive capability (Turrini et al., 2023). However, study showed that physical activity may provide a protection against cognitive decline in older persons as shown by recent study by Handayani et al. (2023) who found that exercise for approximately 30-40 minutes significantly improved memory among the older persons. This is due to the increase in cerebral blood flow and oxygen delivery to the brain, enhancing neuron formation and maintaining the blood volume of the brain (Tarumi & Zhang, 2017).

According to WHO (2016), anxiety and depression are the most common mental disorders, which affect 5% and 7% of the world's older population, respectively. There is available evidence regarding the beneficial effect of physical training as adjunct treatment in psychological problems, mainly depression. A longitudinal study by Joshi et al. (2016), which comprised 3,497 adults aged between 65 and 75 years old, revealed that those who practices the highest levels of physical activity were at lower risk for depression.

Previously, Miller et al. (2019) performed a meta-analysis of randomised controlled trials. They concluded both aerobic and resistance training improve the depressive symptoms among older persons. This meta-analysis corroborates the previous one by Silveira et al. (2013) where they concluded that aerobic exercise together with stretching give an antidepressive effect when it is prescribed on top of standard treatment (Imboden et al., 2020). Another meta-analysis by Kazeminia et al. (2020) concluded being active by engaging in sporting activity reduced anxiety among older adults. Physically active older persons have been associated with the alleviation in incidence of cognitive decline, dementia and Alzheimer (Cunningham et al., 2020). For older persons who are already with dementia, being physically active improves their cognitive as well as non-cognitive functions (Demurtas et al., 2020).

3.3.4 Exercise recommendations for older persons

Life expectancy can be extended by maintaining physical activity as it can influence the development of chronic diseases, slow age-related biological changes and preserve functional capacity. In recommending exercise regimen for older persons, factors like impairments, functional limitation, and disability of older persons such as arthritis, limited range of motion, musculoskeletal discomfort, tendonitis, epicondylitis, back pain, hypoglycaemia and any other discomforts should be considered.

The comprehensive exercise recommended for older persons came from the American College of Sports Medicine (ACSM) and a recent review by Nikita et al. (2022) as well as from ICFSR expert consensus guideline (Izquierdo et al., 2021). Under its expert panel, for healthy adults, ACSM suggests a minimum of 5 days a week of moderate aerobic physical activity or 3 days a week of vigorous aerobic exercise or combination of both. The intensity of the exercise can be based on the 10-point scale. Moderate intensity is referred to as scale 5 to 6 while vigorous intensity is from point 7 to 8. Older persons starting with moderate intensity or those who are already performing moderate intensity should progressively increase to vigorous intensity to gain additional benefits (Izquierdo et al., 2021). Moreover, aerobic exercise is highly recommended for older persons as it improves the heart rate variability (HRV) after physical activity (de Andrade et al., 2023).

Unfortunately, it is common knowledge that who benefits the most from aerobic training also have the most obstacles doing it. Older persons are commonly associated with severe obesity, arthritis, physical disabilities, muscle weakness or sarcopenia, and also diabetes complications. They will find it hard to perform aerobic exercise at the recommended dose. Thus, there is a need to have an alternative form of exercise, which can give similar benefits to this group of individuals. A study by Andriana & Nurdianto (2022) compared the benefits of tai chi exercise with low-intensity steady-state cardio in improving the physical fitness and sleep quality of the older persons. Results of the study revealed that performing 30 minutes tai chi exercise (60%-69% HRmax), 3 times per week for 8 weeks, significantly improved the flexibility, muscle strength in the upper and lower extremities, coordination, balance, and sleep quality. On the other hand, low-intensity steady state cardio (60%-69% HRmax) only benefits in improving the cardiovascular fitness. Thus, it can be concluded that tai chi exercise is better in improving physical fitness and sleep quality for the older persons.

It is recommended for healthy adults to perform resistance training at least two days a week (Fragala et al., 2019; Izquierdo et al., 2021) but ideally three days a week. Training should start at 30 – 40% of 1RM then progress to heavier load, 70 – 80% of 1RM. Training session should be long enough to allow older persons to perform 8 to 10 exercises involving all major upper and lower body muscle groups. Once own body weight cannot provide sufficient resistance, assisted resistance using free weight, weight machine or any type of resistance tube or band should be applied to gain the maximum strength development (Izquierdo et al., 2021). In applying additional resistance/weight other than body weight, the amount to be applied should allow older persons to perform between 8 and 10 repetitions.

In addition to aerobic and resistance exercise, it is also recommended to include flexibility training in the exercise regimen. However, inclusion of this form of exercise should not be a substitute for either aerobic or resistance training. Older persons are recommended for this type of exercise to preserve and improve balance. Participating in both resistance and flexibility training could result in improvement of range of motion (ROM) and would allow older persons to perform activities that require greater ROM around joints (Herriott et al., 2004). ACSM and AHA recommend at least 10 minutes of flexibility activity on top of the general stretching. Flexibility training should involve all major muscle and tendon groups with 10 to 30 s of static stretch with 3–4 repetitions each (Nelson et al., 2007).

Older persons possess a higher risk of falling. Even though data on frequency, intensity and duration of balance training are insufficient, it is recommended for older persons to engage in this type of exercise. It is recommended for older persons to perform this type of training three times a week. This recommendation was based on meta-analysis by Nikita et al. (2022). Considering that some of the older persons may have difficulties to engage in certain type of exercise, yoga could possibly be one of the exercises they can do. A systematic review by Tulloch et al. (2018) revealed that yoga helps to improve the health-related quality of life, as well as the mental well-being of the older persons.

Current technology may have an influence on older persons participation in physical activity. Bangkerd & Sangsawang (2021) developed an augmented reality application for exercise to promote heath among the older persons. The exercise consisted yoga exercise poses and posture that easy to follow.

The intervention resulted improvement in flexibility among the older persons. In addition, a group of researchers (Fu et al., 2022) proposed a conceptual design of an in-home virtual reality (VR) for the older persons, aiming to improve their physical and mental health. Recent review by Heston (2023) indicated that gamification of exercise games is a promising and practical approach to promote regular exercise among the older persons to be physically active and improves their overall health.

For all physical activities and exercise recommendations, older persons are strongly suggested to increase their exercise intensity gradually over time. Besides minimising exercise-related injury, this approach is important to maintain the willingness of older persons to continue exercising. This is particularly important during the initial stage of introducing exercise into previously sedentary people. Every single progression of exercise must be done individually. As aforementioned, regular physical activity is important for healthy aging and it helps in chronic disease management. However, psychological benefits of physical activity should be taken into consideration (Dolenc & Petric, 2018). Thus, it is recommended for older individuals to engage in physical activities which they can adapt according to their needs and perform on regular basis.

3.4 Current Status

It is well known being physically active provide protective effect towards non-communicable disease as well as preserving normal cognitive function. However, according to the report from the latest National Health and Morbidity Survey (NHMS, 2019), one in four Malaysian adults are not physically active. From the same report, more than half of older adults aged 75 years old and above were classified as least active. Ibrahim et al. (2022) reported a cumulative incidence of being physically inactive was 21% among Malaysian older persons.

Ethnicity showed a significant association with physical activity participation. The Chinese participated actively in physical activities (90%), followed by Indians (66.7%) and Malays (30.8%). Finding reported by Zaiton et al. (2012) showed no significant association between ethnicity and physical activity participation in a study on the physical activity among the older persons community dwellers in a suburban district in Selangor.

Chan et al. (2014), found the prevalence of physical inactivity among older persons aged more than 65 years old was reported to be 63.5%. Prevalence among female respondents was higher (70.4%) compared to male subjects (55.2%).

Jasvinder et al. (2014) has done a nationwide community-based survey among 4831 older persons aged >60 years old with face-to-face questionnaires. The prevalence of physical inactivity among older persons was 88%, highest in respondents aged older than 80 years old (95.4%), female (90.1%), other Bumiputra (92.2%), households earning income less than RM1000 (87.9%) and residing in urban locality (88.4%).

The available studies in the country have indicated that overall, there is a poor level of physical activity among older persons in Malaysia. Therefore, it is important to include and highlight physical activity in the Malaysian Dietary Guidelines for older persons to promote and educate the physical activity amongst this aged group. The aim for activity physical recommendation of older person is to encourage them to be physically active daily. At the same time, they should be reminded to minimise sedentary behaviours such like a prolong sitting.

The Physical Activity Pyramid as shown below (Figure 3.1) is a simple and useful guide to guide the older persons to be physically active every day.

PHYSICAL ACTIVITY PYRAMID FOR OLDER PERSONS

Figure 3.1 Physical activity pyramid for older person
Modified from: Malaysian Dietary Guidelines, MDG (2020)

Note:

- Sleep regularly for at least 7 hours every night.
- For activities to be done on 5-7 days a week, examples are listed according to intensity from vigorous at the top to moderate intensity at the bottom. An activity is considered moderate intensity when an individual is able to talk but cannot sing comfortably while doing the activity; vigorous intensity activity, on the other hand, makes an individual "huff and puff".

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3.5 Key Recommendations

Key Recommendation 1

Be physically active as many ways as you can.

How to achieve:

1. Modify your lifestyle by finding opportunities within existing daily routines to increase physical activity.
2. Do household chores and gardening as often as you possibly could.
3. When the surrounding area is safe, walk as much as you possibly could, for example park your vehicle farther and walk to the entrance.
4. Take the stairs than the elevator (if possible). Use the handrail for safety.
5. Engage in any social activity as often as you possibly could.

Key Recommendation 2

Improve or maintain your aerobic fitness.
*Consult a health care professional when planning a physical exercise programme

How to achieve:

1. Accumulate 75 - 150 minutes aerobic activity per week. It can be achieved by:

    a. Perform a continuous 30 minutes moderate intensity aerobic exercise on five days a week. For an easier option to do 5 - 10 minutes bout of exercise, 3 times a day.

    OR

    b. Perform vigorous intensity aerobic exercise for a minimum of 20 minutes on 3 days each week.

    OR

    c. Perform a combination of moderate and vigorous exercises to meet the above recommendations.

2. If your goal is to exercise at moderate intensity, you should produce a noticeable increase in heart rate, breathing and you are still able to talk at this stage. Using a scale 0 (easiest) to 10 (hardest), your effort should be in between 5 to 6.
3. If your goal is to exercise at vigorous intensity, you should produce a larger increase in heart rate and breathing. At this point, talking will be difficult, you need to maintain it till you can.
4. Gradually decrease your intensity before stopping. Using a scale 0 (easiest) to 10 (hardest), your effort should be in between 7 to 8. Because risk of injury increases at high intensity of physical activity, avoid excessive amounts of this type of activity. Example of outdoor activities: brisk walking, tai-chi, jogging, cycling, swimming. Examples of indoor activities: dancing, chairobic (chair assisted aerobic exercise; aerobic exercise while seated on chair), video/online guided exercise.

Key Recommendation 3

Perform activity/exercise to maintain or increase muscular strength and endurance.

How to achieve:

1. Do 10 - 15 minutes warm-up such as walking around, going up and down stairs or static marching.
2. Perform muscular strengthening exercise on alternate days at least twice a week.
3. Perform 8 – 15 muscle strengthening exercises using all major muscle groups (upper body: shoulder, chest, arms, and lower body: thighs, calves). Repeat the same exercises for 3 sets with 8 – 15 reps for each set. Do these exercises in accordance with your pace and pain threshold. (Appendix 3.4).
4. Start with minimum resistance/weight and progressively increase the resistance/weight. Increase the weight when you could perform for 8-12 times easily.
5. Set your goal to reach moderate intensity training. Using a scale 0 (easiest) to 10 (hardest), your effort should be in between 5 to 6.

6. When you are comfortable with the exercise regime and the safety aspect is not compromised, set your goal to reach higher intensity. Using a scale 0 (easiest) to 10 (hardest), your effort should be in between 7 to 8.

Key Recommendation 4

Improve or maintain your body balance at all times to reduce risk of falls.

How to achieve:

1. Perform balance training such as plantar flexion, knee flexion, hip flexion, tandem walk or any other options (Appendix 3.5) for 2 to 3 days a week. Activities such as backward and sideway walks can be done with supervision.

2. Lower body strength exercise is also contributing to balance improvement (Appendix 3.4).

Key Recommendation 5

Perform flexibility exercise to improve or maintain the joint range of motion.

How to achieve:

1. Perform stretching exercises preferably together with aerobic and muscle strengthening exercises.

2. If only stretching exercise is to be performed, do it at least 3 times a week for 10 to 20 minutes each session. (Appendix 3.6).

3. Stretch each muscle group until your maximum point and hold it for 10 to 30 seconds. Repeat the same movement for 3 to 4 times.

Key Recommendation 6

Limit physical inactivity and sedentary habits.

How to achieve:

1. Limit sedentary activities such as watching television, reading newspapers and scrolling electronic gadgets to a maximum of two hours a day.

2. Avoid sitting continuously for more than an hour.

3. If you need to sit down for more than two hours, make an effort every half an hour to move or walk around, perform simple stretching or muscle-strengthening exercise in between.

General notes:

- Stop and seek assistance for any discomfort during any kind of physical activity that is performed.
- Get a partner or exercise in a group for safety purposes and more enjoyable activity.
- Make sure to choose proper shoes for a good balance support and proper clothing for a good heat control.

APPENDIX 3.1

1) EXERCISE SAFETY

Regular exercise is beneficial for everybody, in both physical and psychological aspects. For the older persons, attention to exercise safety is particularly important to avoid accidents.

a) Before exercise

1. Do not exercise with an empty stomach or just after a big meal.
2. Wear comfortable attire, socks and shoes.
3. Wear proper size and non-slippery, shock-absorbing shoes.
4. Always carry water with you.
5. Avoid exercise in extreme weather.
6. Remember to have 5 to 10 minutes warming-up and stretching exercise before starting.
7. Do not forget to repeat the cooling-down and stretching after exercise.

b) Exercise Precautions

1. Choose appropriate exercise according to your capability and interest. Never exceed your limit. Remember that it is NOT the harder the better.
2. If you have an acute medical problem (such as fever, or pain), avoid exercising until you have recovered.
3. If you have a chronic medical condition (such as hypertension, diabetes mellitus, ischaemic heart disease and arthritis), it is better to seek advice from your health and medical care professionals beforehand.
4. If you experience dizziness, shortness of breath, chest pain, vomiting, nausea and severe pain during exercise, stop exercising and seek medical advice as soon as possible.
5. For prolonged exercise like walking, drink water from time to time to supplement the loss of body fluid due to sweating. Do not wait until you feel thirsty.
6. Take appropriate breaks during exercise. Do not over-exert yourself.
7. Exercise with friends. Company provides enjoyment, mutual encouragement and support.

c) Proper management of injuries

1. When there is an accident like sprain or injury, stop exercising right away to prevent any further damage. Keep calm and rest in a safe place.
2. If there is muscle cramp, you can stretch the muscle gently and gently massage it, but do not bounce or stretch it too forcefully.

3. The proper treatment for sprain is RICE, i.e.
   - R (rest the injured part)
   - I (ice -pack application
   - C (compression by applying an elastic bandage)
   - E (elevate the swollen parts)

   Do not massage as it would worsen it.

4. Extreme pain, rapid swelling, severe bruising, decreased range of movement or deformity may be signs of fracture, seek help from a doctor immediately.

2) CONSIDERATIONS FOR SPECIAL SITUATION

i. Initiation of physical activity in inactive older persons

Start out with light-intensity activity that lasts less than 10 minutes, and slowly increase the duration of light-intensity activity and number of days a week the person is active. Light-intensity walking is a good beginning activity, and moderate intensity aerobic activity can be added gradually. Engaging in vigorous intensity activity should initially be avoided to reduce risk of injury.

ii. Resuming activity after an illness or injury

Sometimes, it is necessary to take a break from regular physical activity because of an illness or injury. If this occurs, physical activity should be resumed at a lower level and progressively increased to the previous level of activity.

iii. Increasing activity for weight loss

Some older persons require more physical activity than others to sustain a healthy body weight. If needed, the amount of aerobic physical activity should be gradually increased, and caloric intake should be reduced to achieve energy balance and a healthy weight.

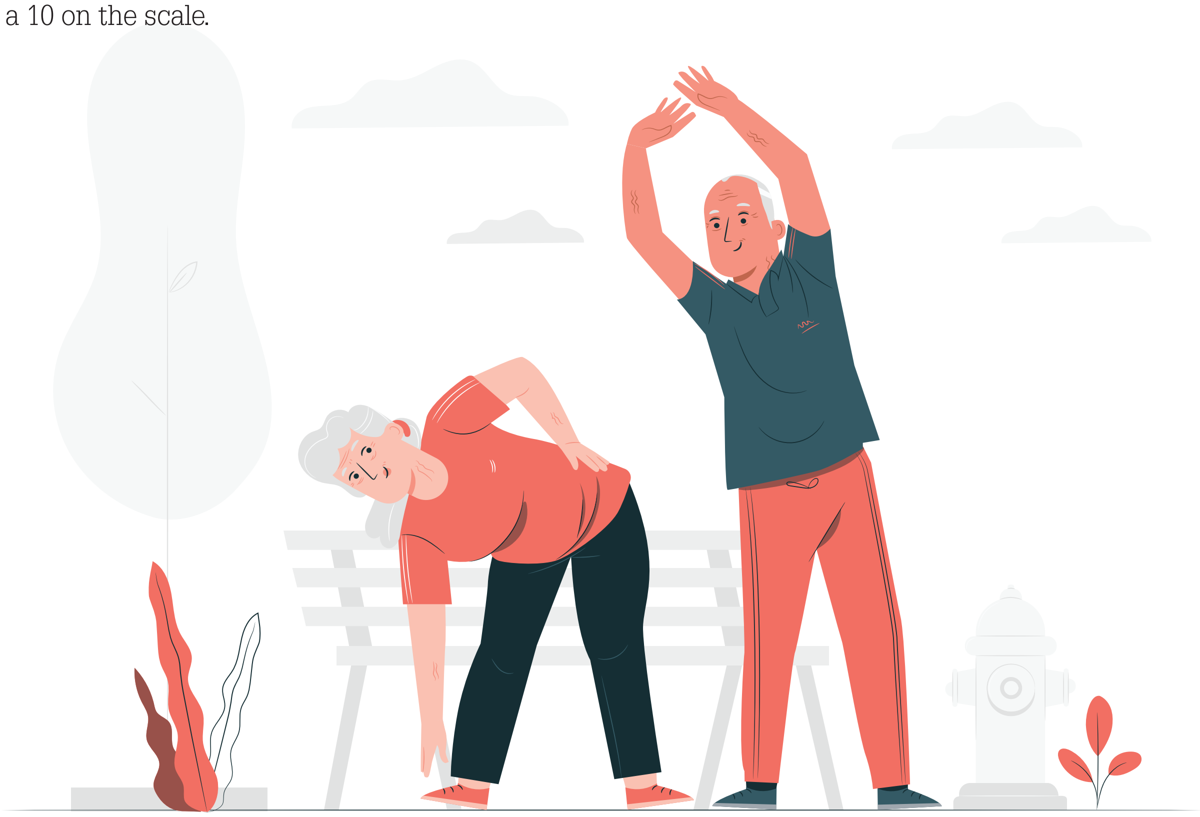
APPENDIX 3.2

BORG SCALE

Number Rating	Verbal Rating	Example
0	Nothing at all	No effort & doing nothing
1	Very light	Your effort is just noticeable
2	Light	
3	Moderate	Still feels like you have enough energy to continue exercising
4	Somewhat hard	
5	Hard	Strong effort needed
6		Very strong effort needed
7	Very hard	You can still go on, but you really have to push yourself. It feels very heavy
8		
9		This is the most strenuous exercise. Almost maximal effort
10	Very, very hard	Absolute maximal effort. Exhaustion.

Rate Perceived Exertion (Borg Scale)

Perceived exertion is assessed by use of a 0-to-10 chart (Modified Borg Scale) to rate the feelings caused by your exertion. Perceived exertion is how hard you feel your body is working. It is based on the physical sensations a person experiences during physical activity, including increased heart rate, increased respiration or breathing rate, increased sweating, and muscle fatigue. For example, quietly sitting in a chair would have a rating of 0. Adding a gentle waving of your arms might increase the effort rating to 0.5. Walking at a pace that you feel is moderate would be given a rating of 3. Remember, the rating of your exertion should be completely independent of the pace you think you are walking; it is dependent solely on the feelings caused by the exertion. Increase the pace to a run and add a hill and you could work your way up to a 10 on the scale.



APPENDIX 3.3

EXAMPLES OF ENDURANCE ACTIVITIES

Examples of activities that are moderate for the average older adult are listed below:

- Swimming
- Bicycling
- Cycling on a stationary bicycle
- Gardening (mowing, raking)
- Walking briskly on a level surface
- Mopping or scrubbing floor
- Golf, without a cart
- Tennis (doubles)
- Volleyball
- Rowing
- Dancing

The following are examples of vigorous activities:

- Climbing stairs or hills
- Brisk bicycling up hills
- Tennis (singles)
- Swimming laps
- Hiking
- Jogging

Adaptation from American Geriatric Society (2001)

APPENDIX 3.4

EXAMPLES OF MUSCLE STRENGTHENING EXERCISE

1) Arm Raise

This upper body exercise strengthens shoulder muscles.

i. Sit in an armless chair with your back supported by the back of the chair.
ii. Keep your feet flat on the floor, even with your shoulders.
iii. Hold weights straight down at your sides, with palms facing inward.
iv. Raise both arms to side, shoulder height.
v. Hold the position for 1 second.
vi. Slowly lower arms to sides. Pause.
vii. Repeat 8 to 15 times.

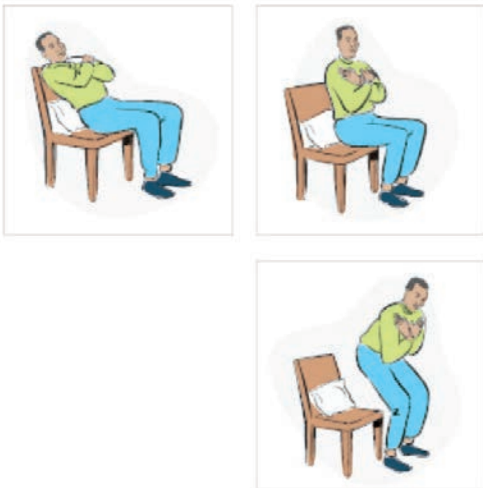
Rest; then do another set of 8 to 15 repetition

2) Chair Stand

This lower body exercise strengthens muscles in abdomen and thighs. Your goal is to do this exercise without using your hands as you become stronger.

i. Place pillows on the back of the chair.
ii. Sit toward the front of the chair, knees bent, feet flat on floor.
iii. Lean back on pillows in a half-reclining position. Keep your back and shoulders straight throughout exercise.
iv. Raise your upper body forward until sitting upright, using hands as little as possible (or not at all, if you can). Your back should no longer lean against pillows.
v. Slowly stand up, using your hands as little as possible.
vi. Slowly sit back down. Pause.
vii. Repeat 8 to 15 times.

Rest; then do another set of 8 to 15 repetitions



3) Biceps Curl

This upper body exercise strengthens upper-arm muscles.

i. Sit in an armless chair with your back supported by the back of the chair.\
ii. Keep your feet flat on the floor, even with your shoulders.
iii. Hold hand weights straight down at your sides, with palms facing inward.
iv. Slowly bend one elbow, lifting weight toward the chest. (Rotate palm to face shoulder while lifting weight.)
v. Hold position for 1 second.
vi. Slowly lower arm to starting position. Pause.
vii. Repeat with another arm.
viii. Alternate arms until you have done 8 to 15 repetitions with each arm.

Rest; then do another set of 8 to 15 alternating repetitions

Adaptation from American Geriatric Society (2001)

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4) Calf Raises

This lower body exercise strengthens ankle and calf muscles. Use ankle weights, if you are ready.

i. Stand straight, feet flat on floor, holding onto a table or chair for balance.
ii. Slowly stand on tiptoe, as high as possible.
iii. Hold position for 1 second.
iv. Slowly lower heels all the way back down. Pause.
v. Do the exercise 8 to 15 times.
vi. Rest; then do another set of 8 to 15 repetitions.

Variation: As you become stronger, do the exercise standing on one leg only, alternating legs for a total of 8 to 15 times on each leg. Rest; then do another set of 8 to 15 alternating repetitions.

5) Triceps Extension

This upper body exercise strengthens the upper-arm muscles the Supporting your arm with your hand throughout the exercise.

i. Sit in a chair with your back supported by the back of the chair.
ii. Keep feet flat on the floor even with shoulders.
iii. Hold a weight in one hand. Raise that arm straight toward the ceiling, palm facing in.
iv. Support this arm, below the elbow, with the other hand.
v. Slowly bend raised arm at elbow, bringing hand weight toward same shoulder.
vi. Slowly straighten your arm toward the ceiling.
vii. Hold position for 1 second.
viii. Slowly bend arm toward shoulder again. Pause.
ix. Repeat the bending and straightening until you have done the exercise 8 to 15 times.
x. Repeat 8 to 15 times with your other arm.

Rest; then do another set of 8 to 15 alternating repetitions.

6) Knee Flexion

This lower body exercise strengthens the muscles in back of thigh. Use ankle weights, if you are ready.

i. Stand straight holding onto a table or chair for balance.
ii. Slowly bend the knee as far as possible. Don't move your upper leg at all; bend your knee only.
iii. Hold position for 1 second.
iv. Slowly lower your foot all the way back down. Pause.
v. Repeat with the other leg.
vi. Alternate legs until you have done 8 to 15 repetitions with each leg.

Rest; then do another set of 8 to 15 alternating repetitions.

Adaptation from American Geriatric Society (2001)

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7) Hip Flexion

This lower body exercise strengthens the thigh and hip muscles. Use ankle weights, if you are ready.

i. Stand straight to the side or behind a chair or table, holding on for balance.
ii. Slowly bend one knee toward the chest, without bending your waist or hips.
iii. Hold position for 1 second.
iv. Slowly lower leg all the way down. Pause.
v. Repeat with the other leg.
vi. Alternate legs until you have done 8 to 15 repetitions with each leg.

Rest; then do another set of 8 to 15 alternating repetitions

8) Shoulder Flexion

This upper body exercise strengthens the shoulder muscles.

i. Sit in an armless chair with your back supported by the back of the chair.
ii. Keep your feet flat on the floor even with your shoulders.
iii. Hold hand weights straight down at your sides, with palms facing inward.
iv. Raise both arms in front of you (keep them straight and rotate so palms face upward) to shoulder height.
v. Hold position for 1 second.
vi. Slowly lower arms to sides. Pause.
vii. Repeat 8 to 15 times.

Rest; then do another set of 8 to 15 repetitions

9) Knee Extension

This lower body exercise strengthens the muscles in front of thigh and shin. Use ankle weights, if you are ready.

i. Sit in a chair. Only the balls of your feet and your toes should rest on the floor. Put a rolled towel under your knees, if needed, to lift your feet. Rest your hands on your thighs or on the sides of the chair.
ii. Slowly extend one leg in front of you as straight as possible.
iii. Flex foot to point toes toward head.
iv. Hold position for 1 to 2 seconds.
v. Slowly lower leg back down. Pause.
vi. Repeat with the other leg.
vii. Alternate legs until you have done 8 to 15 repetitions with each leg.

Rest; then do another set of 8 to 15 alternating repetitions

Adaptation from American Geriatric Society (2001)

10) Hip Extension

This lower body exercise strengthens the buttock and lower-back muscles. Use ankle weights, if you are ready.

i. Stand 12 to 18 inches from a table or chair, feet slightly apart.
ii. Bend forward at hips at about 45-degree angle; hold onto a table or chair for balance.
iii. Slowly lift one leg straight backwards without bending your knee, pointing your toes, or bending your upper body any farther forward.
iv. Hold position for 1 second.
v. Slowly lower leg. Pause.
vi. Repeat with the other leg.
vii. Alternate legs until you have done 8 to 15 repetitions with each leg.

Rest; then do another set of 8 to 15 alternating repetitions.

11) Side Leg Raise

This lower body exercise strengthens the muscles at sides of hips and thighs. Use ankle weights, if you are ready.

i. Stand straight, directly behind a table or chair, feet slightly apart.
ii. Hold onto a table or chair for balance.
iii. Slowly lift one leg 6-12 inches out to the side. Keep your back and both legs straight. Don't point your toes outward; keep them facing forward.
iv. Hold position for 1 second.
v. Slowly lower leg. Pause.
vi. Repeat with the other leg.
vii. Alternate legs until you have done 8 to 15 repetitions with each leg.

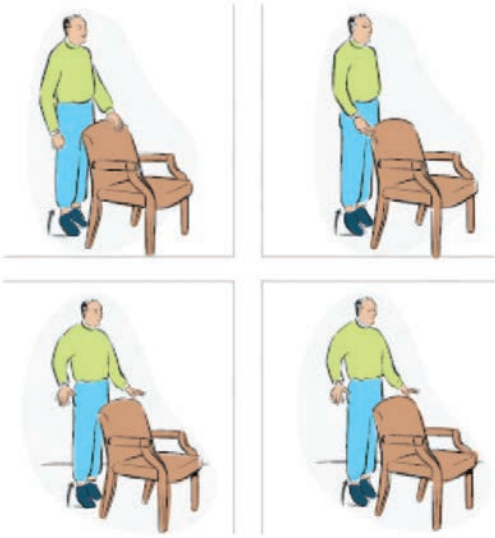
Rest; then do another set of 8 to 15 alternating repetitions.

Adaptation from American Geriatric Society (2001)

APPENDIX 3.5

EXAMPLE OF BALANCE EXERCISE

1) Plantar Flexion



Plantar flexion is already included in your strength exercises. When doing your strength exercises, add these modifications to plantar flexion as you progress: Hold table with one hand, then one fingertip, then no hands; then do exercise with eyes closed, if steady.

Summary:

i. Stand straight; hold onto a table or chair for balance.
ii. Slowly stand on tip toe, as high as possible.
iii. Hold position for 1 second.
iv. Slowly lower heels all the way back down. Pause.
v. Repeat 8 to 15 times.
vi. Rest; then do another set of 8 to 15 repetitions.

Add modifications as you progress

2) Knee Flexion

Do knee flexion as part of your regularly scheduled strength exercises and add these modifications as you progress: Hold table with one hand, then one fingertip, then no hands; then do exercise with eyes closed, if steady.

i. Stand straight; hold onto a table or chair for balance.
ii. Slowly bend the knee as far as possible, so your foot lifts behind you.
iii. Hold position for 1 second.
iv. Slowly lower your foot all the way back down. Pause.
v. Repeat with the other leg.
vi. Alternate legs until you have done 8 to 15 repetitions with each leg.
vii. Rest; then do another set of 8 to 15 alternating repetitions.

Add modifications as you progress

3) Hip Flexion

Do hip flexion as part of your regularly scheduled strength exercises and add these modifications as you progress: Hold table with one hand, then one fingertip, then no hands; then do exercise with eyes closed, if steady.

i. Stand straight; hold onto a table or chair for balance.
ii. Slowly bend one knee toward the chest, without bending your waist or hips.
iii. Hold position for 1 second.
iv. Slowly lower leg all the way down. Pause.
v. Repeat with the other leg.
vi. Alternate legs until you have done 8 to 15 repetitions with each leg.
vii. Rest; then do another set of 8 to 15 alternating repetitions.

Add modifications as you progress.

Adaptation from American Geriatric Society (2001)

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4) Hip Extension

Do hip extension as part of your regularly scheduled strength exercises and add these modifications as you progress: Hold table with one hand, then one fingertip, then no hands; then do exercise with eyes closed, if steady.

i. Stand 12 to 18 inches from a table or chair, feet slightly apart.
ii. Bend forward at hips at about 45-degree angle; hold onto a table or chair for balance.
iii. Slowly lift one leg straight backwards without bending your knee, pointing your toes, or bending your upper body any farther forward.
iv. Hold position for 1 second.
v. Slowly lower leg. Pause.
vi. Repeat with the other leg.
vii. Alternate legs until you have done 8 to 15 repetitions with each leg.
viii. Rest; then do another set of 8 to 15 alternating repetitions.

Add modifications as you progress

5) Side Leg Raise

Do leg raises as part of your regularly scheduled strength exercises, and add these modifications as you progress: Hold table with one hand, then one fingertip, then no hands; then do exercise with eyes closed, if steady.

i. Stand straight, directly behind a table or chair, feet slightly apart.
ii. Hold onto a table or chair for balance.
iii. Slowly lift one leg to the side 6-12 inches out to the side. Keep your back and both legs straight. Don't point your toes outward; keep them facing forward.
iv. Hold position for 1 second.
v. Slowly lower leg all the way down. Pause.
vi. Repeat with the other leg.
vii. Alternate legs until you have done 8 to 15 repetitions with each leg.
viii. Rest; then do another set of 8 to 15 alternating repetitions.

Add modifications as you progress.

6) Tandem Walk

These types of exercises also improve your balance. You can do them almost anytime, anywhere, and as often as you like, as long as you have something sturdy nearby to hold onto if you become unsteady.

Examples:

- Walk heel-to-toe. Position your heel just in front of the toes of the opposite foot each time you take a step. Your heel and toes should touch or almost touch. (See Illustration.)
- Stand on one foot (for example, while waiting in line at the grocery store or at the bus stop). Alternate feet.
- Stand up and sit down without using your hands.

Adaptation from American Geriatric Society (2001)

APPENDIX 3.6

EXAMPLE OF STRETCHING EXERCISE

1) Hamstring

Stretches muscles behind thigh.

i. Sit sideways on a bench or other hard surface (such as two chairs placed side by side.
ii. Keep one leg stretched out on the bench, straight, toes pointing up.
iii. Keep the other leg off the bench, with your foot flat on the floor.
iv. Straighten back.
v. If you feel a stretch at this point, hold the position for 10 to 30 seconds.
vi. If you don't feel a stretch, lean forward from hips (not waist) until you feel stretching in leg on bench, keeping back and shoulders straight. Omit this step if you have had a hip replacement unless the surgeon/therapist approves.
vii. Hold position for 10 to 30 seconds.
viii. Repeat with the other leg.

Repeat 3 to 5 times on each side.

2) Alternative Hamstring Stretch

Stretches muscles in the back of the thigh.

i. Stand behind the chair, holding the back of it with both hands.
ii. Bend forward from the hips (not waist), always keeping back and shoulders straight.
iii. When the upper body is parallel to the floor, hold position for 10 to 30 seconds. You should feel a stretch in the backs of your thighs.

Repeat 3 to 5 times.

3) Calves

Stretches lower leg muscles in two ways: with knee straight and knee bent.

i. Stand with hands against the wall, arms outstretched and elbows straight.
ii. Keeping your left knee slightly bent, toes of right foot slightly turned inward, step back 1-2 feet with right leg, heel and foot flat on floor. You should feel a stretch in your calf muscle, but you shouldn't feel uncomfortable. If you don't feel a stretch, move your foot farther back until you do.
iii. Hold position for 10 to 30 seconds.
iv. Bend the knee of the right leg, keep the heel and foot flat on the floor.
v. Hold position for another 10 to 30 seconds.
vi. Repeat with your left leg.

Repeat 3 to 5 times for each leg

Adaptation from American Geriatric Society (2001)

4) Ankles

Stretches front ankle muscles.

i. Remove your shoes. Sit toward the front edge of a chair and lean back, using pillows to support your back.
ii. Stretch legs out in front of you.
iii. With your heels still on the floor, bend ankles to point feet toward you.
iv. Bend ankles to point feet away from you.
v. If you don't feel the stretch, repeat with your feet slightly off the floor.
vi. Hold the position for 1 second.

Repeat 3 to 5 times.

5) Triceps Stretch

Stretches muscles behind upper arm.

i. Hold one end of a towel in your right hand.
ii. Raise and bend your right arm to drape the towel down back. Keep your right arm in this position and continue holding onto the towel.
iii. Reach behind your lower back and grasp the bottom end of the towel with your left hand.
iv. Climb left hand progressively higher up towel, which also pulls your right arm down. Continue until your hands touch, or as close to that as you can comfortably go.
v. Reverse positions.

Repeat each position 3 to 5 times.

6) Wrist Stretch

Stretches wrist muscles.

i. Place hands together, in a praying position.
ii. Slowly raise elbows so arms are parallel to floor, keeping hands flat against each other.
iii. Hold position for 10 to 30 seconds.

Repeat 3 to 5 times.

Adaptation from American Geriatric Society (2001)

7) Quadriceps

Stretches muscles in front of thighs.

i. Lie on the side of the floor. Your hips should be lined up so that one is directly above the other one.
ii. Rest head on pillow or hand.
iii. Bend knee that is on top.
iv. Reach back and grab the heel of that leg. If you can't reach your heel with your hand, loop a belt over your foot and hold belt ends.
v. Gently pull that leg until the front of the thigh stretches.
vi. Hold position for 10 to 30 seconds.
vii. Reverse position and repeat.

Repeat 3 to 5 times on each side. If the back of your thigh cramps during this exercise, stretch your leg and try again, more slowly.

8) Double Hip Rotation

Stretches outer muscles of hips and thighs. Don't do this exercise if you have had a hip replacement unless your surgeon approves.

i. Lie on the floor on your back, knees bent and feet flat on the floor.
ii. Always keep shoulders on the floor.
iii. Keeping knees bent and together, gently lower legs to one side as far as possible without forcing them.
iv. Hold position for 10 to 30 seconds.
v. Return legs to upright position.
vi. Repeat toward the other side.

Repeat 3 to 5 times on each side

9) Single Hip Rotation

Stretches muscles of pelvis and inner thigh. Don't do this exercise if you have had a hip replacement unless your surgeon approves.

i. Lie on your back on the floor, knees bent and feet flat on the floor.
ii. Keep shoulders on the floor throughout exercise.
iii. Lower one knee slowly to side, keeping the other leg and your pelvis in place.
iv. Hold position for 10 to 30 seconds.
v. Bring the knee back up slowly.
vi. Repeat with another knee.

Repeat 3 to 5 times on each side.

Adaptation from American Geriatric Society (2001)

10) Shoulder Rotation

Stretches shoulder muscles.

i. Lie flat on the floor, pillow under head, legs straight. If your back bothers you, place a rolled towel under your knees.

ii. Stretch arms straight out to the side. Your shoulders and upper arms will remain flat on the floor throughout this exercise.

iii. Bend elbows so that your hands are pointing toward the ceiling. Let your arms slowly roll backwards from the elbow. Stop when you feel a stretch or slight discomfort and stop immediately if you feel a pinching sensation or a sharp pain.

iv. Hold position for 10 to 30 seconds.

v. Slowly raise your arms, still bend at the elbow, to point toward the ceiling again. Then let your arms slowly roll forward, remaining bent at the elbow, to point toward your hips. Stop when you feel a stretch or slight discomfort.

vi. Hold position for 10 to 30 seconds.

vii. Alternate pointing above head, then toward ceiling, then toward hips. Begin and end with pointing-above-head position.

Repeat 3 to 5 times.

Adaptation from American Geriatric Society (2001)

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Key Message 4

Drink plenty of water daily

KM4

Drink plenty of water daily

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Drink plenty of water daily

Prof. Dr. Hamid Jan Jan Mohamed and Dr. Divya Vanoh

4.1 Terminology

Alcoholic beverages
An alcoholic beverage is a liquor that has more than two percent volume per volume of ethanol or alcohol.

Beverages
A beverage refers to any drink that is in liquid form except plain water.

Decaffeinated beverages
A decaffeinated beverage refers to any caffeine containing beverage that has been processed to remove the caffeine component. This can be noticed from the packaging label.

Dehydration
Dehydration is excessive loss of body water which impairs bodily functions and metabolism.

Soft drinks
A soft drink refers to carbonated fruit-flavoured beverage which does not contain alcohol.

Plain Water
Plain water refers to any clear/ colourless and clean water in its original form. It may be filtered water, mineral water or treated tap water which is boiled.

Soup
Soup refers to any clear or creamy liquid dish that is prepared by boiling food such as fish, meat or vegetables in water.

Xerostomia (Dry mouth)
Xerostomia or dry mouth is a symptom that is due to a lack of excretion of saliva. It could be caused by dehydration, side effects of medications or treatments such as chemotherapy or radiotherapy.

4.2 Introduction

Water is a fundamental element for all living organisms, especially in human body. Total water can be obtained from drinking water and also from other liquids such as from fruits and soups. It has various important functions such as carrying nutrients, remove waste products, facilitate metabolic reaction, lubricant around joints, regulation of temperature and maintain blood volume (Lukito et al., 2021). The average total-body water decreases according to age where older people have the lowest total-body water content (Dmitrieva et al., 2011). The decrease in total body water among older people is driven by three main factors which are reduction of lean mass, diminished thirst sensation and inefficient kidney function. All these factors increase the risk of dehydration among older people. This is further aggravated by external factors such as heat wave, poor dietary intake and poor hydration practise (Balmain et al., 2018). Side effects of certain medications may cause xerostomia or dry mouth due to decrease in saliva excretion (Tan et al., 2018). Older people and caregivers are responsible for preventing dehydration by practising appropriate hydration practice and relevant dietary intake.

4.3 Scientific Basis

An average healthy adult has a total-body water content of around 60 percent. Water comprises from 75 percent body weight in infants to 60 percent in adults. However, an average older person has a total-body water of around 55 percent (Serra-Prat et al., 2019). It is evident that as age increases, the body water content decreases which increases the risk of dehydration (Volkert et al., 2022). Ageing related dehydration is common among older persons because of their low water reserves. Loss of 2%-3% of body fluids may lead to physical and cognitive impairment among older persons. Dehydration among older persons is due to lower thirst sensation, decreased renal perfusion, and altered sensitivity to antidiuretic hormone (Picetti et al., 2017). Thus, it may be prudent for the older persons to learn to drink regularly even when they are not thirsty (Volkert et al., 2022).

In the body, water is the matrix in which all life processes occur. It has very important role in hydration and metabolism. Around 6 to 8 glasses of water per day may prevent dehydration as it could provide between 1.5 L to 2.0 L of water, respectively. The Institute of Medicine (2004) recommends adequate total water intake of 3.7 L/day and 2.7 L/day for older persons males and females, respectively. Due the huge variability in water needs based on metabolism, environmental conditions and activities, it is difficult to propose a single level of water intake that would assure adequate hydration and optimum health for half of all apparently healthy persons in all environmental conditions. Thus, an Adequate Intake (AI) is established in place of an EAR for water (Popkin et al., 2010).

Dehydration is dangerous and may be fatal to older persons. The major routes of water loss are from urine, faeces, vomiting and sweat. The risk of dehydration increases especially in hot weather due to sweating or if the older persons is suffering from diarrhoea and vomiting as the latter causes huge water loss. The sign and symptoms of dehydration are decreased skin turgor, dizziness, fatigue, dry mouth, dark coloured urine and less urine output. Older persons are not well-hydrated due to diminished thirst sensation, decreased mobility, reduced body water capacity, inefficient kidney function to retain water and decreased mobility (Schols et al., 2009). Incontinence could also predispose to dehydration as the older persons may limit their fluid intake. It is estimated that around 13% of older persons has health issues related to urinary incontinence (Batmani et al., 2021). Lack of water may cause falls, constipation, infections, impaired cognition, acute confusion, kidney stone, frailty and even death (Edmonds et al., 2021).

Total water is obtained not only from drinking water, but also from other liquid sources including soups and beverages (Zhou, 2022). Table 4.1 shows that wholesome food such as red bean porridge and noodle soup could contribute to almost 200-500 ml of water per meal. In addition, fruits and vegetables such as orange, watermelon and spinach also have high water content. Ideally plain water should be consumed regularly even when not feeling thirsty. Consumption of caffeinated drink such as coffee and tea should be limited as it contains diuretic agents which may increase urination and subsequently water loss. Older people are advised to consume no more than 180 mg of caffeine-containing beverages per day (2 cups of coffee) due to the diuretics effects of caffeine (Reyes et al., 2018). Sweet beverages should be minimised to prevent excess sugar intake and the risk of diabetes development.

Older persons also need to avoid alcohol due to its ability to impair the immune system and increasing the susceptibility to infectious disease (Satre, 2020). Alcohol intake is risky for older people due to the physiological changes associated with ageing such as comorbidities, increased risk of falls, frailty, sarcopenia and polypharmacy (Kelly et al., 2018). Moreover, changes in body composition and metabolism with ageing may interfere with the pharmacokinetics (absorption and metabolism) and pharmacodynamics of alcohol (Crome et al., 2012). Even low levels of alcohol can be harmful for older persons due to the presence of medications-alcohol interaction which may increase the risk of hypoglycemia, hypotension, sedation, gastrointestinal bleeding and liver damage (Holton et al., 2019). Alcoholic beverages must not be consumed to satisfy thirst at any occasion especially after exercising (NCCFN, 2021). The National Institute on Alcohol Abuse and Alcoholism (NIAAA) advises men and women aged 65 years old and above to consume not more than 1 standard drink per day or 7 standard drinks per week. Moreover, the Canadian Centre on Substance Use and Addiction has recommended that older men may drink no more than 1-2 standard drinks per day and not exceeding 7 alcoholic drinks per week, while for older women, 1 standard alcoholic drink per day and not exceeding 5 alcoholic drinks per week. When older people decide to stop alcohol consumption, this must be done gradually to avoid withdrawal syndrome. Older people may experience signs and symptoms such as headache, tremor, automatic hyperactivity, tachypnea, hyperthermia, diaphoresis, hallucinations, hypertension and seizures (Jesse et al., 2017). The mortality rate from alcohol withdrawal is higher if untreated. Older persons are advised to reduce alcohol consumption gradually and get advice from healthcare professionals. They may join various support groups for guidance to stop drinking.

Table 4.1 Water content values of selected food

Food	Weight of edible portion in household measure/ as purchased (g)	Water (ml)
Ice cube	30	30
Chicken rice	250	162
Red bean porridge	230	203
Soup noodle	563	478
Watermelon	133	124
Orange	131	114
Spinach	100	91.3

Source: Nutrient Composition of Malaysian Food (1997)

Raising awareness of the older persons, their families, and caregivers on the potential risks of dehydration, its consequences and normal fluid requirement is fundamental to prevent dehydration (Beck et al., 2021). By having water readily available all day, for example, by placing water bottle at chairside or bedside, could help to increase the accessibility and frequency of water consumption for the older people. Also, consuming water from bottle with volume level indicator could provide accurate estimate of water consumed by the older persons in a day. This information could also be used to monitor fluid intake throughout the day. Encouragement of 'wet' food intake such as porridge, noodle soup and watermelon were also found to be useful to maintain or increase fluid intake among older people (Woodward, 2007). Dipping biscuits or bread in liquid such as milk could improve the fluid intake of the older persons. Preventive measures of dehydration in older persons should be made known not only to the older persons itself but also to their caregivers.

However, older persons with heart failure, chronic liver diseases, undergoing hemodialysis which require fluid restriction are advised to drink water according to their doctor`s recommendations.

4.4 Current Status

The prevalence of older persons not having sufficient intake of plain water is reported to be in the increasing trend from 27.1% in National Health and Morbidity Survey (NHMS, 2015) to 30.2% in NHMS 2018. Based on NHMS 2018, the lack of sufficient water intake was higher among older persons residing in the in the rural areas (37.2%) as compared to urban (27.6%) (IPH, 2018).

Older women are encouraged to drink at least 1.6L per day and men 2.5L per day. Fluid requirement increases in case of excessive water loss due to fever, diarrhea, vomiting, or severe hemorrhage (Volkert et al., 2022).

4.5 Key Recommendations

Key Recommendation 1

Drink adequate plain water even when not feeling thirsty.

How to achieve:

1. Drink 6 to 8 glasses of plain water per day. Increase the amount of water in hot weather, during illness and during and after exercise.
2. Add mint, lemon slices, or lemongrass in plain water to increase water intake.
3. Sip water regularly and/or suck on ice cubes to moisten mouth
4. Ensure plain water is readily available on table or at bedside.
5. Remember to drink plain water frequently by setting reminders in phone, notes on tables, mirror, or refrigerator.
6. Use bottle with volume indicator to monitor water intake

Key Recommendation 2

Obtain fluid from other food sources.

How to achieve:

1. Obtain fluid from other food sources such as soup, coconut water, juices and beverages low in fat, salt and sugar.
2. Limit consumption of caffeinated beverages such as teh tarik, tea and coffee to only 2 cups (200 ml × 2) per day. Decaffeinated beverages are better choice.
3. Limit carbonated beverages, energy drinks and other sugar-sweetened beverages to avoid dehydration.
4. Avoid drinking tea or coffee with your main meal for enhancing nutrient absorption.
5. Choose more frequently high water content fruit and vegetables such as watermelon, apples, oranges, pineapple, pear, cabbage, spinach, tomatoes, cucumber.

Additional Consideration

Older persons are recommended to avoid all forms of alcoholic beverages. They may choose tea, milk, coconut water, smoothies, or non-alcoholic sparkling juice instead of alcoholic beverages. For older adults who practice consuming alcoholic beverages, avoid keeping alcoholic beverages at home. Gradually reduce alcohol intake to avoid withdrawal syndrome symptoms such as anxiety, tachycardia, hallucination and seizures (Schwebach & Vempati, 2023).

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Key Message 5

Consume foods and beverages with low fat, salt and sugar

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Consume foods and beverages with low fat, salt and sugar

Dr. Yasmin Ooi Beng Houi, Dr. Khor Ban Hock, Dr. Shashikala Sivapathy and Mr. Mohammad Affendy Mohd Akhir

5.1 Terminology

5.1.1 Fats and Oils

Cholesterol
Cholesterol is a waxy, fat-like substance that is found in food from animal sources. The human body can synthesise the cholesterol that it needs.

Fatty acid
Fatty acid is a carboxylic acid consisting of a hydrocarbon chain with a terminal carboxyl group [-C(O)OH] and a methyl group at the other end (-CH3). The nutritional implications of a fatty acid depend on the length of the hydrocarbon chain, the presence of double bond in the hydrocarbon chain, and the presence of trans configuration of the double bond.

Saturated fatty acids
Saturated fatty acids (SFAs) are fatty acids that do not have a double bond in the hydrocarbon chain. SFAs common to the human diet are built of 12 to 18 carbon-chain lengths, namely lauric acid (C12:0), myristic acid (C14:0), palmitic acid (C16:0) and stearic acid (C18:0).

Monounsaturated fatty acids
Monounsaturated fatty acids (MUFAs) are fatty acids that have only one double bond in the hydrocarbon chain. The common MUFA in human diet is oleic acid (C18:1n-9).

Polyunsaturated fatty acids

Polyunsaturated fatty acids (PUFAs) are fatty acids that have more than one double bond in the hydrocarbon chain. PUFAs are further categorized into omega-3 PUFAs and omega-6 PUFAs, depending on the position of double bond counted from the methyl end of the hydrocarbon chain. Omega-3 PUFAs include alpha-linolenic acid (ALA, C18:3n-3), eicosapentaenoic acid (EPA, C20:5n-3) and docosahexaenoic acid (DHA, C22:6n-3). ALA is mainly found in plant sources while EPA and DHA are found in marine sources. A common omega-6 PUFA in the human diet is linoleic acid (LA, C18:2n-6).

Essential fatty acids

Essential fatty acids (EFAs) are fatty acids that cannot be synthesized by the human body, and therefore must be obtained from the diet. The two EFAs in human diet are ALA and LA.

Trans fatty acids

Trans fatty acids (TFAs) are fatty acids that have at least one double bond in the trans configuration. TFAs are produced in industrial process through partial hydrogenation to solidify liquid plant oils. Elaidic acid (C18:1n-9t), a common industrial TFA is produced during formulation of margarines, vanaspati (vegetable ghee), shortenings, and bakery products. Naturally occurring TFAs include vaccenic acid (C18:1n-7t) found in human milk, fat of ruminants, and dairy products.

5.1.2 Sodium and salts

Coarse salt

It is a course-grain salt that is either sold by weight in markets or as novelty salts in supermarkets. Each grain is larger than the grains of table salt. Examples are *garam kasar, garam bukit*, rock salt, sea salt and kosher salt. Like table salt, coarse salt also contains sodium.

Iodised salt

It is a form of salt that has been fortified with iodine. According to the Food Regulations 1985, iodised salt shall contain potassium iodide or iodate or sodium iodide or iodate equivalent to not less than 20mg/kg and not more than 40mg/kg iodine (MOH, 1985).

Low sodium, very low sodium and sodium-free food

Low sodium, very low sodium and sodium-free foods are defined as foods with sodium concentration not more than 0.12 g/100 g (solid) or 0.06 g/100 ml (liquid), 0.04 g/100 g (solid) or 0.02 g/100 ml (liquid) and 0.005 g/100 g (solid) or 0.005 g/100 ml (liquid) of food respectively (MOH, 2010).

Salt and sodium

The terms salt and sodium are often used interchangeably in leaflets, magazines, and online. The most commonly used salt in food is an inorganic compound consisting of sodium and chloride ions, i.e., NaCl, One gram of salt (NaCl) contains about 400 mg of sodium (Na). One mmol of sodium is equivalent to 23mg sodium (NaCl consists about 40% Na). Thus, 1 teaspoon or 5g salt provides 2000 mg of sodium. Sodium is also present in other forms, such as monosodium glutamate, sodium nitrate, and sodium benzoate. Sodium is also present naturally in plants and animals which contribute towards sodium intake.

Salt substitutes

It is referred to as light salts. They typically replace all or some of the sodium with another mineral, such as potassium or magnesium.

Sauce

Sauce is liquid or sometimes semi solid food served on food as a relish (served as an accompaniment to food), as gravy (kuah) or food used as a flavourful seasoning in preparing other foods. This includes soy sauce (fermented soy beans), oyster sauce, tomato and chilli sauces, fish sauce (made from fermented fish), prawn sauce (cencaluk), teriyaki sauce, and Worchester sauce. These sauces often contain a high amount of sodium.

Table salt

It is a fine-grained salt that often contains an anti-caking ingredient, such as calcium silicate, to keep it free-flowing. It is available iodised or non-iodised. This type of salt is mainly used in cooking and at the table. It is also known as refined salt.

5.1.3 Sugars

Added sugars
Added sugars, also known as extrinsic sugars are sugars that are usually added to foods. The terms refined, added and extrinsic sugars are sometimes used to denote sucrose, glucose and fructose used in the food industry and in the home. There is no difference between sugars that occur naturally in the food and refined sugars that are added to food on metabolism and health outcomes. Sugars that are present naturally within the cellular structure of food, e.g. sugars found in fruits and vegetables, are also known as intrinsic sugars. Foods with high added sugar content often have a lower nutrient content but are energy dense. The term no added sugar means no sugars have been added during the manufacturing process; it does not mean that no sugar is present, since most foods contain sugar in some form.

Glycaemic index
Glycaemic index, often abbreviated to GI is a numerical value associated with food that indicates its effect on a person's post prandial blood glucose. The GI concept is used by dietitians counselling their patients on the choice of carbohydrate foods. Criticism of GI include its inability to reflect rate of rise in post prandial blood glucose.

Simple carbohydrates
Simple carbohydrates are either mono-saccharide or disaccharide sugar molecules. Examples of commonly found mono-saccharide is glucose, and disaccharide is sucrose. Table sugar and castor sugar are sucrose. Simple carbohydrates are very rapidly digested.

5.2 Introduction

Fats and oils are part of a healthy diet if consumed within the recommended amount. They are important sources of energy, providing 9 kcal per gram. They also provide EFAs that cannot be synthesised by the human body, namely ALA and LA. In addition, fats and oils are essential for absorption of fat-soluble vitamins. Fats and oils are triacylglycerol molecules carrying fatty acids, which could be SFAs, MUFAs, and PUFAs, with some minor components such as phospholipids and steroids. Cholesterol is a form of steroid. However, overconsumption of fats and oils can cause excessive energy intake and therefore lead to weight gain. Therefore the Malaysian Dietary Guidelines 2020 recommends adults to reduce intake of foods high in fat and limit saturated fat intake (NCCFN, 2020). The MDG 2020's target age group is 18 to 59 years. Older persons might experience xerostomia, a condition which causes dry mouth, and dysgeusia, where the taste of food is unpleasantly distorted, particularly in older persons who are on multiple long-term medication (De Mendonça Guimarães et al., 2023; Drożak et al., 2021). In such situations, food with a smooth texture and fatty food might increase food intake (Cichero, 2017) in individuals who are at risk of under-eating. Therefore, the present Dietary Guidelines for Older Persons continues the dietary choices recommended by the MDG 2020 by recommending older persons to consume foods and beverages with low fat.

The recommendations in this guideline is also applicable for older persons with diabetes. Based on findings derived from analysis of the European Prospective Investigation in Cancer and Nutrition (EPIC), which is a cohort with 6,384 participants with diabetes and 258,911 participants without known diabetes, the multivariate adjusted mortality risk among individuals with diabetes compared with those without was increased (HR: 1.62, 95% CI: 1.51 – 1.75) (Sluik et al., 2014). Sluik et al. (2014) reported that among individuals with diabetes, intake of fruits, legumes, nuts, seeds, pasta, poultry and vegetable oil was related to a lower mortality risk, and intake of butter and margarine was related to an increased mortality risk. These associations were significantly different in magnitude form those in diabetes-free individuals, but directions were similar. Therefore, dietary and lifestyle advice with respect to mortality for individuals with diabetes should not differ from recommendations for the general population. Shi et al. (2015) reported that among elderly Chinese aged 80 years old and over, daily consumers of sugar had an adjusted hazard ratio for all-cause mortality of 1.04 (95% CI: 0.97 – 1.12) compared to non-consumers.

Sodium chloride is the most commonly consumed salt in Malaysia, most of which are iodised. Since 30 September 2019, only iodised salt could be sold for human consumption throughout the country when Regulation 285 regarding iodised refined salt or iodised salt under the Food Regulations 1985 was gazetted on 12 November 2018 (Federal Government Gazette, 2018). Where the Universal Salt Iodinisation is gazetted, Sabah since 1999 and Sarawak since 2008, only iodised salt could be sold for human consumption (Codling et al., 2017). Most Malaysians do not add salt to food at the table. Intake of sodium is most commonly through addition of salt and seasonings during cooking, through salted processed products, in the form of sauce and seasonings used as a dip at the table. Coarse salt, including *garam bukit* is often used to preserve fish and meats.

Other than breakdown of complex starches into sugar in the body, Malaysians usual consumption of added sugar is in the form of sucrose. Sugar is added to most home-made and bought sugar sweetened beverages. Sugar is also present in traditional cakes *(kuih),* cakes, pastries, desserts, and sometimes added to cooking as part of seasoning.

5.3 Scientific Basis

5.3.1 Evidence for dietary fat recommendations

Total dietary fat

The daily diet of older persons should not be devoid of fat. The Recommended Nutrient Intakes (RNI) 2017 recommends that 25% – 30% of total energy intake (TEI) is from fat (NCCFN, 2017). Based on a 1,500 kcal/day, that would be a fat intake of 42 – 50g/day. A diet that is devoid of fat may reduce palatability, and possibly in addition to presence of xerostomia and dysgeusia, might result in poor oral intake and therefore inadequate nutrient intake in older persons, particularly in individuals who have had a drastic change in social or family environments. In addition, decreasing dietary fat intakes may result in concomitant increase in carbohydrate intake, which contributes to increased risk of obesity and diabetes mellitus (Bahl, 2015; DiNicolantonio, 2014). Fat intake that is too low may contribute to unfavourable changes in plasma LDL-cholesterol and triglycerides (Graham et al., 2007). An overly restrictive low fat diet may cause deficiency in essential fatty acid (EFA) intakes. EFAs generate longer chain arachidonic, eicosapentaenoic (EPA) and docosahexaenoic (DHA) fatty acids which are required for regulation of blood clotting, blood pressure and cell membrane formations (Hornstra & Lussenburg, 1975). In contrast, a diet with excessive daily consumption of dietary fat, with regards to total amount of fat and type of fat, contributes to excessive total energy intake and adversely affects cardiometabolic health. Therefore, it is imperative to maintain the dietary fat intake within the recommended amount (NCCFN, 2017).

Replacement of saturated fat with unsaturated fat

The World Health Organization (WHO) recommends to reduce SFA intake to less than 10% of TEI, and to reduce TFA intake to less than 1% of TEI (WHO 2023), and the dietary fats are preferably to be made up of PUFAs (up to 10% of TEI) or MUFAs (10% – 15% of TEI) (WHO, 2007a). The RNI 2017 also recommends a distribution of SFA to be less than 10% of TEI, MUFAs to be between 12% to 15% of TEI, n-6 PUFAs to be between 3% to 7% of TEI and n-3 PUFAs to be between 0.3% to 1.2% of TEI. The recommendations by the WHO and RNI 2017 to reduce SFAs to less than 10% of TEI are based on the evidence that SFAs increase low-density lipoprotein cholesterol (LDL-C), a well-established surrogate marker for atherosclerotic cardiovascular diseases. A systematic review and multiple regression analysis reported favourable improvement in total and LDL-C and triglyceride levels when SFAs were replaced with MUFAs and PUFAs. When the effects of the individual SFA on the serum lipoprotein profile were evaluated, it was shown that lauric, myristic or palmitic acid raised serum total, LDL-C and high density lipoprotein cholesterol (HDL-C) levels, and lowered triglyceride levels, while stearic acid did not have a significant effect on these serum lipid levels. Lauric acid alone reduced the total cholesterol to HDL-C ratio as well as the LDL-C to HDL-C ratio (Mensink, 2016).

A recent cross-sectional study showed that dyslipidaemia was prevalent among Malaysian older persons (≥ 60 years old), with 64.9% had elevated total cholesterol (> 5.2 mmol/l), 56.7% had elevated LDL-cholesterol (> 3.4 mmol/l), 40.1% had elevated triglycerides (> 1.7 mmol/l), 33.9% had low HDL-cholesterol (<1 mmol/l in males or < 1.2 mmol/l in females), and 57.3% had elevated non HDL-cholesterol (> 4.2 mmol/l) (Mohamed-Yassin et al., 2021). In Malaysia, the major sources of SFAs in the diet is palm oil and coconut oil in the form of coconut milk *(santan)*. Palm oil contains almost equal amount of saturated and unsaturated fatty acids (40% palmitic acid, 4% stearic acid, 43% oleic acid). Coconut milk contains 92% saturated fatty acids, mostly in the form of lauric acid (49%) and myristic acid (18%). A large proportion of vegetable oil used by Malaysians is palm oil, which contribute to 3% – 4% of TEI for EFA, 12% of TEI for SFA, and 14% of TEI for MUFA (NCCFN, 2017).

Despite the well-established effect of SFAs on total and LDL-C levels, controversies exist whether SFAs are associated with an increased risk of cardiovascular diseases and mortality. Several meta-analyses of prospective cohort studies concluded that dietary SFA is not associated with an increased risk of cardiovascular outcomes (Chowdhury et al., 2014; Siri-Tarino et al., 2010; de Souza et al., 2015). The Prospective Urban and Rural Epidemiological (PURE) study, a multinational study that recruited individuals aged 35 years old – 70 years old from 18 countries (including Malaysia), reported that dietary SFA intake was not associated with an increased risk of cardiovascular outcomes. Instead, dietary SFA intake was associated with a lower risk of mortality and stroke (Dehghan et al., 2017). On the contrary, a recent meta-analysis by Kim et al. (2021) showed that every 5% increment in dietary SFA was associated with 5% higher risk of cardiovascular mortality.

Specific to the elderly population, the Health ABC study showed that there was no association between dietary SFA intake and cardiovascular disease in elderly aged 70 years old – 79 years old (Houston et al., 2011). Contrarily, a 10-year prospective cohort study reported that dietary SFA intake was significantly associated with atherosclerotic vascular disease mortality in elderly women (Blekkenhorst et al., 2015). A prospective cohort study included 371,159 participants aged 50 years old – 71 years old showed that isocaloric replacement of SFA with other macronutrients such as unsaturated fat, low-quality carbohydrate, high-quality carbohydrate, animal protein, and plant protein were associated with lower total mortality, cardiovascular mortality, and cancer mortality (Zhao et al., 2023).

Similarly, meta-analyses of randomized controlled trials also reported mixed findings on the effect of dietary SFA reduction on cardiovascular outcomes. The meta-analyses by Skeaff & Miller (2009), Mozaffarian et al., (2009), Sacks et al., (2017), and Hooper et al., (2020) reported favourable cardiovascular outcomes with reduction of dietary SFA intakes. Contrarily, the meta-analyses by Ramsden et al., (2013), Harcombe et al., (2016), and Hamley (2017) did not demonstrated significant reduction in cardiovascular events with the reduction of dietary SFA intake. The disparities in the findings from these meta-analyses are attributed by the number of randomized controlled trials included in the meta-analyses (Heileson, 2019).

Beyond the cardiometabolic outcomes, the type of dietary fat is also associated with other health outcomes in older persons. A prospective cohort study reported that dietary SFA intake was associated with both higher frailty and mortality among subjects aged ≥ 50 years above (Jayanama et al., 2020). The Singapore Chinese Health Study also showed that substitutions of SFAs with MUFAs or PUFAs were associated with a lower risk of cognitive impairment in Chinese elderly (Jiang et al., 2020). Another 3-year prospective cohort study reported that greater dietary MUFA intake was associated with less cognitive decline, and this effect was not seen for SFA (Naqvi et al., 2011). A prospective cohort study consisting of 1,385 middle-aged and elderly people also reported that a higher intake of unsaturated fatty acids was associated with better cognitive function (Tao et al., 2019). Griffin et al., (2006) showed that 1 g/d of dietary intake of n-3 long chain PUFAs from foods promotes favourable alterations in lipoprotein particle size that could be caused by a decrease in basal and postprandial plasma triacylglycerol. A review showed that DHA supports cognitive function and delays dementia in the elderly (Mohajeri et al., 2015; Ruxton et al., 2016). However Jiao et al. (2014) showed that DHA supplementation does not improve cognitive performance in the elderly. This conclusion has been criticised for the small number of studies reviewed (Murphy & Winwood, 2015).

There are evidence to suggest that excessive levels of n-6 PUFAs relative to n-3 PUFAs increases the probability of diseases that have a prothrombotic, proinflammatory and proconstrictive pathogenesis (Patterson et al., 2012; McDaniel et al., 2013). Animal model studies suggest that PUFA intake may differentially affect those with metabolic syndrome (Burghardt et al., 2010). Arachidonic acid (C20:4n-6) conversion to n-6 precursors is part of the inflammatory response. Comparative studies suggest that an n-6: n-3 ratio of less than 4:1 may contribute to health (Simpoulos, 2008). A high ratio of n-6: n-3 possibly increases the risk of prostate cancer in men (Williams et al., 2011). However there are also evidence to show that deceasing n-6: n-3 PUFA ratio by altering n-3 PUFAs, does not influence insulin sensitivity or postheparin plasma lipase activities in older men and women (Griffin et al., 2006), and that dietary intake of n-6 PUFAs in men and women over 55 years was observed to be inversely associated with C-reactive protein, which is a marker for inflammation (Muka et al., 2015).

Limit intake of trans fat

TFAs raise LDL-cholesterol: HDL-cholesterol ratio about two times higher than C12 – 16 SFAs (Ascherio et al., 1999). Mozaffarian & Clarke (2009) concluded from a meta-analyses of 4 cohorts that an additional 2%TEI TFA resulted in a 23% increase in CVD risk. Two main sources of TFAs are residual TFAs from manufactured products that have undergone hydrogenation process, and naturally occurring ruminant TFAs in milk and dairy products. Systematic reviews and meta analyses have shown contradictory conclusions for the relationship between ruminant TFAs and coronary heart disease (CHD) risks. Brouwer et al. (2010) showed all TFAs irrespective of industrial or ruminant TFAs raised the ratio of plasma LDL to HDL cholesterol. Bendsen et al. (2011) & de Souza et al. (2015), were not able to show any relationship between ruminant TFAs and CHD, although lower intake levels of ruminant TFAs might have contributed to this conclusion (Bensen et al., 2011; de Souza et al., 2015). Intake of trans fatty acids (TFAs) from partially hydrogenated vegetable fats might not be of high concern in Malaysia because the main source of fat intake is from palm oil, which does not need to undergo a hydrogenation process. Palm oil's SFA content, higher melting points and structure of triacylglycerol negates the need for hydrogenation in the food industry. There should be some caution when deep-frying foods at high temperature using all types of oil, including palm oil as it has been shown that cooking at high temperature (>250^{o}C) leads to formation of TFAs (Chen et al., 2014). In a small study in a home for older persons in Ipoh, Malaysia, 81.5% of older persons residents consumed fried or oily foods > 4 times per week (Suganthan et al., 2019).

Caution should be exercised if consumers include margarines, vegetable ghee, partially hydrogenated edible oils, and bakery products containing partially hydrogenated fats as they may contain TFAs. A recent Spanish study showed that TFAs are still present in low amounts in margarines, at 0.68% for store bought and 0.43% for premium brands of margarine. These amounts are lower than the maximum permitted content of TFA in fats in Europe, and could therefore be considered as "trans free products" (Astiasarán et al., 2017). The limit on TFAs in countries where legislation exists is 2g per 100g of fat or oil (WHO Regional Office for Europe, 2015). There are no reported studies on the accumulated amount of ingested TFAs from a diet with a high intake frequency of these products.

Dietary cholesterol for older persons

Over a 9-year follow-up period, Houston et al. (2011) showed that higher dietary cholesterol and frequent egg consumption were significantly associated with increased risk of CVD among the US elderly aged 70 years old to 79 years old (n= 1947) with type 2 diabetes mellitus (DM). An analysis of 16,594 adults aged ≥ 60 years old from the China National Nutrition and Health Survey 2010 – 2012 showed that consumption of eggs, red meats and seafood were the top contributors to dietary cholesterol intake at 57.7%, 24.0% and 10.9% respectively. In these elderly Chinese, serum total cholesterol and LDL-cholesterol were significantly related to dietary cholesterol intake; each 100 mg increase in dietary cholesterol intake led to a 0.035 mmol/L (p=0.001) increase in serum total cholesterol and a 0.038 mmol/L increase in LDL-cholesterol (p<0.001) (Pang et al., 2017). Spence et al. (2012) showed that patients (n= 1262, 61.5 ± 14.8 years) attending vascular prevention clinics who consumed < 2 eggs per week had significantly less plaque area than those consuming ≥ 3 eggs per week.

However, Voutilainen et al. (2013) demonstrated that egg consumption including its yolk, at mean 3.9 eggs per week (n= 1941, 51.5 years) was not statistically associated with increased carotid atherosclerosis. Shin et al. (2013) reported no association between egg consumption and CVD in a healthy population, from a meta-analysis of 22 prospective cohorts. Consumption of egg was compared between ≥ 1 egg/day and < 1 egg/week. However Shin et al. (2013) reported an increased risk of CVD in diabetic patients who consumed ≥ 1 egg/day compared to those consuming < 1 egg/week. Two randomised controlled cross-over studies involving healthy young men and young and middle-aged women demonstrated that dietary cholesterol in whole egg was not well absorbed in men and women who had triacylglycerol lipoprotein fractions measured hourly from baseline to 10 hours after ingestion of a test meal of either whole cooked eggs or control food. These findings indicated that the dietary cholesterol present in the whole cooked eggs did not acutely influence plasma total cholesterol concentration (Kim & Campbell, 2018). Intervention studies have not found significant effects of increasing egg consumption on risk markers for CVD and DM type 2 in healthy participants and participants with DM type 2 (Geiker et al., 2018).

Results from a meta-analysis on seven prospective studies, showed that in a sub group of the pooled participants, women, those aged ≥ 60 years old, and those with BMI ≥ 24kg/m^2 seemed to have a relative risk of stroke for higher cholesterol intake (Kim & Campbell, 2018). Dietary cholesterol was also shown to be positively associated with risk of stroke and cerebral infarction in a 10.4 years follow-up of the Swedish Mammography Cohort (n = 34,670 women, aged 49 – 83 years) (Larsson et al., 2012). However, they also showed that total fat, saturated fat, monounsaturated fat, polyunsaturated fat, α-linolenic acid, and omega-6 PUFA intakes were not associated with stroke, while intake of long-chain omega-3 PUFAs was inversely associated with risk of total stroke. The Adult Treatment Plan (ATP) IV guidelines of the American Heart Association, the USDA 2015 and the RNI 2017 have removed the restriction on dietary cholesterol intake for a healthy population (Eckel et al., 2013). In the present dietary guidelines, egg is not given a particular mention in this recommendation. Despite the lack of evidence on the benefits from dietary cholesterol reduction, individuals should still be aware that foods that have high dietary cholesterol also have substantial content of SFA, which are known to increase TC and LDL-cholesterol (Fattore et al., 2014; Sun et al., 2015) and triacylglycerol levels (Kim & Campbell, 2018). The science advisory from the American Heart Association on dietary cholesterol and cardiovascular risk states that for older normocholesterolemic patients, given the nutritional benefits and convenience of eggs, consumption of up to 2 eggs per day is acceptable within the context of a heart-healthy dietary pattern (Carson et al., 2020).

Dairy fat for older persons

Full fat dairy products such as butter, cheese, and whole milk contain a high amount of SFA, and therefore dietary guidelines for older persons have recommended consumption of reduced-fat milk and yogurt as a means to reduce saturated fat intake (Dorrington et al., 2022; Lysons et al., 2022). However, recent scientific evidence has demonstrated that SFAs from dairy products have varying effects on health outcomes compared to SFA of other food sources because a greater proportion of short- and medium-chain SFA present in dairy. A review by Mozaffarian (2019) inclusive of several prospective cohort studies and meta-analyses suggests that the consumption of whole-fat dairy is not associated with weight gain and increased risks of diabetes mellitus, cardiovascular disease, and mortality. A cross-sectional analysis of 226 institutionalised older adults showed that saturated fat from dairy products was associated with a higher HDL-C level and a lower TC:HDL-C ratio (Liu et al., 2019). A prospective cohort study involving 2,907 adults aged ≥ 65 years old showed that plasma phospholipid dairy fatty acids, namely pentadecanoic acid (C15:0), heptadecanoic acid (C17:0), and trans-palmitoleic acid (trans-16:1n-7) were not associated with total mortality and incident cardiovascular disease (de Oliveira Otto et al., 2018).

On the other hand, several studies have suggested that the consumption of low/reduced-fat dairy may offer additional health benefits to the older persons population. The PREDIMED study showed that the consumption of low-fat dairy products was associated with a lower systolic blood pressure in 2,290 older persons with a high cardiovascular risk at 12-month follow up (Toledo et al., 2008). Another prospective cohort study on 1,871 community-dwelling adults aged ≥ 60 years old showed that a higher consumption of low-fat milk and yogurt was associated with a lower risk of frailty and weight loss, while whole milk or yogurt was not associated with incident frailty (Lana et al., 2015). Another prospective cohort study with a median follow-up of 3.2 year showed that consumption of a greater amount of low-fat dairy products, yogurt (all types), and low-fat milk was associated with a reduced risk of metabolic syndrome in an elderly Mediterranean population aged 55 years old – 80 years old with an underlying high cardiovascular disease risk (Babio et al., 2015).

Evidence for dietary sodium recommendations

The evidence linking dietary salt intake and blood pressure is almost 5,000 years old: "If too much salt is used for food, the pulse hardens…" (Huang Ti Nei Ching Su Wen [The Yellow Emperor's Classic of Internal Medicine], 2698-2598 BC) (Delahaye, 2013). Lower sodium consumption is associated with lower cardiovascular disease risk (Cook et al., 2016; Engberink et al., 2017). Two of the most important findings which demonstrated a dose-effect relationship between sodium intake and systolic blood pressure come from the Prospective Urban Rural Epidemiology (PURE) ongoing in 21 countries. Adults aged 35 years old - 70 years old without cardiovascular disease from the general population were chosen. Morning fasting urine to estimate 24 h sodium and potassium excretion was used as a surrogate for intake. The study determined the relationships between sodium and potassium intake and blood pressure (Mente et al., 2018).

United Kingdom Scientific Advisory Committee on Nutrition reported salt intake to be positively associated with risk of hypertension, stroke and ischemic heart disease mortality (SACN, 2017). A meta-analysis of 11 RCTs in subjects aged 60 years old and above similarly found sodium chloride intake to be positively associated with systolic and diastolic blood pressure (SACN, 2003). Higher sodium intake has also been associated with increased carotid intima-media thickness and atherosclerotic plaque evidence (Mazza et al., 2018). Nonetheless, salt enhances dietary palatability, helping prevent protein energy malnutrition, which is prevalent among older adults (BAPEN, 2010).

Population observations of middle-aged to elderly adults have demonstrated that reductions in dietary salt intakes are linked to improvements in blood pressure (Kabir et al., 2015). Cross-sectional studies such as the Shandong Ministry of Health Action on Salt Reduction and Hypertension (SMASH) on 1,948 Chinese adults aged 18 years old to 69 years old (Yan et al., 2015) and the Nagahama study involving 9,144 individuals aged 54 ± 13 years old showed that salt intake is associated with increased blood pressure (Tabara et al., 2015). In overweight or obese individuals, there is a dose-effect relationship between sodium intake and systolic blood pressure (Yan et al., 2015). Analysis of data from the National Diet and Nutrition Survey 2008/2009 to 2011/2012 of 784 adults aged 49 ± 17 years old showed that salt intake is significantly associated with an increased risk of obesity (Ma, He & MacGregor, 2015). A meta-analysis showed that a 10 mmol/d increase in dietary sodium was associated with a significant 1% increase in CVD mortality (p=0.016), (relative risk = 1.12, 95%CI: 1.06 – 1.19) (Poggio et al., 2015). A randomised controlled cross-over intervention showed that adults with untreated elevated blood pressure had a substantial increase in blood pressure after receiving a 3 g/d supplemental sodium equivalent to 7.6 g/d of salt for 4 weeks. Their 24 hours blood pressure increased significantly by 7.5/2.7 mmHg (Gijsbers et al., 2015).

Malaysian Dietary Guidelines and WHO recommendations on sodium consumption for adults, which is 2000 mg/day is equivalent to < 5g (90 mmol) of salt per day (NCCFN, 2020; WHO, 2003; WHO, 2007b; WHO, 2012). The Dietary Guidelines for Americans 2020 recommend intake of 2300 mg/day for adults and over older persons (USDA & USDHHS, 2020). The Australian and New Zealand Nutrient Reference Values recommend an Adequate Intake of 460 – 920mg/day (20 – 40 mmol) of sodium for adult men and women; and an Upper Level of 2300 mg/day (100 mmol) (NHMRC, 2017). Therefore, this Dietary Guideline advises that older persons ask for less salt in the cooking when eating out, and to avoid adding soy sauce to food that already has soy sauce or other sauces. A longitudinal study of 8959 Chinese aged 80 years old and above found a positive association between intake of salt-preserved vegetables and mortality. Daily consumers of salt-preserved vegetables had an adjusted hazard ratio for all-cause mortality of 1.10 (95% CI: 1.06 – 1.19), and occasional consumers 1.12 (95% CI: 1.06 – 1.19) compared to those who never consumed it (Shi et al., 2015). Therefore, this guideline advises older persons to prefer fresh fruits and vegetables instead of pickled fruits and vegetables.

Used within the recommended limit, salt can help older persons to maintain adequate food and nutrient intake. As part of the ageing process, the salivary glands tend to produce less saliva with age, so older persons may experience xerostamia. Xerostamia may also develop as side effect of medications or radiation treatment for cancer. Many medications block the acetylcholine pathway which signals the muscles and glands, including the salivary glands. The sense of thirst in older persons also becomes less acute. As part of the ageing process, they would also experience gradual loss in taste and smell. Impairment of taste and smell perception can lead to decreased food intake in older persons. A 10-year follow up data from a prospective cohort of 2,642 elderly adults (age: 73.6 ± 2.9 years) showed that consuming more than 2300 mg/day of sodium was not associated with significantly higher mortality in adjusted models. The ten-year mortality was not significantly lower in the group receiving 1500 to 2300 mg/d of sodium than in the group receiving < 1500 mg/d and the group receiving > 2300 mg/d (p=0.07). Sodium intake was assessed by a food frequency questionnaire (Kalogeropoulos et al., 2015). However their findings do not change the current recommendations for sodium intake, and more evidence is required about the role of sodium on health outcomes in older adults.

In Southeast Asia, salted and fermented foods are part of the population's food habits, in addition to a variety of sauces that contain salt such as oyster sauce, soy sauce, fish sauce (*budu*), shrimp paste, tomato ketchup and chilli sauce (Amarra & Khor, 2015). Despite the process of urbanisation in these countries, increased salt consumption has been shown to come from both home-cooked foods and foods eaten away from home (Brown et al., 2009). In terms of salt usage, the public health message should be to continuously monitor the salt concentration of dishes to maintain a daily intake that does not exceed intake recommendations. A double-blind randomised controlled community study in Japan with participants aged 40 – 75 years old (n=50) demonstrated that monitoring salt concentration of home-cooked dishes had a potentially stronger salt-reducing effect than usage of low-sodium seasoning without regard to the amount of seasoning (Nakadate et al., 2018). In Malaysia, a qualitative study which included focus group discussion involving policy makers from relevant government ministries, non-governmental organisations and food science / food technology researchers and food industries identified that barriers to reduction of salt intake are perceived poor consumer acceptance, lack of knowledge and resources, challenges in reformulation of food products, and unavailability of guidelines and salt targets. These are despite implementation of awareness campaigns and education, sodium content labelling, product reformulation strategies (Harun et al., 2023). This Dietary Guidelines aim to provide suggestions for individuals to reduce their salt intake.

5.3.2 Evidence for dietary sugar recommendations

Added sugar and cardiovascular risks

Yudkin (1972) called sugar "pure, white and deadly" and suggested that it is associated with obesity, heart disease, hyperactivity, diabetes and dental caries. Forty years later, Te Morenga et al., (2014) demonstrated via meta-analyses of 39 randomized controlled trials, that dietary sugars significantly influence blood pressure and serum lipids. Their analyses showed that higher sugar intakes significantly raised triglyceride concentrations by a mean difference of 0.11 mmol/L (95% CI: 0.07 – 0.15, $p < 0.001$), raised total cholesterol by a mean difference of 0.16 mmol/L (95% CI: 0.10 – 0.24, $p < 0.001$), raised LDL-cholesterol by a mean difference of 0.12 mmol/L (95% CI: 0.05 – 0.19, $p = 0.001$), and raised HDL-cholesterol by a mean difference of 0.02 mmol/L (95% CI: 0.00 – 0.03, $p = 0.03$). A separate systematic review of cohort studies showed a 'U' or 'J' shaped relationship between added sugar intake and all-cause and cardiovascular mortality (Song et al., 2022). Song et al., (2022) reported that increased mortality was observed in higher added sugar intake and a lower intake of less than 5% of energy, albeit noting discrepancies between gender, age groups and countries.

In a randomised controlled trial with healthy overweight participants, Raben et al. (2011) showed that after ten weeks of intervention, triglyceride and leptin levels were significantly higher in the group receiving a diet high in sucrose versus the group receiving artificial sweeteners. Stanhope et al. (2009) showed that overweight/obese adults aged about 50 years old showed an increase in triglyceride levels by approximately 10% after ten weeks of intervention drinking glucose-sweetened beverages providing 25% of energy requirements. Stanhope et al. (2009) also showed that visceral adipose volume increased significantly in subjects consuming fructose sweetened beverages providing 25% of kcal. The fructose group also showed significant increases in fasting apoB, LDL, small dense LDL, and oxidized LDL. These studies showed that a diet high in sugar intake increases risk of these diseases in the long term.

Malik & Hu (2022) in their review showed that sugar-sweetened beverages were the major source of added sugars in the diet. A meta analyses of 11 studies evaluating the consumption of sugar-sweetened beverages (SSBs) and 8 studies evaluating the consumption of low-calorie sweetened beverages (LCSBs) showed that 1 serving / day increment of SSBs was associated with an 8% (RR: 1.08, 95% CI: 1.02 – 1.14, I2 = 43.04%) higher risk of CVD incidence and an 8% (RR: 1.08, 95% CI: 1.04 – 1.13, I2 = 40.6%) higher risk of CVD mortality. Habitual consumption of SSBs was associated with a higher risk of CVD morbidity and mortality in a dose-response manner. Yin et al., (2021) predicted that consumption of SSBs may be linked to 9.3% (95%CI: 6.6 – 11.9) of predicted CVD incidence in the USA from 2015 to 2025 among adults aged 40 years old – 77 years old in 2015 – 2016. Higher intake levels of low-calorie sweetened beverages (> 2 servings / day) appeared to be significantly associated with CVD mortality in a nonlinear manner ($P = 0.003$ for nonlinearity); however LCSBs association with higher risk of CVD morbidity and mortality may be complicated by reverse causation and residual confounding (Yin et al., 2021). A Malaysian cross-sectional study (n = 1,057, aged ≥ 60 years) showed that total sugar intake at 50th, 75th and 100th percentile increased total cholesterol risk by 1.7 fold ($p<0.01$), 1.5 fold ($p<0.05$) and 2.3 fold ($p<0.001$) respectively. Intake of added sugar at 100th percentile increased risks of high total cholesterol by two-fold ($p<0.001$) and triglyceride by 1.8 fold ($p<0.001$) (Zainuddin et al., 2018).

Zhang et al., (2021) analysed the association of SSB and artificially sweetened beverage (ASB) intakes with mortality using the US National Health and Nutrition Examination Survey (NHANES 1999 – 2014, n = 31,402). NHANES collected SSB and ASB intake using 24-hour dietary recalls. They reported that multivariate hazard ratio associated with each additional serving/day of SSB was 1.05 (95%CI: 1.01

– 1.09) for all-cause mortality and 1.11 (95% CI: 1.03 – 1.21) for heart disease mortality. Substituting one serving/day of SSB by an equivalent amount of ASBs, unsweetened coffees and teas, and plain water was associated with a 4% – 7% lower risk of all-cause mortality.

Added sugar, obesity and metabolic syndrome

Meta-analyses have consistently showed positive associations between intake of SSBs and vascular risk factors, such as body weight gain (Malik et al., 2013) and type 2 diabetes (Imamura et al., 2015). Malik et al. (2013) in their meta-analyses of 6 RCTS and prospective cohort studies involving adults in Europe and United States showed increases in body weight when SSBs were added (random and fixed effects: 0.85 kg; 95% CI: 0.50, 1.20 kg). Servio et al., (2014) in their meta-analysis, showed that sugar consumption was a significant predictor of obesity ($p= 0.03$), together with physical inactivity ($p= 0.003$) and cereal consumption ($p <0.001$). Te Morenga et al., (2012) in their meta-analysis of adults eating ad libitum, showed that groups with highest intake of sugar sweetened beverages gained 1.55 kg body weight over one year compared with groups with the lowest intake because the sugars modify energy intake and thus body weight. Raben et al. (2002) reported that a high-sugar diet increases body weight and a low-sugar diet decreases weight in overweight adults over a 10-week intervention. The high sugar group gained an average of 1.6 kg over 10 weeks, of which 1.3 kg was a gain in fat mass. The low-sugar diet group lost an average of 1.0 kg over the same period, of which 0.7 kg was fat free mass and 0.3 kg was fat mass.

Higgins & Mattes (2019) showed that in a 12-week randomised controlled intervention designed to compare the effects of low calorie sweeteners (aspartame, saccharin, sucralose and rebaudioside) and sucrose on body weight changes in adults aged 18 years old – 60 years old with overweight or obesity ($n= 154$), participants assigned to sucralose had negative body weight changes at 12 weeks and the change was significantly lower compared with other low calorie sweeteners (weight difference $\geq 1.37 \pm 0.52$ kg, $P \leq 0.008$). Energy intake decreased with sucralose consumption ($p = 0.002$). Body weight of participants assigned to the sucrose (Δweight = $+1.85 \pm 0.36$ kg) and saccharin (Δweight = $+1.18 \pm 0.36$ kg) group increased significantly ($p \leq 0.02$) across the 12-week intervention.

A meta-analysis showed an increased risk of about 20% for metabolic syndrome comparing the highest to the lowest categories of sugar-sweetened beverages (Malik et al., 2010). Using the Framingham Offspring Study cohort, Yoshida et al. (2007) showed that middle- and older-aged adults who consumed sugar-sweetened drinks had a significant increase in fasting insulin. Those who did not consume such drinks had fasting insulin of 188 pmol/L vs those who consumed at least two servings/day had 206 pmol/L ($p <0.001$). Sugar-sweetened drinks contain high amounts of fructose corn syrup (Malik & Fu, 2012). Stanhope (2012) showed that fructose decreases insulin sensitivity in older, overweight and obese individuals. Fructose in sugar-sweetened beverages increases hepatic lipogenesis and insulin resistance. Insulin resistance and β-cell dysfunction occur before onset of type 2 diabetes mellitus (DM). Therefore, increased consumption of energy yielding sweetened beverages, which contain rapidly absorbed simple sugars may contribute to an increased risk of type 2 DM. Malik & Fu (2012) showed using meta-analysis that pooled data from 310,819 participants, that individuals who consumed 1–2 servings/day of sugar-sweetened beverages had a 26% greater risk (RR 1.26, 95%CI: 1.12 – 1.41) of developing type 2 DM compared to those who consumed none to <1 serving per month. One serving/ day of sugar-sweetened beverage was associated with a 15% increase in risk for diabetes (RR 1.15, 95%CI: 1.11 – 1.20) (Malik et al., 2010). Evidence from systematic review and meta-analysis of 13 prospective studies ($n = 49,591$) showed that adverse associations were only demonstrated for SSBs and incident metabolic syndrome (RR for 355 mL/d, 1.14; 95% CI, 1.05-1.23). There was no adverse association between other major food sources of fructose containing sugars such as yogurt, fruit, 100% fruit juice and mixed fruit juice (Semnani-Azad et al., 2020). Based on these evidence, the focus of Key Recommendation 5 in this Dietary Guidelines is to reduce consumption of sugary beverages.

Greenwood et al. (2014) in their meta-analysis showed that consumers of sugar-sweetened soft drinks had RR 1.20/330 ml per day (95% CI: 1.12 – 1.29, $p<0.001$) compared to consumers of artificially sweetened soft drinks had RR of 1.13/330 ml per d (95% CI: 1.02 – 1.25, $p=0.02$) for type 2 DM. Imamura et al., (2015) in their meta-analysis showed that higher consumption of sugar-sweetened beverages was associated with a greater incidence of type 2 diabetes, by 18% per one serving/day (95%CI: 9 – 28), and for artificially sweetened beverages, 25% (95%CI: 18 – 33). Their findings suggest that there may be detrimental effects of sugar-sweetened beverage independent of obesity. They also suggested that findings for artificially sweetened beverages and type 2 DM may have publication bias, and therefore caution should be exercised in their interpretation (Imamura et al., 2015).

On average, sugar-sweetened beverages contain 140 to 150 calories and 35 to 37.5 g sugar per approximately 350 ml (Malik & Fu, 2012). Therefore, in this Dietary Guideline, older persons are advised to choose plain water over various sugar-sweetened beverages that are available in the market, i.e., carbonated and non-carbonated drinks, cordials, and beverages prepared at home or ordered when eating out. Sugars in sugar-sweetened beverages and those added into food acutely increase blood glucose levels and have a high glycaemic index. Long-term exposure to high glycaemic index (GI) is a risk factor for type 2 diabetes (Livesey et al., 2013). However, a systematic review of low GI or low glycaemic load (GL) diet showed these diets may result in little to no difference in fasting blood glucose level compared to higher GI or GL diets (mean difference: 0.12 mmol/L, 95% CI 0.03 to 0.21; $I2 = 0\%$; 6 studies, 344 participants; low-certainty evidence) (Chekima et al., 2023).

Evidence from these form the basis of Key Recommendation 5, which recommends older persons to reduce consumption of sugary beverages. In this Key Recommendation, the first two suggestions were for older persons to choose plain water and to train themselves to prefer beverages without any additional sugar or sweeteners. Recognising that loss of taste or change of taste might occur in older persons, and these might negatively affect intake, subsequent suggestions provided limits for various added sugars and sugar reduction strategies.

Added sugar and increased blood pressure

Raben et al. (2002) showed that blood pressure increased significantly in the high-sugar diet group compared to the low-sugar diet. Between-group difference was 6.9 mmHg for systolic and 5.3 mmHg for diastolic blood pressure. Sucrose intake was a significant predictor of changes in systolic blood pressure. Two separate studies also showed greater risk of developing hypertension among consumers of sugar-sweetened beverages compared to non-consumers, RR 1.18, 95% CI: 0.96 – 1.44 (Dhingra et al., 2007) and RR 1.10; 95% CI: 0.87 – 1.39 (Nettleton et al., 2009). These two teams of researchers also reported that frequent sugar consumers had marginally increased risk for developing hypertriglyceridemia compared to infrequent consumers, RR 1.25, 95% CI: 1.04 – 1.51, and RR 1.25, 95% CI: 0.98 – 1.57 respectively, and increased risk for low HDL-cholesterol, RR 1.32, 95% CI: 1.06 – 1.64, and RR 1.24, 95% CI: 0.95 – 1.61 respectively.

A Malaysian cross-sectional study ($n = 1,057$, aged ≥ 60 years) showed that total sugar intake at 50th percentile increased risk of blood pressure by 0.68 fold ($p<0.05$) (Zainuddin et al., 2018). Another cross-sectional study in the United States ($n=128$, aged 65 – 80 years) showed significant association between intake of added sugar and blood pressure in females after controlling for age, income, body mass index, physical activity levels, total energy intake and usage of blood pressure medication (Mansoori et al., 2019). Mansoori and colleagues' multiple linear regression model predicted that a decrease of 2.3 ± 0.5 teaspoons of added sugar would result in a 8.4 mmHg drop in systolic blood pressure and a 3.7 mmHg drop in diastolic blood pressure. Their model showed that fruit consumption did not show a negative association, instead showing a 0.71 cup increase in whole fruit consumption was associated with a 2.8 mmHg drop in diastolic blood pressure.

5.4 Current Status

Fat intake

The nationally representative findings from MANS 2014 showed that median fat intake for all adults was 46.61 g/day for men and 45.99 g/day for women (IPH, 2014), compared to MANS 2003 which showed that median intake was 50g/day for men and 41 g/day for women (Mirnalini et al., 2008). Among the 50 – 59 year olds, median fat intake was 44.17g/day for men and 45.30g/day for women in MANS 2014 (IPH, 2014) compared to 46g/day for men and 38g/day for women in MANS 2003 (Mirnalini et al., 2008). Elderly men decreased their fat intake but elderly women increased their fat intake in the time between MANS 2003 and MANS 2014. MANS 2014 also reported similar median fat intake in Peninsular Malaysia (46.55 g/day) and Sabah and Sarawak regions (46.33 g/day). Urban dwellers had higher median fat intake (48.7 g/day) compared to rural dwellers (44.46 g/day). Malaysians of Chinese ethnicity had the highest median fat intake (53,31 gg/day) compared to Indians (47.22 g/day), Malays (45.33 g/day), Bumiputera Sarawakians (45.32 g/day) and Bumiputera Sabahans (44.73 g/day).

The TSC for RNI 2017 recommends that total fat intake should not exceed 30% of total energy intake. MANS 2014 showed that median fat intake for adults of all age groups was 46.43g/day, which contributed approximately 29% of median energy intake (IPH, 2014). This was an increase from MANS 2003 which showed that fat intake contributed 27% of median energy intake (Kandiah et al., 2008). Both these two methodologically homogenous surveys sampled Malaysians aged 18 to 59 years. There is as yet no nationally representative study on the nutritional status of elderly Malaysians. Several small studies showed contradictory results in fat intake of the older persons. Some studies showed intakes below 30% of energy intake, others showed intakes above 30% of energy intake. The groups of people who had reported fat intakes > 30% of energy intake were Chinese premenopausal women, Malays, and rural women with varying levels of food security (Shahar et al., 2018). The TSC noted that the intake of PUFAs from the typical Malaysian diet remains low. The TSC recommends that Malaysians consume more sources of PUFAs, and to use a cooking oil that is a blend of palm olein and other vegetable oils which are good sources of PUFAs (NCCFN, 2017). Karupaiah et al., (2016) reported that the ratio of n-6: n-3 for a typical Malaysian diet is 24.2 ± 9.6 from chemical analyses. Ng (1995) reported a ratio of 10 from calculations.

Earlier observations reported that elderly Malaysians (age: 60 to 93 years, n=344) in Johor, Negeri Sembilan and Melaka had energy intakes below the Malaysian RDA. The elderly in the 60 – 69 years age group consumed 33.56 ± 14.68 g/day; the 70 – 79 years group consumed 32.02 ± 15.13g/day; and those ≥ 80 years consumed 31.37 ± 14.33g/day. These intake levels were 19.98% to 25.07% of total energy intake (Suriah et al., 1996). A survey on 1,081 elderly individuals in publicly funded homes in Peninsular Malaysia showed that under-eating was a problem, with 14.3 percent had BMI < 18.5 kg/m^2 and almost one-third of respondents had BMI<20 kg/m2 (Visvanathan & Zaiton, 2006). Using a Nutritional Health Checklist (NHC), 32.1 % of these individuals were at moderate risk and 26.6 % were at high risk of undernutrition (Visvanathan et al., 2005). Individuals with low BMIs are very likely to also have sarcopenia (Iannuzzi-Sucich et al., 2002), which in turn is associated with poor health outcomes (Janssen et al., 2002; Janssen et al., 2004a; Janssen et al., 2000b). A study on 47 Malaysians aged ≥ 60 years (69.23 ± 8.63y) who were undernourished (BMI < 22 kg/m^2) showed that they scored poorly on the physical performance tests, had depression and a high risk of falls. Their biochemistry results showed that 10.9 percent were anaemic, 73% had hypoalbuminemia (< 3.0g/dL), and 21.7 percent were at risk of protein energy malnutrition (prealbumin 100 – 170 mg/L) (Singh et al., 2014).

Whilst inadequate dietary consumption is a problem for some older persons, over consumption is also a problem for some others. The National Health and Morbidity Survey, NHMS 2019 showed that prevalence of raised blood cholesterol among Malaysians aged 18 years and above was 38.1% (95%CI: 36.15 – 40.00); known hypercholesterolaemia was 13.5% (95%CI: 12.51 – 14.51) and undiagnosed hypercholesterolaemia was 24.6% (95%CI: 23.03 – 26.19). Older people aged 50 – 54 years had the highest prevalence of unknown raised cholesterol, 36.0% (95%CI: 31.93 – 40.28). Therefore, it is important to increase public health strategies to ensure health surveillance and nutritional and lifestyle preventative advice reach middle aged and older Malaysians. The prevalence of raised blood cholesterol was at least 60% in all age groups above 50 years and above. By ethnicity, Malays had the highest prevalence of raised blood cholesterol, 43.5% (95%CI: 41.46 – 46.57). The prevalence in other ethnic groups was in the range of 33.5% (95%CI: 29/29 – 38.07) among Bumiputera Sabah and 38.1% (95%CI: 32.85 – 43.58) among Indians (IPH, 2019).

The National Health and Morbidity Survey, NHMS 2015, reported an overall prevalence of hypercholesterolaemia (known and undiagnosed) of 47.7% (95% CI: 46.5 – 48.9) among adults aged 18 years and over. The prevalence of overall hypercholesterolaemia in NHMS 2015 was highest in the 55 – 59 years age group, at 68.8% (95%CI: 65.9 – 71.5). The prevalence of undiagnosed hypercholesterolaemia in NHMS 2015 was highest in adults aged 55–59 years, at 48.5% (95%CI: 45.3 – 51.8). The prevalence of hypercholesterolaemia in Malaysia increased from 20.7% in NHMS 2006 to 32.6% in NHMS 2011 to 47.7% in NHMS 2015. In NHMS 2019, the overall prevalence of hypercholesterolemia reduced to 38.1%. The ratio of known to undiagnosed hypercholesterolaemia was almost 1:2 (IPH, 2011; IPH, 2015; IPH, 2019). The NHMS data could be interpreted as current status of dietary intake is associated with high risk levels of NCD risks, and these risks cannot be ignored. Hence, guidance on dietary intake for the elolder persons should be given and implemented effectively.

Sodium intake

The nationally representative findings from MANS 2014 showed that median sodium intake for all adults was 1970 mg/day for men and 1914 mg/day for women (IPH, 2014), compared to MANS 2003 which showed median intake was 2588 mg/day for men and 2072 mg /day for women (Mirnalini et al., 2008). Among the 50 – 59 year olds, median sodium intake was 1829 mg/day for men and 1951 mg/day for women in MANS 2014 (IPH, 2014) compared to 2322 mg/day for men and 1850 mg/day for women in MANS 2003 (Mirnalini et al., 2008). Sodium intake decreased for elderly men but increased for women from 2003 to 2014. In both surveys, sodium intake was estimated from data from 24 hour food recalls. MANS 2014 also reported that median sodium intake was higher in Sabah and Sarawak (2007 mg/day) compared to Peninsular Malaysia (1934 mg/day). Urban dwellers had higher median sodium intake (2026 mg/day) compared to rural dwellers (1822 mg/day). Malaysians of Chinese ethnicity had the highest median sodium intake (2251 mg/day) compared to Bumiputera Sabahans (2026 mg/day), Indians (1870 mg/day), Malays (1839 mg/day) and Bumiputera Sarawakians (1776 mg/day).

However, the MySalt 2015 study involving 1,568 health staff aged 18 – 59 years, which measured sodium intake using 24-hour urine excretion analysis showed that the current sodium intake was 2860 mg/day (95% CI: 2776 – 2944) corresponding to 7.15g sodium chloride (IPH, 2016). Adults aged 40 years and above had mean sodium of 2838 mg (95%CI: 2697 – 2980). This study found that the main sources of sodium intake were light soy sauce (225.3 ± 564.0mg/d), fried rice (162.8 ± 356.3mg/d), egg omelette (156.6 ± 233.8 mg/d), nasi lemak (151.7 ± 222.9 mg/d), dark soy sauce (140.8 ± 585.3mg/d), and roti canai (139.0 ± 241.0mg/d). Most of the respondents in this study reported that salt is added to food cooked at home, but not added to food at the table. A majority of them reported that they took regular action to control salt intake by not adding salt at the table and avoiding processed food. It is important to note that despite this awareness and action, salt intake exceeded the RNI 2017 recommendation of 1500 mg/d for men and women aged 51–69 years, and 1200 mg/d for men and women aged ≥ 70 years (NCCFN, 2017). Usage of salt within the recommended amounts, and in combination with herbs and spices, could contribute towards adequate nutritional intake among the older persons, as it increases palatability of foods.

Over-consumption of salt is associated with increased blood pressure. NHMS (2019) showed a plateau in the prevalence of hypertension among Malaysians aged 18 years and above at 30.0%; known hypertension 15.9% (95% CI: 14.79 – 17.05), unknown raised blood pressure, 14.1% (95% CI: 13.13 – 15.19). Among those with unknown raised blood pressure, the highest prevalence was in the 45 – 49 years, 22.6% (95% CI: 18.48 – 27.35). (IPH, 2019). In comparison, NHMS 2015 showed overall prevalence of hypertension (known and undiagnosed) was 30.3% (95% CI: 29.3 – 31.2) (IPH, 2015). The prevalence was 32.7% (95% CI: 31.6 – 33.7) in NHMS 2011 (IPH, 2011). The age group with the highest prevalence of hypertension was the 75 years and older (81.7%, 95% CI: 77.03 – 85.55) in NHMS 2019 (IPH, 2019), the 70 – 74 years (75.4%, 95% CI: 70.5 – 79.7) in NHMS 2015 (IPH, 2015), and the 65 – 69 years (74.1%, 95% CI: 69.6 – 78.3) in NHMS 2011 (IPH, 2011).

Despite MANS 2014 showing Bumiputera Sarawakians had the lowest median sodium intake (1776 mg/day) (IPH, 2014), NHMS 2015 showed that Bumiputeras other than Malays had the highest prevalence of hypertension (33.4%, 95%CI: 30.6 – 36.3) (IPH, 2015) and NHMS 2019 again showed Bumiputera Sarawakians with the highest prevalence of hypertension (46.8%, 95%CI: 39.65 – 54.16) (IPH, 2019). NHMS 2015 grouped Bumiputera Sabahans and Sarawakians in the same category. Other predictors of hypertension such as age, obesity, physical activity, alcohol consumption and stress notwithstanding, dietary sodium consumption is the main predictor of raised blood pressure, with a dose-response relationship demonstrated in a cohort study (n = 4523) (Takase et al., 2015). Therefore, sodium intake in MANS 2014 might have been under-reported. Health professionals, particularly nutritionists and dietitians, family members, and older persons should make an effort to reduce sodium intake. Key Recommendation 3 of this Dietary Guideline provides suggestions to reduce sodium intake by demonstrating the various dietary sources of sodium and ways to reduce sodium consumption.

Sugar intake

The nationally representative data from MANS 2014 showed that mean sugar intake (white, brown and palm sugar (Melaka) was 28.28 g/day with a prevalence of 58.2% for men and 22.21 g/day with a prevalence of 53.3% for women. Mean intake of sweetened condensed milk or creamer was 57.00 g/day for all adults with a prevalence of 26.2% for men and 41.80 g/day with a prevalence of 20.6% for women (IPH, 2014), compared to MANS 2003 which showed that Malaysian adults consumed 4 teaspoons of sugar per day (28g) with a prevalence of 59%. Among those in the 50 – 59 years age group in MANS 2003, 58.2 percent consumed sugar daily with a frequency of 1.9 per day (Norimah et al., 2008). MANS 2014 also showed the prevalence of Malaysians aged 50 – 59 years consuming confectionary is 2.87% (95%CI: 2.60 – 3.13). The relative standard error was >25%, suggesting caution should be applied as there might have been high sampling error (IPH, 2014). A more recent Malaysian cross-sectional study (n = 1,057, aged ≥ 60 years) conducted in Johor, Perak, Kelantan and Selangor from March to September 2016 showed that mean total sugar intake was 40.5 ± 32.9 g/day (≈ 8 teaspoons), added sugar intake was 33.0 ± 30.9 g/day (≈ 6 teaspoons) and natural sugar was 7.4 ± 10.4 g/day (≈ 2 teaspoons) (Zainuddin et al., 2018). Natural sweeteners such as steviol glycosides and monk fruit (luo han guo) fruit extracts are high intensity sweeteners that are generally recognised as safe (GRAS). However expert opinion favours limiting consumption of any sweetener may well be better (Mooradian et al., 2017).

Sugar was usually added to beverages such as tea, coffee and chocolate-based drinks. Sugar is also contained in cordials, syrups, cocktails, sweetened boxed drinks, sweetened carbonated drinks dan sweetened dairy beverages. Sugar was also found in local cakes (kuih). Sugars in sweetened beverages and local kuih were not quantified in both the MANS 2014 and MANS 2003 studies. Zainuddin et al., (2018) reported that most added sugar came from sweetened beverages (27.7 ± 28.5 g/day), followed by fruits (6.1 ± 9.1 g/day), traditional kuih (3.9 ± 6.7 g/day), ready-to-eat sugar (1.5 ± 3.4 g/day) and dairy beverages (1.3 ± 4.2 g/day)

In a systematic review on the impact of energy intake and health status found that adults drinking sugar-sweetened beverages (SSBs) had a 7.8 percent higher energy intake than those who drank plain water. Amarra et al., (2016) in their review of MANS 2003 data from food frequency questionnaires, suggested that mean consumption of sweetened condensed milk was 30g/d (equivalent to 16 g sugar) and 21 g/d of table sugar (sucrose), which together were below the WHO recommendation of 50 g/d of sugar for every 2000 kcal/d. However, they cautioned that more accurate data using biomarkers of sugar intake should be carried out (Amarra, Khor & Chan, 2016). In an earlier study on elderly Malays aged > 60years in rural areas in the east coast of Peninsular Malaysia, Shahar et al., (2000) reported that sugar contributed 14% of total energy intake. The TSC for RNI 2017 recommends that added sugar should be less than 10% of total energy intake (NCCFN, 2017).

In Malaysia, the overall prevalence of diabetes mellitus (known and undiagnosed) among adults aged 18 years and above increased from 15.2% (95% CI: 14.2 – 16.1) in NHMS 2011 (IPH, 2011) to 17.5% (95%CI: 16.6–18.3) in NHMS 2015 (IPH, 2015) to 18.3% (95% CI: 17.08 – 19.58) in NHMS 2019 (IPH, 2019). The NHMS 2019 showed that was no difference in prevalence of diabetes between urban and rural populations. The 65 – 69 year age group showed the highest prevalence, 43.4% (95% CI: 37.37 – 49.65) (IPH, 2019). In NHMS 2015, the age group with the highest prevalence was the 70 – 74 years, 27.9% (95% CI: 22.7 – 33.9) (IPH, 2015). In NHMS 2011, the age group with the highest prevalence was 65 – 69 years, 36.6% (95% CI: 32.0 – 41.4) (IPH, 2011). Given the increasing prevalence of diabetes, this Dietary Guideline provides suggestions to decrease sugar intake and preference for sweetness, taking into account the food preferences of Malaysians with regards to foods containing sugar.

Nutrition information on food label

Nutrition information on food label might be helpful in aiding people in making their decision regarding sugar intake. Food labelling includes any written, printed or graphic matter that are present on the label, accompanies the food, or is displayed near the food, including that for the purpose of promoting its sale (MOH, 1985). Food label provides information to consumers to better understand and make the right choice based on the food labelling information. Nutrition labelling and claims regulations have been gazetted in Malaysia in 2003 and several small scale studies have been conducted since 2010 to determine consumer understanding and utilisation of nutrition labelling. Nutrient content of a food product are declared as a table in one section of a food label, commonly known as a nutrition information panel (NIP). As for the claims, three (3) types on nutrition claims are permitted in Malaysia, namely nutrient content claim, nutrient comparative claim and nutrient function claim (MOH, 2010).

Findings from Malaysia Adult Nutrition Survey (MANS), a nationwide cross-sectional study among respondents aged 18-59 years old in non-institutionalized living quarters, found that only 4 out of 10 Malaysia adults reported reading labels and the most frequently read information was expiry date (Ambak et al., 2018). From the same study, males were found 1.52 times more likely not to read labels. MANS (2014) showed that among those who read nutrition label when buying or receiving food (45.0%, n = 129), the nutrition information that was most commonly read was carbohydrate, including sugar [21.6%, 95%CI: 19.44 – 24.0)] (IPH 2014), followed by fat content [19.9%, (95% CI: 17.9 – 22.1)], total energy [14.5% (95%CI: 12.9 – 16.2)], salt / sodium [6.7% (95% CI: 4.6 – 6.7)], and food additive [6.2% (995%CI: 5.0 – 7.5)] (IPH, 2004).

There has been no published national study focusing on nutrition labelling among older persons in Malaysia except from the Third National Health and Morbidity Survey III (NHMS III) undertaken in 2006 (IPH, 2008). In this survey, a total of 4,429 older persons aged 60 years and above participated and those with formal education were found more likely to understand nutrition labelling as compared to those without formal education. This finding implied that education programmes for older persons must be planned to create awareness of reading nutrition information using their preferred mass media such as electronic media, talks and counselling. Although food label is beneficial as a source of information about a product, it is only one of the educational tools in guiding food choices. Other reliable sources of nutrition information should be used to underpin nutrition knowledge about food and nutrition and their role in health and disease.

5.5 Key Recommendations

Key Recommendation 1

Reduce consumption of foods high in fat, particularly foods high in saturated fatty acids (e.g. animal fat and skin, full fat dairy products, coconut milk, and coconut oil) and trans fatty acids.

How to Achieve:

1. Choose foods prepared or cook with cooking methods using little fat or oils such as stir-frying, grilling, pan-frying, and air-frying.
2. Reduce consumption of deep-fried foods (e.g., *keropok, vadai, yue char kuay, keropok lekor*) or fried battered foods (e.g., *pisang goreng, tempura*).
3. Avoid the practice of adding excessive high-fat or coconut milk-based gravy (*kuah banjir*) on to rice or other cereal products such as *roti canai* or *chapati*.
4. Reduce consumption of foods added with butter, cheese or cream cheese (e.g., *pisang goreng* cheese and potato wedges added with cheese).
5. Remove chicken skin and trim visible fat from meat when preparing food.
6. Limit intake of foods rich in trans fatty acids (e.g., hard margarine and shortening) and foods prepared with these ingredients such as pastries, bakery products, fried food, and fast food.

Key Recommendation 2

Consume foods rich in polyunsaturated fatty acids.

How to Achieve:

1. Consume more foods that are rich in omega-3 polyunsaturated fatty acids from either plant (e.g., flaxseed, chia seed, walnut, sesame, cashew nut, and pistachio) or animal sources (e.g., *siakap, cencaru, and kembung*).

2. Mix an equal amount of palm olein oil with any omega-6 polyunsaturated fatty acids rich oil and use it for all types of cooking except deep-frying.

Key Recommendation 3

Reduce salt intake.

How to Achieve:

1. All types of salt (e.g., salt, rock salt, sea salt, Himalayan salt) contain sodium. Do not consume more than 1 teaspoon (5g) daily.

2. Use fresh or dry herbs and spices (e.g., lemon, garlic, ginger, onions, lemongrass and white pepper), and stock from boiled chicken or anchovies to replace commercially flavoured cubes, ready-to-use seasonings, salt, monosodium glutamate (MSG), soy sauce, oyster sauce, fish or prawn sauce (e.g., *budu, cencaluk, belacan*).

3. Soak anchovies, salted fish and dried shrimp in water to reduce the salt.

4. Limit intake of smoked, fermented, preserved foods (e.g., pickled fruits and vegetables) with high salt content (e.g., salted fish, salted egg, salted vegetables and fruit pickles), high salt snacks (e.g., salted nuts, potato crisps and fish/ prawn crackers, *papadum*), processed foods (e.g., canned sardines, sausages, chicken nuggets, meatball and burger patty), fast foods, instant noodles, instant soup and other types of highly seasoned foods (e.g., dishes made from salted egg powder).

5. Request for less salt when eating out or choose food with MyChoice logo when having takeaways or food deliveries.

6. Request for a less salt product from the producer of ready-to-eat homemade products sold online or other form of sale points.

7. Read food labels and Nutrition Information Panel on sodium (salt) and choose no added/ reduced/ less/ light/ low/ very low/ free sodium (salt), or those with Healthier Choice Logo (HCL) if available. Compare with other available brands of the same product and choose the one with the lower sodium content.

8. Limit the gravy or sauces added into food eaten at home or when eating out (*kuah banjir*). Do not finish the soup in noodle dishes that contain a lot of salt or seasoning.

Key Recommendation 4

Reduce consumption of sugary foods.

How to Achieve:

1. Consume foods containing sugar such as *kuih* and cakes less frequently.

2. Choose *kuih* and cakes with less sugar.

3. Replace sweet dessert such as *kuih* and cakes with healthier options such as low sodium high fibre biscuits, unsalted roasted nuts, and fresh fruits.

4. When enjoying a sweet dessert such as chocolates, *kesari* and *pengat*/ tong sui occasionally, limit it to one small piece or one small bowl which are made to be less sweet.

5. Read food label to identify foods that are high in sugar. Identify and choose product with low sugar label.

Key Recommendation 5

Reduce consumption of sugary beverages.

How to Achieve:

1. Choose plain water rather than carbonated and non-carbonated sugary drinks (e.g., soft drinks, syrup, and cordial).

2. Train yourself to prefer beverages without any additional sugar or sweeteners. Inform family members and carers that you prefer beverages that way.

3. If using table sugar, sweetened condensed milk or sweetened condensed creamer, limit to not more than one teaspoon of sugar or one dessert spoon of condensed milk/creamer per cup of beverage.

4. When ordering drinks, ask for less sugar or less sweetened condensed milk or sweetened condensed creamer.

5. When ordering tea or coffee, ask to replace sweetened condensed milk/ evaporated milk/ creamer without added sugar.

6. When ordered beverages arrive as too sweet, either ask for it to be remade to less sweet, or discard half its content and add plain water.

7. Reduce the consumption of beverages and products containing high sugar such as packed, bottled and canned drinks including cordial, *air batu campur* (ABC), and *cendol*.

8. If you choose instant beverages (premix), choose the ones without added sugar or creamer.

Key Recommendation 6

Use nutrition information on food label effectively.

How to achieve:

1. When purchasing products, obtain information from Nutrition Information Panel (NIP) on the amount of energy and other nutrients.

2. Familiar yourself with nutrition information and consider how the energy and nutrients contained in the food contribute to your total nutrient intake of the day.

3. Make a comparison of all the nutrients on the label of the different brands available for the same product.

4. Make use of nutrient content claim wisely. Look out for the following keywords and consider them as a healthier option: "low in", "free of", "zero", "no" or words of similar meaning for product that contains certain nutrients that are to be reduced in intake such as sugar, sodium and cholesterol. Meanwhile, "source of", "high in", "rich in" or words of similar meaning indicate product that contains certain nutrients that are encouraged to be consumed such as calcium, dietary fibre and protein.

5. Make use of nutrient comparative claim wisely. Look out for the following keywords and consider them as a healthier option: "reduced", "less than", "light", "fewer" or words of similar meaning for product that has a new formulation with lower or reduced amount of certain nutrients. Meanwhile, "increased", "more", "extra" or words of similar meaning indicate product that has a new formulation with increased or extra amount of certain nutrients.

6. Front-Of-Pack Nutrition Labelling (FOP-NL) such as Energy Icon and Healthier Choice Logo (HCL) can be used as an additional guide to make informed choices on healthier options.

7. Make use of the ingredient list on food label. Ingredients are listed in descending order by weight; the first few ingredients listed are the main ingredients of that particular product. If sugar/ or salt are listed in the beginning of ingredients list, the product is high in sugar/ or salt.

8. Familiarise yourself with different terms of an ingredient. For example, sugar may be listed as sucrose, glucose, fructose or corn syrup. Salt may be listed as sodium chloride, baking soda, baking powder or monosodium glutamate (MSG). Fat may be listed as oil, butter or margarine.

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Consume foods and beverages with low fats, salts and sugar

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Key Message 6

Practise food safety when purchasing, preparing, cooking and storing food

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Practise food safety when purchasing, preparing, cooking and storing food

Ms. Roziaton Abdul Wahab, Prof. Dr. Wan Rosli Wan Ishak, Assoc. Prof. Dr. Chai Lay Ching, Ms. Nurul Huda Aziz and Mr. Mohd. Nurul Haryadie Mazuki

6.1 Terminology

Cholera
Cholera is an acute diarrheal infection caused by toxigenic bacterium *Vibrio cholerae* through contaminated food and water. It has a short incubation period of 2 hours to 5 days. Cholera causes mild to severe symptoms. In the severe case, Cholera causes profuse watery diarrhea that could lead to dehydration and even death if untreated. Cholera is affecting both children and adults, particularly in area with inadequate access to clean water and sanitation facilities (WHO, 2017).

Cross-contamination
The transmission of microorganisms from one object to another is known as cross-contamination. Cross-contamination of ready-to-eat food can happen in many ways such as via contaminated food handler hands, contaminated utensil, chopping board and direct transfer from raw food ingredients.

Dysentery
Dysentery is bloody diarrhoea, i.e. any diarrhoeal episode in which the loose or watery stools contain visible red blood. Patients typically experience mild to severe abdominal pain or stomach cramps. In some cases, untreated dysentery can be life-threatening, especially if the infected person cannot replace lost fluids fast enough. Dysentery is most often caused by Shigella species (bacillary

dysentery) or *Entamoeba histolytica* (amoebic dysentery). These pathogens typically reach the large intestine after entering orally, through ingestion of contaminated food or water, oral contact with contaminated objects or hands, and so on.

Food poisoning
Acute onset of vomiting and/or diarrhoea and/or other symptoms associated with ingestion of food. May also presented with neurological symptoms such as paresthesia, motor weakness and cranial nerve palsies.

Food safety control measures
Any action or activity that can be used to prevent foodborne illnesses.

Gastrointestinal illness
A gastrointestinal (GI) illness is an infection that primarily affects the gastrointestinal system, most commonly the stomach, small intestine, and large intestine. Bacteria or viruses cause most gastrointestinal illnesses, although sometimes toxins created by bacteria can also cause GI illness.

Hepatitis A
Hepatitis A is a viral liver disease that can cause mild to severe illness. Symptoms of hepatitis A range from mild to severe, and can include fever, malaise, loss of appetite, diarrhoea, nausea, abdominal discomfort, dark-coloured urine and jaundice. The hepatitis A virus (HAV) is transmitted through ingestion of contaminated food and water or through direct contact with an infectious person. Safe water supply, food safety, improved sanitation, hand washing and the hepatitis A vaccine are the most effective ways to combat the disease.

Immuno-compromised
Immuno-compromised is a condition where the immune system is not functioning for certain number of reasons. Immuno-compromised patients are very susceptible to infection and vulnerable to the rapid spread of diseases such as cancer. There can be several reasons why a patient might be immuno-compromised but in any event it is the result of a lack of white blood cells or the inability of a patient to produce antibodies.

Malnutrition
The condition that occur as a result of eating a diet that is deficient in particular nutrients or excessive in certain nutrients (too high in intake), or in the improper proportions of inadequate calories or specific diet components.

Malnutrition refers to deficiencies, excesses or imbalances in a person's intake of energy and/or nutrients. The term malnutrition covers two (2) broad groups of conditions which are undernutrition and overnutrition. Undernutrition is a condition where micronutrient deficiencies or insufficiencies happened (a lack of important vitamins and minerals).While overnutrition is a condition which consist of overweight, obesity and diet-related non-communicable diseases such as heart disease, stroke, diabetes and cancer (WHO, 2006).

Pasteurization
Pasteurization is the process of heating a food, usually a liquid, to a specified temperature for a predefined length of time and then cooling it immediately. Pasteurization aims to slow down microbial food spoilage in food and hence prolong shelf-life.

Pathogen
A pathogen, or also known as an infectious agent, is a biological agent that causes disease or illness in its host. They include a few types of bacteria, viruses, protozoa, and other organisms. Pathogens could spread through contaminated water, food, air and vectors to cause waterborne, foodborne, airborne and vector-borne diseases, respectively.

Safe water
Water that is safe to drink or used for food preparation without causing health problems.

Treated water
Treated water is drinking water that has been chemically treated, filtered, or boiled to eliminate infectious bacteria, viruses, and parasites (such as *Giardia lamblia*). These organisms can cause diarrhea and mild to severe illness.

Typhoid/ Paratyphoid
Typhoid and paratyphoid fevers are caused by the bacteria *Salmonella Typhi* and *Salmonella Paratyphi,* respectively. Bacteria causing typhoid and paratyphoid are passed in the faeces and urine of infected people. Symptoms can be mild or severe and include sustained high fever, malaise, anorexia, headache, constipation or diarrhoea, rose-coloured spots on the chest area and enlarged spleen and liver. Most people show symptoms 1-3 weeks after exposure. Paratyphoid fever has similar symptoms to typhoid fever but is generally a milder disease.

People become infected by eating or drinking food or beverages that have been handled by an infected person, or by drinking water tainted by sewage carrying the bacterium.

Clean water, hygiene and good sanitation prevent the spread of typhoid and paratyphoid. In addition, typhoid vaccine can prevent typhoid. There are two vaccines to prevent typhoid. One is an inactivated (killed) vaccine given as a shot. The other is a live, attenuated (weakened) vaccine which is taken orally (by mouth). Typhoid vaccination is currently mandatory for all food handlers under the Malaysian Food Act 1983 and Food Hygiene Regulations 2009.

6.2 Introduction

Unsafe food poses global health threats, endangering everyone. Infants, young children, pregnant women, older persons and those with an underlying illness are particularly vulnerable. Every year 220 million children contract diarrhoeal diseases and 96000 die. Unsafe food creates a vicious cycle of diarrhoea and malnutrition, threatening the nutritional status of the most vulnerable (WHO, 2020).

Food borne illness is caused by presence of foodborne hazard in a food or beverages and is defined as a biological, chemical or physical agents in, or condition of, food, with the potential to cause and adverse health effect. Many of these hazards have long been recognized and addressed by food safety control measures. These include biological hazards such as infectious bacteria, toxin-producing organism, moulds, parasites and viruses, chemical hazard such as natural toxins, food additives, pesticides residues, veterinary drug residues, environmental contaminants and allergen as well as physical hazard such as metal, glass, stone and bone chips (WHO, 2015).

Changes in organ and body systems as well as eating lifestyle are major factors which influence food safety among older persons. As individuals age, body function and immune system reduce and not at par as compared with the younger people. In addition, poor nutrition and poor blood circulation may result in a weakened immune system in older persons. These changes often make older persons more susceptible to contracting a food borne illness. For example, their stomach and intestinal tract may hold on to foods for a longer period of time; their liver and kidneys may not readily rid their bodies of toxins; and their sense of taste or smell may be altered. Older persons are more likely to have a lengthier illness, undergo hospitalization, or even die if suffering from food borne illness (Buzby, 2002).

Access to food among older persons may vary and each may contribute to food safety issues. While preparing food by themselves, older persons may have a problem with sense of taste, smell and sight and they can't determine whether the food especially raw materials are good or spoiled and prepared in a safe way.

Besides self-cooking, eating out is very convenient especially for older persons who are not able to cook. However, not all food premise operators implement food safety and hygienic measures including premise hygiene, safe food handling and suitable storage temperature. The dining area of a restaurant may look good and welcoming, but the kitchen may not be in the same condition. The Food Safety and Quality Division (FSQD), Ministry of Health conducts premise inspection as routine activity to ensure premise hygiene is in compliance with the Food Act 1983 and Food Hygiene Regulations 2009. In 2020, FSQD reported a total of 67,976 food premises have been inspected and 1,751 out of the total were in unsanitary condition and have been ordered to be closed under Section 11, Food Act 1983 (FSQD MOH, 2020).

Take-away food is also popular nowadays. However, lack of knowledge in terms of storage methods and temperature as well as reheating food can lead to microbial growth and multiplication that could cause food borne illness.

Ordering food online or by phone calls is a new trend in getting foods. Food handlers may not practice food safety and hygiene while preparing, storing and transporting food. Transporting food by operators over a long period of time may allow microbial growth which can affect safety of the foods.

Attending feasts and gatherings such as *kenduri* are common practices among the Malaysians, especially older persons. However, the foods are cooked much earlier prior to serving time, in large quantities and stored at room temperature. Besides, some of the food handlers especially the ones who never attend food handler's training may not have sufficient knowledge in food safety and hygiene.

In other situations, there are many older persons who stay at older persons care homes or nursing homes. They rely on health care workers to prepare food for them. Some of these workers may not have been trained in food safety. In addition, the kitchens in these care homes may have not been inspected by the health officers. Furthermore, food borne infections may easily spread among the residents of these homes due to the close confinement with others who may be ill.

Food handlers play a major role in transmitting pathogens passively from contaminated sources such as transmitting pathogens from raw meat to a ready to eat food. Food handlers may also carry some human specific foodborne pathogens such as Hepatitis A, noroviruses, typhoidal Salmonella, *Staphylococcus aureus* and Shigella sp. in their hands, cuts or sores, mouth, skin and hair. Food handlers may also shed foodborne pathogens, such as E.coli O157:H7 and non-typhoidal Salmonella during the infectiousness period or less important during recovery period of a gastrointestinal sickness.

6.3 Scientific Basis

Older persons population is considered as immuno-compromised group and likely to experience severe complications due to food borne illness. Gastrointestinal problems and vomiting are most common acute problems experienced by this group. Even though death is relatively rare due to food borne illness, complications and symptoms experienced by this group is mainly a concern. There are a few factors that contribute susceptibility and severity of food borne pathogens among older persons (Patricia et al., 2006).

1) The aging immune system – Immune system is decreased rapidly in older persons due to aging process. Intestinal motility and mucosal immune function decrease with normal aging, increasing susceptibility to systemic infection via the gut. Prolong used of antibiotic will lose the competitiveness of natural micro flora due to the overgrowth of pathogens and this condition will contribute to the food borne infections when they are exposed to newly emerging or genetically mutated pathogens.

2) Chronic diseases – Older persons population are at increased vulnerability because of immune suppression associated with aging. Diseases such as high blood pressure and diabetes and therapeutic regimens will increase their exposure to food borne illness.

3) Nutritional status and physical activity – Studies have showed that older persons are more likely to experience malnutrition, which may also increase susceptibility to infection. A deficiency of protein, zinc, selenium, iron, copper, vitamins A, C, E, or B-6, or folic acid can lead to impaired immune function. Malnutrition in older persons is caused by several factors including physical impairments, early satiety, taste and smell changes and difficulty in chewing and swallowing food. Furthermore, older persons population will reduce their physical activity and sedentary lifestyle will affect the immune system which decreasing of a few antigens such as IL-2 and IL-4.

Hence, older persons are more vulnerable to foodborne illnesses due to their generally weakened immune system, malnutrition and other chronic diseases they may suffer. Poor food handling and practices may pose a higher risk to older persons population.

Pathogens based on types of Malaysian food, symptoms and onset time after ingesting are listed in Table 6.1.

Table 6.1 Pathogens based on common types of Malaysian food, symptoms and onset time after ingesting

Name of pathogen	Food and causes linked to outbreaks	Possible symptoms (from most to least common)	Onset time after ingesting
Bacillus cereus (bacteria)	Rice, starchy foods such as potato and pasta, meat, casseroles, vegetable dishes, foods containing spices e.g.: steamed rice, fried rice, *nasi lemak, roti canai,* noodles.	Two different forms of food poisoning: diarrhoea, abdominal pain, occasional nausea.	1 to 6 hours (vomiting) 10 -12 hours (diarrhoea)
Campylobacter (bacteria)	Undercooked chicken, unpasteurized milk, undercooked chicken liver, untreated drinking water, raw vegetables e.g.: *ulam-ulaman,* raw milk, undercooked grilled or fried chicken.	Muscle pain, headache, fever followed by diarrhoea (can be bloody), abdominal pain, nausea.	2 to 5 days, range 1 to 10 days
Clostridium perfringens (bacteria)	Meat dishes, especially rolled roasts, stuffed meat, soups, stews, gravies, pies e.g.: chicken/meat curry, chicken/ meat soup	Severe abdominal pain, watery diarrhoea, occasional vomiting and nausea.	10 to 12 hours, range 6 to 24 hours
Cryptosporidium parvum (parasite)	Untreated drinking water, raw fruits and vegetables, fresh fruit juice, unpasteurized milk, salads.	Watery diarrhoea, vomiting, stomach cramps, weight loss.	3 to 11 days
E. coli (O157:H7) (bacteria)	Undercooked minced meat, unpasteurized milk, lettuce, sprouts, unpasteurized fruit juices, raw fruits and vegetables. e.g.: *ulam, kerabu,* blanched/ stir-fried vegetable, *sambal* belacan and air asam, minced meat.	Severe abdominal pain, watery (then bloody) diarrhoea, occasional vomiting.	1 to 8 days
Hepatitis A (virus)	Contact with infected people, shellfish, infected food handlers.	Fever, malaise, nausea, vomiting, loss of appetite, abdominal pain, jaundice.	10 to 50 days
Listeria monocytogenes (bacteria - can grow slowly in refrigerator temperature)	Raw milk, cheese, ice cream, raw vegetables, sausages, improper heating of meat and meat products, smoked seafood. e.g.: milk, yoghurt, grilled fish	Fever, chill, headache, backache, sometimes stomach upset, abdominal pain and diarrhoea.	12 hours

Name of pathogen	Food and causes linked to outbreaks	Possible symptoms (from most to least common)	Onset time after ingesting
Salmonella (bacteria)	Raw meats, poultry, unpasteurized milk and dairy products, seafood, fresh produce (including sprouts), and foods handled by infected food handlers. e.g.: undercooked food – hard/ half boiled/ fried egg, fish/ poultry/ meat with gravy, fresh milk, yoghurt, *ulam* and food that involve direct handling. e.g.: *Kebabs,* Sandwiches	Nausea, vomiting, abdominal cramps, diarrhoea, fever, headache.	6 hours to 2 days
Staphylococcus aureus (bacteria)	Ham, cooked meats, yoghurt, chicken salad, pasta dishes, bakery products (especially cream-filled), cheese e.g.: chicken/ meat curry, chicken/ meat soup, yoghurt	Nausea, vomiting, abdominal cramps, occasional diarrhoea.	2 to 4 hours, range 30 minutes to 7 hours
Vibrio parahaemolyticus (bacteria)	Raw seafood such as oysters, clams, crabs, shrimp, fish e.g.: seafood curry/ grilled	Diarrhoea, abdominal pain, nausea, vomiting, headache, fever, chills.	4 hours to 4 days
Shigella (bacteria)	Potato salad, tuna, shrimp, pasta, raw vegetables, water and ice e.g.: *ulam, kerabu,* water, ice cube	Diarrhoea (often bloody), fever, stomach cramps	12 to 50 hours

Sources:
1. Ministry of Primary Industries, New Zealand, 2015
2. U.S Department of Agriculture and Food and Drug Administration, 2011
3. Ministry of Health Malaysia, 2009

6.4 Current Status

A food borne disease outbreak is occurrence of two or more cases of similar illness resulting from ingestion of a common food. It is rather common in Malaysia that many such outbreaks go unreported or not investigated. The outbreak of food borne disease still occurs in certain high-risk areas where they must be not properly managed. It is a major obstacle in outbreak investigation when in almost all cases, the contaminated food could not be traced (Shafira Ezat, 2013).

The increase of food poisoning cases might indicate that the food handlers have been neglecting the importance of safe food handling in the kitchen. Mortality has been shown to be associated with cholera from 2005 to 2011, but none was reported for the next two years. In 2012, none of the diseases cause mortality, however in 2013 food poisoning and typhoid were shown to cause death of Malaysian citizen.

Malaysian government has implemented few rules to ensure owners of food premises abide by the rules to avoid illness and outbreak. The hygiene level of the kitchen is the most important aspect that should be given attention because it will reflect the safety of the food to be consumed. In addition, the temperature in the kitchen is usually higher than the dining area which makes it a perfect condition to promote bacterial growth. It has also been proven that foodborne bacteria can grow on most of the surfaces in the kitchen like cutting board, cloth, sink, cleaning sponge and knives (Kusumaningrum et al., 2003; Mattick et al., 2003). Cross contamination to food could occur if those items are not properly clean and food handlers neglect the correct way of food preparation.

The main reason for foodborne illness in Malaysia is insanitary food handling procedures which contribute to 50% of the cases (MOH, 2007) for instance, the preparation of food in advance, inappropriate ways of cooling and insufficient temperature during reheating of food (Beumer & Kusumaningrum, 2003). All of these mishandlings permit the growth of microbial pathogens because they fail to kill the bacteria or they help to keep the bacteria dormant before they reach sufficient temperature to multiply. It is known that temperature play important part to determine the microbial activity and shelf life of food product (Aung & Chang, 2014). Hence, temperature manipulation, as well as hygiene and sanitation of kitchen environment should be the highest priority to ensure safe foods with low risk of contamination are served to consumers.

The food borne disease outbreak occurred mainly due to insanitary food handling procedures which had more than 50% of the poisoning episode due to inappropriate food handling methods, meals prepared too early, food kept in the ambient temperature until served and unhygienic practices (MOH, 2007).

Incident rate of food borne and water borne illness cases in Malaysia is increasing especially related to food poisoning. It is due to lack of awareness on good hygiene practice and poor handling practices among food handlers and consumers. However, the mortality rates of food borne and water borne illness cases in Malaysia are small which are mainly caused by cholera and food poisoning.

While foodborne illness data among older persons are lacking in Malaysia, studies in United States of America showed that mortality rate for infections caused by *Listeria monocytogenes* is highest among older persons due to consumption of raw milk products, smoked seafood, soft cheeses, refrigerated pate or meat spreads, and ready-to-eat foods, such as hot dogs and luncheon meats (Buzby, 2002). In FoodNet sites from 1996–1999, hospitalizations for Salmonella (cause by undercooked dairy products) and Campylobacter (caused by water) infection were most likely to occur among older persons. It was reported that 47% of persons aged 60 years old or older were hospitalized (Kennedy et al., 2004) due to Salmonella and 27% of persons aged 60 years old or older were hospitalized (3.5 times) more likely than persons aged 10 years old to 49 years old because of Campylobacter infection (Samuel et al., 2004). In recent multistate outbreak of Salmonella infections associated with peanut butter and peanut butter-containing products, over 500 persons were infected, 116 patients were hospitalized, and the infection may have contributed to 8 deaths in patients aged 59 years old and older. This recent outbreak underscores the fact that older adults are particularly susceptible to life-threatening illness as a result of food-borne infections.

As we get older, our olfactory function declines. Not only we lose our sense of smell, we lose our ability to discriminate between smells too. Chewing problems associated with tooth loss and dentures can also interfere with taste sensations, along with the reduction in saliva production. Relatives, neighbours or care taker need closer involvement to check for spoiled food that if eaten could lead to food poisoning (Boyce & Shore, 2006).

6.5 Key Recommendations

Key Recommendation 1

Practice good personal hygiene.

How to achieve:

1. Wash hands with soap and water after going to toilet.
2. Wash hands before handling food and during food preparation.
3. Handle ready to eat food with utensils rather than hands as much as possible. If using gloves, make sure it is single use.
4. Wash all surfaces and equipment used for food preparation with clean water and detergent and disinfect, if necessary.
5. Protect kitchen areas and food from insects, pests and other animals by keeping the kitchen and surrounding areas cleaned.
6. Cover cuts or wounds on hands with clean bandages and gloves when handling food.
7. Avoid preparing food if feeling ill (e.g : diarrhoea).
8. Seek for treatment from healthcare provider if experience symptoms of food poisoning such as diarrhoea, vomiting, abdominal pain and fever.

Key Recommendation 2

Avoid cross-contamination when preparing food.

How to achieve:

1. Separate raw meat, poultry and seafood from other foods such as fruits and vegetables.
2. Use separate equipment and utensils such as knives and cutting boards for handling raw foods and cooked food. Change kitchen towels daily or after every meal prepared with raw meat and poultry. It is recommended to use paper towel.
3. Clean kitchen towel after use.
4. Sanitizes plates and utensils with hot water,especially if there are sick persons at home.
5. Do not thaw frozen foods at room temperature. The recommended ways to thaw frozen foods are as below:
   a. Ideally, plan ahead to leave enough time and space to defrost small amounts of food in the fridge.
   b. If defrosting food in the chiller is not possible, put it in a container and then place it under running water.
   c. Food can be defrosted in microwave using 'defrost' setting.
6. Wash fruits and vegetables (e.g: *ulam*) thoroughly before consumption.

Key Recommendation 3

Storing food safely.

How to achieve:

1. Promptly refrigerate or freeze perishable leftovers. Store food in covered containers for one serving in refrigerator or freezer.
2. Store food in covered containers separately to avoid contact between raw and cooked foods inside refrigerator or freezer.
3. Set the temperature of refrigerator to <4°C or <-10°C for freezer. Avoid storing too many foods in the refrigerator or freezer as it will cause temperature to rise and lead to microbial growth.
4. Store cooked foods not longer than 3 days in refrigerator and 2 months in freezer (state the date when food was cooked).
5. Read the label for proper storage method after opening. Check the condition of food (e.g. mould-growth, color or smell changes etc.) before eating.

6. Store raw eggs at room temperature or in refrigerator. If store in refrigerator, the eggs must be stored in a covered container separately from other foods to avoid cross contamination. In addition, eggs must be washed prior cooking to avoid cross contamination.

Key Recommendation 4

Cook foods thoroughly.

How to achieve:

1. Cook foods thoroughly, especially meat, poultry, eggs and seafood.

2. Boil food like soup consistently and evenly.

3. Cook eggs until the yolk and white are firm. If possible, avoid half-boiled eggs. The egg which is not well cooked may still contain harmful bacteria.

4. Reheat cooked foods thoroughly and avoid reheating the same foods more than once.

Key Recommendation 5

Serve foods under safe conditions.

How to achieve:

1. Eat food immediately after serve. Do not leave cooked foods at room temperature for more than 4 hours.

2. Keep hot foods hot or on a warmer (above 60 °C) prior to serving. Keep cold foods cold or in the refrigerator or freezer (<4°C / <-10°C).

3. Place cooked foods on clean containers.

4. Make sure food is covered appropriately

5. Use clean utensil to cut and serve ready-to-eat food.

6. Practice visual, smell and taste before serving and/or eating food. Check the food for signs of visual decay. Smell the food to test whether it smells as it usually would. Taste a small amount of the food to check whether it's turned unusually tangy or sour.

7. If possible, avoid serving and eating raw foods because the risk to become contaminated with pathogenic bacteria is high and may cause severe health implication (e.g. raw milk, half cooked meat).

Key Recommendation 6

Use treated water to cook food

How to achieve:

1. Use treated water to cook food.

2. If using well water or ground water, the water need to be boiled prior to usage.

Key Recommendation 7

Choose or purchase safe foods.

How to achieve:

1. Select fresh and clean foods, check for mold and soft spots when choosing vegetable and fruit. For chicken and beef, practice smell and visual. Red meat should be bright red and chicken meat should be pink. –Choose fresh food with certified logo such as Good Agricultural Practices (GAP) and Veterinary Health Mark (DVS) for food safety assurance.

2. Buy processed food with certified food safety assurance programme logo such as MeSTI (Makanan Selamat, Tanggungjawab Industri), GMP (Good Manufacturing Practices), HACCP (Hazard Analysis and Critical Control Point), and ISO 22000 (Food Safety Management System).

3. Choose safe food such as pasteurized milk instead of raw milk.

4. Choose canned foods that are not rusted, dented or bulging. For pre-packaged foods, ensure that they are properly sealed, not torn or leaking.

5. Do not use food beyond its expiry date.

6. Be very careful when buying food from online platform. Buy food from trustworthy manufacturer and sources.

Key Recommendation 8

When eating out, choose safe and clean premises.

How to achieve:

1. When eating out, choose safe and clean premises. If possible, choose premise with grade A recognition from competent authority.

2. Choose premises that are in clean and tidy conditions with running pipe water, proper drainage system, and covered rubbish bins. The premises should be free from signs or presence of pets, rodents, pests and insects.

3. Choose premises that use cutleries and utensils that are in clean and good condition..

4. Choose premises that serve appropriate covered foods and beverages.

5. Choose premises where the staffs practice good personal hygiene and habits at work.

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Malaysian Dietary Guidelines for Older Persons Focus Group Discussion on the Key Messages, Key Recommendations & How to Achieve (5th January 2023)

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